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32384 CHARLES COUNTY POMONKEY PUMP STATION IMPROVEMENTS

PLC/RTU CONTROL PANEL RECORD SUBMITTAL

SECTION 17441 - CONTROLS SYSTEM HARDWARE

November 11, 2026

ENGINEER: WRA

CONTRACTOR: JOHNSTON CONSTRUCTION

32384 CHARLES COUNTY POMONKEY PUMP STATION IMPROVEMENTS

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32384 CHARLES COUNTY POMONKEY PUMP STATION IMPROVEMENTS

General Comments:

1. The digital process meter has been submitted under a separate cover.

Resubmittal #17250-01-02 Comment Confirmations:

1. Control Panel is approved. Final panel layout and code compliance remain the Contractor's responsibility.

   Sherwood Logan Response:

   Acknowledged.

2. R2 S&L Response: The digital chart recorder has been removed from the BOM, panel, and the wiring diagrams for the enclosure.

   R2 WRA Response: Page 185 (of 384) currently lists “RSG35” as a BOM item. Revise as required and provide record copy

   Sherwood Logan Response:

   The appropriate line containing the chart recorder has been removed from the drawings BOM.

Parts Index:

SHT	MANUFACTURER	MODEL	DESCRIPTION
1	SCE	SCE-60EL3616SS6LPPL	NEMA 4X Wall Mount Enclosure
1	SCE	SCE-DF60EL36LP	Dead Front Panel
1	SCE	SCE-60P36	Enclosure Backpanel
1	SCE	SCE-ELMFK4	Wall Mount Foot Kit
1	SCE	SCE-BVKD	NEMA 4X Vent/Drain
2	Jasco	26741	LED Light Fixture
3	Schneider Electric	9001SKT38LWW31	PTT Light, 120VAC LED, White
4	Schneider Electric	9001SKR1B	Pushbutton, Black
4	Schneider Electric	9001KA2	1 NO Contact
5	Citel	DS42S-120/G	Transient Voltage Surge Suppressor
6	Phoenix Contact	2866776	24VDC Power Supply
7	Phoenix Contact	2320157	24VDC Power Supply Redundancy Module
8	Weidmuller	7940102584	Circuit Breaker, 120VAC, 20A
8	Weidmuller	7940102579	Circuit Breaker, 120VAC, 1A
8	Weidmuller	7940102582	Circuit Breaker, 120VAC, 15A
8	Weidmuller	7940102572	Circuit Breaker, 120VAC, 2A
8	Weidmuller	7940102624	Circuit Breaker, 24VDC, 1A
9	Hubbel	GFRST15W	Duplex GFCI Receptacle, 15A
10	Schneider Electric	RXM2AB2F7	Miniature Relay, 120VAC, 8 Blade
10	Schneider Electric	RXZE2M114	Miniature Relay Socket, 120VAC, DIN
11	Ventev	T09060010006HM	Cellular Antenna
12	Moxa	EDS-205	Unmanaged Ethernet Switch, 5 Port
13	Mersen	MI5HF25V2/100	Miniature Fuse, Fast Acting, 2A
13	Mersen	MI5HF25V0.5/100	Miniature Fuse, Fast Acting, 0.5A
13	Mersen	GGM3	Fast Blow Fuse, 3A
13	Phoenix Contact	3046100	Fuse Terminal Block, 120VAC
14	Hoffman	ALFSWD	Remote Door Switch
15	Modicon	BMXCPS3500H	Power Supply Module
16	Schneider Electric	HMIDT752	Operator Interface Terminal
16	Schneider Electric	HMIG2U	Display Base Unit

SHT	MANUFACTURER	MODEL	DESCRIPTION
16	Schneider Electric	VJDBTPRO3P	HMI Software
17	Modicon	BMXP342020H	Processor
17	Modicon	CEXSPUCZSGPAZZ	Modicon PLC Software
18	Modicon	BMXXBP1200H	Backplane, 12 Slots
19	Modicon	BMXFTB2000	Terminal Block, 20 Pin
19	Modicon	BMXFTB2800	Terminal Block, 20 Pin
20	Modicon	BMXDAI1604H	Discrete Input Module, 120VAC
21	Modicon	BMXDRA0815H	Discrete Output Module
22	Modicon	BMXAMI0810H	Analog Input Module
23	Modicon	BMXAMO0410H	Analog Output Module
24	Modicon	BMXDDI1602H	Discrete Input Module, 24VDC
24	Phoenix Contact	2865366	IS Barrier/Splitter
25	Polyphaser	IS-50NX-C2	Coaxial RF Surge Protector
27	Ventev	LMR195NMSM-6	SMA Jumper Cable
26	Amphenol	LDF4-50A	Coaxial Cable
28	Finder	7T.81.0.000.2403	Enclosure Heater Thermostat, NC
28	SCE	SCE-TSH75	Enclosure Heater
29	Phoenix Contact	3044076	Terminal Block
29	Phoenix Contact	3044092	Grounding Terminal Block
29	Phoenix Contact	3047028	Terminal End Barrier
29	Phoenix Contact	0800886	Terminal Anchor

32384 CHARLES COUNTY POMONKEY PUMP STATION IMPROVEMENTS

Spare Parts Index:

SHT	MANUFACTURER	MODEL	DESCRIPTION	QTY
15	Modicon	BMXCPS3500H	Power Supply Module	1
20	Modicon	BMXDAI1604H	Discrete Input Module, 120VAC	1
21	Modicon	BMXDRA0815H	Discrete Output Module	1
22	Modicon	BMXAMI0810H	Analog Input Module	1

32384 CHARLES COUNTY POMONKEY PUMP STATION IMPROVEMENTS

Data Sheet 1

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
1	Mfg: SCE: NEMA 4X Wall Mount Enclosure Model Number: SCE-60EL3616SS6LPPL Tags/Service: ENC / PLC/RTU Control Panel Specifications: • 0.075″ stainless steel type 304 • Seams continuously welded and ground smooth • Stainless steel concealed hinges • Black zinc die cast keylocking/padlocking handles • 3-point latching mechanism • Pour in place oil and water resistant gasket • Ground stud on door and body • NEMA 4X rated
1	Mfg: SCE: Dead Front Panel Model Number: SCE-DF60EL36LP Tags/Service: ENC / PLC/RTU Control Panel Specifications: • 0.075″ carbon steel

	• Designed for use in wall mount enclosures with SCE concealed hinges • Quarter turn wing knob closures • Powder coated white finish inside and out
1	Mfg: SCE: Enclosure Backpanel Model Number: SCE-60P36 Tags/Service: ENC / PLC/RTU Control Panel Specifications: • Bent Subpanel • 57″H x 33″W x 0.88″D dimensions • Powder coated white finish
1	Mfg: SCE: Wall Mount Foot Kit Model Number: SCE-ELMFK4 Tags/Service: ENC / PLC/RTU Control Panel Specifications: • Chemical and Corrosion resistant polyamide material • NEMA 4X rated • Comes with stainless steel bolt and zinc nuts • Kit of four
1	Mfg: SCE: NEMA 4X Vent/Drain Model Number: SCE-BVKD Tags/Service:

	DRAIN / PLC/RTU Control Panel Specifications: • NEMA 4X rated • Installed on bottom of enclosure as dual drain and vent • 10 inch·lb installation torque

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95 Midland Road Saginaw, MI 48638-5770
(800) 234-6871 - Fax: (989) 799-4524
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WALL MOUNT ENCLOSURE - NEMA 4X

SCE-60EL3616SSLPPL

TAGS: ENC



Product Specifications:

Part Number: SCE-60EL3616SSLPPL
Description: S.S. LPPL Enclosure
Height: 60.00"
Width: 36.00"
Depth: 16.00"
Price Code: S9
List Price: $3,934.42
Catalog Page: 254
Est. Ship Weight: 161.70 lbs

Construction
- 0.075 In. stainless steel Type 304.
- Seams continuously welded and ground smooth.
- Flange trough collar around all sides of door opening.
- Collar studs 3/8-16 provided for mounting optional panels.
- Mounting holes in back of the enclosure for wall mounting.
- Mounting hardware, sealing washers and hole plugs included.
- Stainless steel concealed hinges.
- Removable and interchangeable doors.
- Black zinc die cast keylocking/padlocking handles.
- 3-point latching mechanism.
- Removable print pocket.
- Pour in place oil & water resistant gasket
- Ground stud on door and body.

Application
Designed to house electrical and electronic controls, instrumentation and components in indoor or outdoor locations. For outdoor application a drip shield and drain vent is recommended.

For details about the design, performance expectations, applications and design suggestions - See Design Considerations www.saginawcontrol.com/instman/considerations.pdf

Finish
#4 Brushed finish on all exterior surfaces. Optional sub-panels are powder coated white.

Industry Standards - (IS6)
- NEMA Type 3R, 4, 4X, 12 and Type 13
- UL Listed Type 3R, 4, 4X and 12
- CSA Type 4, 4X and 12
- IEC 60529
- IP 66

Notes
Special Instructions apply for IS3, IS4 and IS6 to maintain the environmental rating of Type 3R for these parts. Instructions are located on the enclosure door. Drip shield is required on IS3, drip shield is recommended on IS4 and IS6. Drain holes are required on all.

Optional Accessories
- SCE-60P36 Subpanel, Bent
- SCE-60P36GALV Subpanel, Bent Galvanized
- SCE-BVK Breather Vent
- SCE-DF60EL36LP Panel, Dead Front (Wall Mount)
- SCE-DS36SS Shield, S.S. Drip
- SCE-ELMFK4 Foot Kit, EL Mounting (4pc.)
- SCE-ELMFK4SS6-OS Foot Kit, S.S. EL Mounting (4pc.)
- SCE-ELSP3 KIT, Swing-Out Panel (20 High & Up)
- SCE-RD60EL36SSLPPL Door, Replacement

Similar Part Numbers
- SCE-48EL3612SSLPPLS.S. LPPL Enclosure
- SCE-48EL3616SSLPPLS.S. LPPL Enclosure
- SCE-60EL3612SSLPPLS.S. LPPL Enclosure
- SCE-60EL3624SSLPPLS.S. LPPL Enclosure
- SCE-72EL3624SSLPPLS.S. LPPL Enclosure

Installation Information
- Sealing Washer Specifications
- EL Flush Mount Frame
- Swing Panel ELSP3 for Encl. Height > 16
- Drip Shield Kit Assembly
- Service Parts Wall Mount Enclosures
- Dead Front Wall Mount Installation Instructions. EL-LP REV. 2
- Dead Front Wall Mount With 3 Point Latching Hardware. EL-LPPL REV. 2

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DEAD FRONT PANEL

TAGS: ENC

SCE-DF60EL36LP



Product Specifications:

Part Number: SCE-DF60EL36LP
Description: Panel, Dead Front (Wall Mount)
Height: 56.00"
Width: 32.70"
Depth: 0.83"
Price Code: P2
List Price: $277.67
Catalog Page: 454
Est. Ship Weight: 88.00 lbs

Construction
- 0.075 In. Carbon Steel.

Application
Designed for use in Wall Mount Enclosures with SCE concealed hinges.

Dead front panels are designed as a fixed depth panel to be mounted directly behind the existing hinged door. This allows space to mount: gauges, meters and push buttons directly behind the front door making them easily visible when mounting them through the main door is not an option. Dead front panels also provide a barrier against incidental contact with live components during routine monitoring or maintenance of equipment.

Dead front panels are installed by mounting hinges back to back with the existing door hinge of the enclosure and only adding two additional holes on the latching side of the enclosure to mount the latch brackets. Dead Front panels include quarter turn wing knobs for easy entry to the interior and mounting hardware to maintain the environmental rating of the enclosure.

Finish
Powder coated white inside and out.

Industry Standards - (IS17)
- NEMA Not Applicable
- UL Not Applicable
- CSA N/A

Notes
Cutout will need to be added for clearance of Defeater and defeater hook when use in Flange mount disconnect enclosure

Optional Accessories
SCE-115259 Extension, 2 inch Dead Front

Similar Part Numbers
SCE-DF48EL36LPPanel, Dead Front (Wall Mount)
SCE-DF72EL36LPPanel, Dead Front (Wall Mount)

Installation Information
- Dead Front Wall Mount < 20 In Height Installation Instructions. EL-LP REV. 2
- Dead Front Wall Mount Installation Instructions. EL-LP REV. 2
- Dead Front Wall Mount Installation Instructions. H-LP REV. 2
- Dead Front Wall Mount Installation Instructions. XEL-LP REV. 2
- Dead Front Wall Mount With 3 Point Latching Hardware. EL-LPPL REV. 2

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ENCLOSURE BACKPANEL

SCE-60P36

TAGS: ENC

Product Specifications:

Part Number: SCE-60P36
Description: Subpanel, Bent
Height: 57.00"
Width: 33.00"
Depth: 0.88"
Price Code: P3
List Price: $234.55
Catalog Page: 445
Est. Ship Weight: 58.00 lbs

Finish
Powder Coated White.

Industry Standards - (IS17)
- NEMA Not Applicable
- UL Not Applicable
- CSA N/A

Similar Part Numbers
SCE-48P24Subpanel, Bent
SCE-48P30Subpanel, Bent
SCE-48P36Subpanel, Bent
SCE-48P42Subpanel, Bent
SCE-60BFR42Subpanel, Bent
SCE-60P24Subpanel, Bent
SCE-60P30Subpanel, Bent
SCE-60P42Subpanel, Bent
SCE-60P48Subpanel, Bent
SCE-60PC36Subpanel, Bent

Installation Information
- Sub-Plate Layout & Grounding for 10-32

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WALL MOUNT FOOT KIT

TAGS: ENC

SCE-ELMFK4



Product Specifications:

Part Number: SCE-ELMFK4
Description: Foot Kit, EL Mounting (4pc.)
Height: 5.00"
Width: 3.00"
Depth: 1.25"
Price Code: P2
List Price: $21.86
Catalog Page: 437
Est. Ship Weight: 2.00 lbs

Application
Made for use with all SCE Enviroline enclosures and two-door wall-mount enclosures. Chemical and Corrosion resistant Polyamide material rated for type 3, 3R, 4, 4X and 12. Available in kits of four, six, 25 and 50. Product comes with stainless steel bolt and zinc nuts to maintain type 3, 3R, 4, 4X and 12 rating of your enclosure.

Industry Standards - (IS17)
- NEMA Not Applicable
- UL Not Applicable
- CSA N/A

Similar Part Numbers
- SCE-ELMFK4-25Foot Kit, EL Mounting (4pc.) - Bulk
- SCE-ELMFK4-50Kit, EL Mounting Foot - Bulk
- SCE-ELMFK6Foot Kit, EL Mounting (6pc.)

Installation Information
- Mounting Foot Kit for Enviroline Enclosures
- Mounting Foot Kit for WFLP Enclosures

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NEMA 4X VENT/DRAIN

TAGS: DRAIN

SCE-BVKD



Product Specifications:

Part Number: SCE-BVKD
Description: Drain, Breather Vent Kit
Height: 2.00"
Width: 2.00"
Depth: 2.00"
Price Code: P2
List Price: $11.70
Catalog Page: 397
Est. Ship Weight: 0.25 lbs
UL Recognized Type: 1, 2, 3, 3R, 3RX, 3S, 4, 4X & 12
UL File: E69392
Installation Torque: 10 Inch Lb
Location used: Bottom of Enclosure
IEC 60529: IP65

Application
Designed for indoor or outdoor applications to handle the large negative pressure differential caused in a tightly sealed enclosure in extreme weather conditions or wash down applications that draw water into the enclosure. Adding a Breather Vent will help to equalize this pressure differential. Breather vents are made of corrosion and UV resistant Polycarbonate material.

Options
SCE-BVK can be installed on the side of your enclosure as a breather or installed on the bottom as a drain and vent. SCE-BVKD can be installed on the bottom of an enclosure as a drain and vent.

Industry Standards - (IS24)
- UL Component Recognized

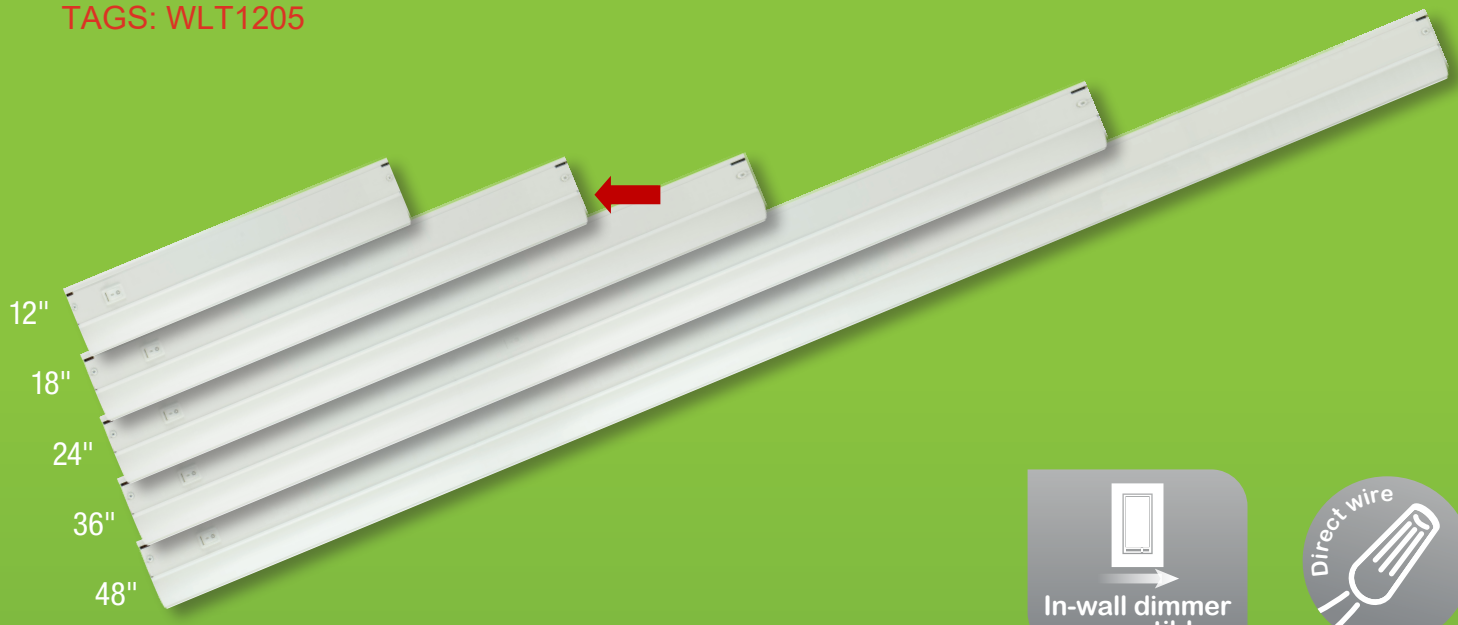
Similar Part Numbers
SCE-BVKBreather Vent

Installation Information
- SCE-BVK, Breather Vent Kit
- SCE-BVKD, Breather Vent Kit Drain

Data Sheet 2

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
1	Mfg: Jasco: LED Light Fixture Model Number: 26741 Tags/Service: WLT1205 / PLC/RTU Control Panel Specifications: • 120VAC 60Hz supply voltage • 3000K color temperature • 773 lumen brightness • 18″L x 3.25″W x 1.33″D dimension • 50,000 hours of LED life

TAGS: WLT1205







PREMIUM LED DIRECT WIRE DIMMABLE LIGHT FIXTURES

- Ideal for designer kitchen, home office or studio
- Full range front phase dimmability when used with compatible in-wall dimmer
- High quality steel housing
- Fixture thickness: 1.33"
- No bulbs to replace – clean white light
- 50,000 hours of LED life
- Energy efficient – longer life
- Evenly diffused light output
- Quick connect wiring and convenient mounting system for easy installation
- 120 VAC 60Hz
- Energy Star rated
- UL listed

Item Number	Dimensions	Color Temp	Brightness
26739 26763 *(4Pc Tray)*	12" L x 3.25" W x 1.33" D	3000K	496 Lumens
26741 26764 *(4Pc Tray)*	18" L x 3.25" W x 1.33" D	3000K	773 Lumens
26742 26765 *(4Pc Tray)*	24" L x 3.25" W x 1.33" D	3000K	961 Lumens
29434 29435 *(4Pc Tray)*	36" L x 3.25" W x 1.33" D	3000K	1348 Lumens
30890 30891 *(4Pc Tray)*	48" L x 3.25" W x 1.33" D	3000K	1798 Lumens

Data Sheet 3

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
1	Mfg: Schneider Electric: PTT Light, 120VAC LED, White Model Number: 9001SKT38LWW31 Tags/Service: LT1204 / PLC/RTU Control Panel Specifications: • 120VAC supply voltage • Plastic bezel materal • White polycarbonate fresnel lens • 1 C/O contact type • White LED light source

Product data sheet

PTT LIGHT, 120VAC LED, WHITE

Specifications

TAGS: LT1204



Push-to-test pilot light, Harmony 9001SK, plastic, polycarbonate, fresnel lens, white, 30mm, LED white, 120V

9001SKT38LWW31

Product availability: Stock - Normally stocked in distribution facility

Main

Range of Product	Harmony 9001SK
Product or Component Type	Push-to-test pilot light
Lens type	Fresnel plastic
Light block supply	Full voltage 120 V

Complementary

Bezel material	Plastic
Mounting diameter	1.2 in (30 mm)
Shape of signaling unit head	Round
Cap/operator or lens colour	White
Light source	White LED
Bulb base	BA 9s
Operating Mode	Push-to-test
Lens type	Fresnel
[Us] rated supply voltage	120 V
Light source colour	White
Contacts type and composition	1 C/O
Lens material	Polycarbonate
Size	30 mm
Light source	LED

Environment

Protective treatment	TC
IP Degree of Protection	IP65/IP66

Ordering and shipping details

Category	US10CS121429
Discount Schedule	0CS1
GTIN	785901546443
Returnability	Yes
Country of origin	MX

Price is “List Price” and may be subject to a trade discount – check with your local distributor or retailer for actual price.

Disclaimer: This documentation is not intended as a substitute for and is not to be used for determining suitability or reliability of these products for specific user applications

Packing Units

Unit Type of Package 1	PCE
Nbr. of units in pkg.	1
Package 1 Height	6.350 in (16.129 cm)
Package 1 Width	2.50 in (6.35 cm)
Package 1 Length	16.129 in (40.968 cm)
Package weight(Lbs)	4.160 oz (117.934 g)

Contractual warranty

Warranty (in months)	18

Data Sheet 4

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
1	Mfg: Schneider Electric: Pushbutton, Black Model Number: 9001SKR1B Tags/Service: PB1219 / PLC/RTU Control Panel Specifications: • Plastic bezel material • Octagonal signaling unit head without light source • Black operator color • Flush operator profile • Spring return operator • NEMA 4X protection rating
1	Mfg: Schneider Electric: 1 NO Contact Model Number: 9001KA2 Tags/Service: PB1219 / PLC/RTU Control Panel Specifications: • Silver alloy contact material • 10A conventional free air thermal current • 3A at 240VAC rated operational current • 60kA short-circuit making capacity at 120VAC • 6kA rated breaking capacity at 120VAC • −13...158°F ambient air temperature for operation

Product data sheet

PUSHBUTTON, BLACK

Specifications

TAGS: PB1219



Push button head, Harmony 9001SK, plastic, flush, black, 30mm, spring return

9001SKR1B

Product availability: Stock - Normally stocked in distribution facility

Main

Range of Product	Harmony 9001SK
Product or Component Type	Head for push-button
Device short name	9001SK

Complementary

Bezel material	Plastic
Mounting diameter	1.2 in (30 mm)
Shape of signaling unit head	Octagonal
Light source	without
Light block supply	Without supply
Contacts type and composition	Without contact block
Product Certifications	CE CSA LR24590 class 3211 03 UL listed file E42259 CCN NKCR
NEMA degree of protection	NEMA 3R/4X/13
Cap/operator or lens colour	Black
Operator profile	Flush
Type of operator	Spring return
Size	30 mm

Environment

IP Degree of Protection	IP65/IP66

Ordering and shipping details

Category	US10CS121429
Discount Schedule	0CS1
GTIN	785901793878
Returnability	Yes
Country of origin	MX

Packing Units

Unit Type of Package 1	PCE
Nbr. of units in pkg.	1

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Package 1 Height	2.800 in (7.112 cm)
Package 1 Width	2.300 in (5.842 cm)
Package 1 Length	2.00 in (5.08 cm)
Package weight(Lbs)	1.600 oz (45.359 g)

Contractual warranty

Warranty (in months)	18

Product data sheet

1 NO CONTACT

Specifications

TAGS: PB1219



30mm Push Button, Types K, SK or KX, contact block, 1 NO

9001KA2

Product availability: Stock - Normally stocked in distribution facility

Main

Range of Product	Harmony 9001K Harmony 9001SK
Product or Component Type	Contact block
Device short name	9001K
Contacts type and composition	1 NO

Complementary

Terminals description ISO n°1	(1-2)NC
Net Weight	0.051 lb(US) (0.023 kg)
Contact operation	Standard
Positive opening	Without
Connections - terminals	Screw clamp terminals, 1 x 0.22...2 x 2.5 mm² IEC 60947-1
Tightening torque	7.08 lbf.in (0.8 N.m) IEC 60947-1
Shape of screw head	Cross slotted
Contacts material	Silver alloy contacts
Short-circuit protection	10 A cartridge fuse IEC 60947-5-1
[Ith] conventional free air thermal current	10 A
[Ui] rated insulation voltage	250 V 3)IEC 60947-1
[Uimp] rated impulse withstand voltage	2.5 kV IEC 60947-1
[Ie] rated operational current	3 A 240 V, AC-15, A600-Q600 NEMA ~~0.55 A 125 V, DC-13, A600-Q600 NEMA~~
[Icm] rated short-circuit making capacity	~~12 kA 600 V AC-15, 7200 VA~~ ~~15 kA 480 V AC-15, 7200 VA~~ ~~30 kA 240 V AC-15, 7200 VA~~ 60 kA 120 V AC-15, 7200 VA ~~0.27 kA 250 V DC-13, 7200 VA~~ ~~0.55 kA 125 V DC-13, 7200 VA~~ ~~0.1 kA 600 V DC-13~~
Rated breaking capacity	~~3 kA 240 V AC-15, 720 VA~~ 6 kA 120 V AC-15, 720 VA ~~1.2 kA 600 V AC-15, 720 VA~~ ~~1.5 kA 480 V AC-15, 720 VA~~ ~~0.1 kA 600 V DC-13~~ ~~0.27 kA 250 V DC-13~~ ~~0.55 kA 125 V DC-13~~
[Us] rated supply voltage	0...600 V
Connections - Terminals	Screw-clamp terminals

Price is “List Price” and may be subject to a trade discount – check with your local distributor or retailer for actual price.

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Environment

Protective treatment	TC
Ambient Air Temperature for Storage	-40…158 °F (-40…70 °C)
Ambient Air Temperature for Operation	-13…158 °F (-25…70 °C)
Electrical shock protection class	Class II conforming to IEC 61140
Standards	IEC 60947-5-4 JIS C 852 IEC 60947-5-1 JIS C 4520 CSA C22.2 No 14 UL 508 IEC 60947-1 UL 60947-5-1
Product Certifications	UL 508 NEMA
Vibration resistance	7 gn (f= 2…500 Hz) conforming to IEC 60068-2-6
Shock resistance	50 gn IEC 60068-2-27

Ordering and shipping details

Category	US10CS121434
Discount Schedule	0CS1
GTIN	785901880011
Returnability	Yes
Country of origin	MX

Packing Units

Unit Type of Package 1	PCE
Nbr. of units in pkg.	1
Package 1 Height	1.181 in (3.000 cm)
Package 1 Width	1.575 in (4.000 cm)
Package 1 Length	1.969 in (5.000 cm)
Package weight(Lbs)	0.776 oz (22.000 g)
Unit Type of Package 2	S01
Number of Units in Package 2	100
Package 2 Height	5.906 in (15.000 cm)
Package 2 Width	5.906 in (15.000 cm)
Package 2 Length	15.748 in (40.000 cm)
Package 2 Weight	5.269 lb(US) (2.390 kg)

Contractual warranty

Warranty (in months)	18

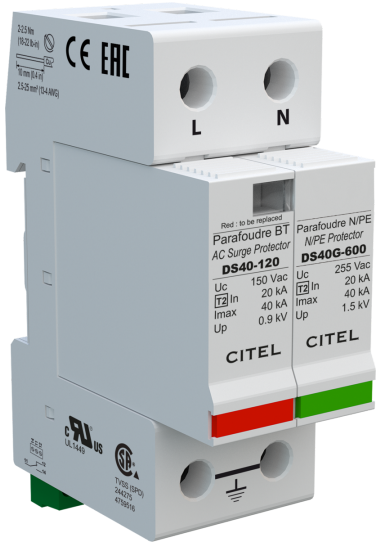
Data Sheet 5

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
1	Mfg: Citel: Transient Voltage Surge Suppressor Model Number: DS42S-120/G Tags/Service: SPD1202 / PLC/RTU Control Panel Specifications: • 120VAC single phase network • Common/differential modes of protection • 40kA total maximum discharge current • −35...60°C ambient temperature range • Disconnection via fuse-link fail-safe behavior • DIN rail mounting • Thermoplastic UL94-V0 material

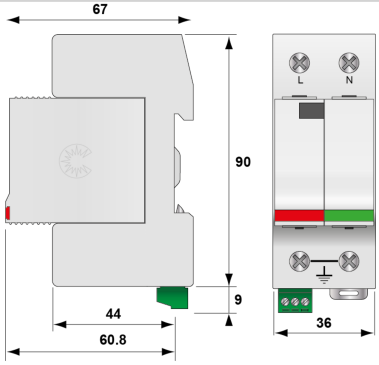
DS42S-120/G

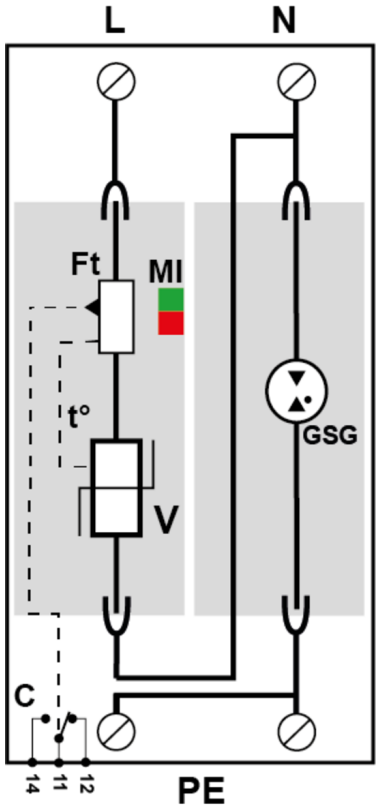
TRANS. VOLTAGE SURGE SUPPRESSOR

TAGS: SPD1202



- MOV+GSG Technology
- UL1449 TYPE 4CA
- 40kA Surge Current Rating per module
- Visual fault indicator and remote contacts
- 10-Year warranty





V: High-energy varistor
GSG: Specific gas tube
Ft: Thermal fuse
C: Remote signaling contact
t°: Thermal disconnection system
MI: Disconnection indicator

Electrical Characteristics		
Network		120 V single-phase
Total Maximum discharge current *max. total withstand @ 8/20 µs*	Imax Total	40 kA
Protection mode(s)		Common/Differential mode
Residual voltage at 5 kA *@ 5 kA (8/20µs)*	Up-5kA	0.6 kV
Protection level L/N *@ In (8/20µs)*	Up L/N	0.9 kV
Protection level N/PE *@ In (8/20µs)*	Up N/PE	1.5 kV
POWER SPD TYPE		UL1449 5th Ed. TYPE 4CA
VOLTS	(V)	240/120
AC/DC/DC PV/RF		AC
PHASE	(PH)	1S
AMPS	(A)	N/A
AMBIENT MIN	(C)	-35
AMBIENT MAX	(C)	+60
MODES		~~L-L~~, L-N, ~~L-G, N-G~~
MLV	(V)	1702/1040/1690/1090
MCOV	(V)	300/150/150/255
IN *15 impulses 8/20µs*	(kA)	20
SCCR	(kA)	n/a
IMAX *8/20µs*	(kA)	40
limp *10/350µs*	(kA)	4
Mechanical Characteristics		
Technology		MOV+GDT
Connection to Network		By screw terminals: 2.5-25mm² / by bus
Format		Plug-in modular box
Failsafe mode		Disconnection from AC network
TECHNOLOGY		MOV+GSG
NETWORK CONFIGURATION		2 pole, 2W+G, Single/Spilt Phase
CONNECTION METHOD		SPD-Screw Terminal (8-12AWG), Terminal blocks 6AWG
MOUNTING		Din Rail
MATERIAL		Thermoplastic UL94-V0
NEMA RATING (IP RATING)		NEMA 2 (IP20)
FAIL-SAFE BEHAVIOR		Disconnection via fuse-link
REAL-TIME DIAGNOSTICS		Visual indicator and remote contacts
DIMENSIONS		See diagram
WEIGHT		0.60 lbs
SPARE PART		DSM40-120, DSM40G-600
Standards		
Certification		cRUus
UL STANDARD		UL1449 5th Edition
UL CATEGORY		VZCA2, VZCA8
UL FILE NUMBER		E326289
STANDARDS		NOM-003-SCFI-2014, NOM-001-SCFI-1993
ENVIRONMENTAL STANDARDS		ROHS
Part number		
461631		

Data Sheet 6

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
2	Mfg: Phoenix Contact: 24VDC Power Supply Model Number: 2866776 Tags/Service: PWS1220, PWS1225 / PLC/RTU Control Panel Specifications: • 100VAC…240VAC nominal input voltage range • 569VA nominal power consumption • 92% efficiency typical (at 120VAC) • 24VDC ± 1% nominal output voltage • 20A nominal output current • 8W maximum no-load power dissipation • DIN rail mounting Power supply load calculations PWS1220 and PWS1225 are redundant

Description	Current Draw (Amps)
Ethernet Switch	0.0417
Modem	0.0417
OIT	0.7083
Level Display	1
Total Current Draw	1.7917

- Total current capacity $= \frac{1.7917}{20}$, 8.96% used

QUINT-PS/1AC/24DC/20 - Power supply

2866776

https://www.phoenixcontact.com/us/products/2866776

24VDC POWER SUPPLY

TAGS: PWS1220, PWS1225

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Primary-switched power supply unit QUINT POWER, Screw connection, DIN rail mounting, SFB Technology (Selective Fuse Breaking), input: 1-phase, output: 24 V DC / 20 A, adjustable from 18 V DC ... 29.5 V DC

Product description

QUINT POWER power supplies with maximum functionality
QUINT POWER circuit breakers magnetically and therefore quickly trip at six times the nominal current, for selective and therefore cost-effective system protection. The high level of system availability is additionally ensured, thanks to preventive function monitoring, as it reports critical operating states before errors occur.
Reliable starting of heavy loads takes place via the static power reserve POWER BOOST. Thanks to the adjustable voltage, all ranges between 5 V DC ... 56 V DC are covered.

Your advantages

- Reliable starting of difficult loads with the static POWER BOOST power reserve with up to 1.5 times the nominal current permanently
- Fast tripping of standard circuit breakers with dynamic power reserve SFB (selective fuse breaking) technology with up to 6 times the nominal current for 12 ms
- For superior system availability
- Preventive function monitoring

Technical data

Input data

AC operation

Nominal input voltage range	100 V AC ... 240 V AC
Input voltage range	100 V AC ... 240 V AC -15 % ... +10 %
Derating $I_{Stat.\ Boost}$	< 100 V AC (1 %/V)
Input voltage range AC	85 V AC ... 264 V AC
~~Input voltage range DC~~	~~90 V DC ... 350 V DC~~
Electric strength, max.	300 V AC
Typical national grid voltage	120 V AC
	230 V AC
Voltage type of supply voltage	AC
Inrush current	< 20 A
Inrush current integral (I^2t)	< 3.2 A^2s
Inrush current limitation	20 A
AC frequency range	45 Hz ... 65 Hz
Frequency range DC	0 Hz
Mains buffering time	> 32 ms (120 V AC)
	> 32 ms (230 V AC)
Current consumption	~~7 A (100 V AC)~~
	5.8 A (120 V AC)
	~~3 A (230 V AC)~~
	~~3.1 A (240 V AC)~~
Nominal power consumption	569 VA
Protective circuit	Transient surge protection; Varistor
Power factor (cos phi)	0.89
Typical response time	< 0.6 s
Input fuse	12 A (slow-blow, internal)
Permissible backup fuse	B10 B16 AC:
Permissible DC backup fuse	DC: Connect a suitable fuse upstream
Recommended breaker for input protection	10 A ... 16 A (Characteristics B, C, D, K)
Discharge current to PE	< 3.5 mA

DC operation

Nominal input voltage range	110 V DC ... 250 V DC
Input voltage range	110 V DC ... 250 V DC -18 % ... +40 %
Derating $I_{Stat.\ Boost}$	< 110 V DC (1 %/V)
Current consumption	6.3 A (110 V DC)
	~~2.8 A (220 V DC)~~

Output data

Efficiency	typ. 92 % (120 V AC)

	~~typ. 92.7 % (230 V AC)~~
Output characteristic	U/I
Nominal output voltage	24 V DC ±1 %
Setting range of the output voltage (U_{Set})	18 V DC ... 29.5 V DC (> 24 V DC, constant capacity restricted)
Nominal output current (I_N)	20 A
POWER BOOST (I_{Boost})	26 A (-25 °C ... 40 °C permanent, U_{OUT} = 24 V DC)
Static Boost ($I_{Stat.Boost}$)	26 A
Selective Fuse Breaking (I_{SFB})	120 A (12 ms)
Magnetic circuit breaker tripping	B2 / B4 / B6 / B10 / B16 / C2 / C4 / C6
Derating	60 °C ... 70 °C (2.5 %/K)
Feedback voltage resistance	≤ 35 V DC
Protection against overvoltage at the output (OVP)	≤ 35 V DC
Control deviation	< 1 % (change in load, static 10 % ... 90 %)
	< 2 % (change in load, dynamic 10 % ... 90 %)
	< 0.1 % (change in input voltage ±10 %)
Residual ripple	< 30 mV_{PP} (with nominal values)
Short-circuit-proof	yes
Output power	480 W
	624 W
Maximum no-load power dissipation	8 W
Power loss nominal load max.	40 W
Rise time	< 0.1 s (U_{OUT} (10 % ... 90 %))
Connection in parallel	yes, for redundancy and increased capacity
Connection in series	yes

Signal: DC OK active

Output description	U_{OUT} > 0.9 x U_N: High signal
Switching voltage range	18 V DC ... 24 V DC
Maximum inrush current	≤ 20 mA (short-circuit-proof)
Continuous load current	≤ 20 mA

Signal: DC OK floating

Output description	Relay contact, U_{OUT} > 0.9 x U_N: Contact closed
Maximum switching voltage	30 V AC
	24 V DC
Maximum inrush current	0.5 A
	1 A
Continuous load current	≤ 1 A

Signal: POWER BOOST, active

Output description	I_{OUT} < I_N: High signal
Switching voltage range	18 V DC ... 24 V DC
Output voltage	+ 24 V DC
Maximum inrush current	20 mA (short-circuit-proof)
Continuous load current	≤ 20 mA

Connection data

Input

Connection method	Screw connection
Conductor cross-section, rigid min.	0.2 mm²
Conductor cross-section, rigid max.	6 mm²
Conductor cross-section flexible min.	0.2 mm²
Conductor cross-section flexible max.	4 mm²
Conductor cross-section AWG min.	18
Conductor cross-section AWG max.	10
Stripping length	7 mm
Screw thread	M4
Tightening torque, min	0.5 Nm
Tightening torque max	0.6 Nm

Output

Connection method	Screw connection
Conductor cross-section, rigid min.	0.2 mm²
Conductor cross-section, rigid max.	6 mm²
Conductor cross-section flexible min.	0.2 mm²
Conductor cross-section flexible max.	4 mm²
Conductor cross-section AWG min.	12
Conductor cross-section AWG max.	10
Stripping length	7 mm
Screw thread	M4
Tightening torque, min	0.5 Nm
Tightening torque max	0.6 Nm

Signal

Connection method	Screw connection
Conductor cross-section, rigid min.	0.2 mm²
Conductor cross-section, rigid max.	6 mm²
Conductor cross-section flexible min.	0.2 mm²
Conductor cross-section flexible max.	4 mm²
Conductor cross-section AWG min.	18
Conductor cross-section AWG max.	10
Screw thread	M4
Tightening torque, min	0.5 Nm
Tightening torque max	0.6 Nm

Signaling

Types of signaling	LED
	Active switching output
	Relay contact

Signal output: DC OK active

Status display	U_{OUT} > 0.9 x U_N: "DC OK" LED green
Note on status display	U_{OUT} < 0.9 x U_N: Flashing "DC OK" LED
	I_{OUT} < I_N: LED ON

Signal output: DC OK floating

Status display	U_{OUT} > 0.9 x U_N: "DC OK" LED green
Note on status display	U_{OUT} < 0.9 x U_N: Flashing "DC OK" LED

Signal output: POWER BOOST, active

Status display	I_{OUT} > I_N: LED "BOOST" yellow

Electrical properties

Number of phases	1
Insulation voltage input/output	4 kV AC (type test)
	2 kV AC (routine test)
Insulation voltage output / PE	500 V DC (routine test)
Insulation voltage input / PE	3.5 kV AC (type test)
	2 kV AC (routine test)

Product properties

Product family	QUINT POWER
MTBF (IEC 61709, SN 29500)	> 900000 h (25 °C)
	> 520000 h (40 °C)
	> 235000 h (60 °C)

Insulation characteristics

Protection class	I
Degree of pollution	2

Dimensions

Width	90 mm
Height	130 mm
Depth	125 mm

Installation dimensions

Installation distance right/left	5 mm / 5 mm
Installation distance top/bottom	50 mm / 50 mm

Alternative assembly

Width	122 mm
Height	130 mm
Depth	93 mm

Mounting

Mounting type	DIN rail mounting

Assembly note	alignable: P_N ≥50%, 5 mm horizontally, 15 mm next to active components, 50 mm vertically alignable: P_N <50%, 0 mm horizontally, 40 mm vertically top, 20 mm vertically bottom
Mounting position	horizontal DIN rail NS 35, EN 60715
With protective coating	no

Material specifications

Housing material	Metal
Hood version	Galvanized sheet steel, free from chrome (VI)
Side element version	Aluminum

Environmental and real-life conditions

Ambient conditions

Degree of protection	IP20
Ambient temperature (operation)	-25 °C ... 70 °C (> 60 °C Derating: 2,5 %/K)
Ambient temperature (storage/transport)	-40 °C ... 85 °C
Ambient temperature (start-up type tested)	-40 °C
Maximum altitude	≤ 5000 m (> 2000 m, observe derating)
Climatic class	3K22 (in accordance with EN 60721-3-3)
Max. permissible relative humidity (operation)	≤ 95 % (at 25 °C, non-condensing)
Shock	18 ms, 30g, in each space direction (according to IEC 60068-2-27)
Vibration (operation)	< 15 Hz, amplitude ±2.5 mm (according to IEC 60068-2-6)
	15 Hz ... 150 Hz, 2.3g, 90 min.
Temp code	T4 (-25 ... +70 °C; > 60 °C, Derating: 2,5 %/K)

Standards and regulations

Rail applications	EN 50121-4
	EN 50121-3-2
HART FSK Physical Layer Test Specification Compliance	Output voltage U_{Out} compliant
Standard – Limitation of mains harmonic currents	EN 61000-3-2
Standard - Electrical safety	IEC 61010-2-201 (SELV)
Standard - Equipment safety	BG (design tested)
Standard - Approval for medical use	IEC 60601-1, 2 x MOOP
Standard – Safety extra-low voltage	IEC 61010-1 (SELV)
	IEC 61010-2-201 (PELV)
Standard - Safe isolation	IEC 61010-2-201
Standard - safety for equipment for measurement, control, and laboratory use	IEC 61010-1
Approval - requirement of the semiconductor industry with regard to mains voltage dips	SEMI F47-0706 Compliance Certificate

Overvoltage category

EN 61010-1	II (≤ 5000 m)
EN 62477-1	III (≤ 2000 m)

EN 61558-2-16	II (≤ 4000 m)

Approvals

CSA	CAN/CSA-C22.2 No. 60950-1-07
	CSA-C22.2 No. 107.1-01
Shipbuilding approval	DNV GL (EMC B, only with upstream filter), ABS, LR, RINA, NK, BV
SIQ	BG (type approved)
UL approvals	UL Listed UL 508
	UL/C-UL Recognized UL 60950-1
	UL 121201 & CSA C22.2 No. 213-17 Class I, Division 2, Groups A, B, C, D T4 (Hazardous Location)

EMC data

Electromagnetic compatibility	Conformance with EMC Directive 2014/30/EU
Low Voltage Directive	Conformance with Low Voltage Directive 2014/35/EC
EMC requirements for noise emission	EN 61000-6-3
	EN 61000-6-4
EMC requirements for noise immunity	EN 61000-6-1
	EN 61000-6-2

Noise emission

Standards/regulations	EN 55011 (EN 55022)

Electrostatic discharge

Standards/regulations	EN 61000-4-2

Electrostatic discharge

Contact discharge	8 kV (Test Level 4)
Discharge in air	15 kV (Test Level 4)
Comments	Criterion A

Electromagnetic HF field

Standards/regulations	EN 61000-4-3

Electromagnetic HF field

Frequency range	80 MHz ... 1 GHz
Test field strength	20 V/m (Test Level 3)
Frequency range	1 GHz ... 2 GHz
Test field strength	10 V/m (Test Level 3)
Frequency range	2 GHz ... 3 GHz
Test field strength	10 V/m (Test Level 3)
Comments	Criterion A

Fast transients (burst)

Standards/regulations	EN 61000-4-4

Fast transients (burst)

Input	4 kV (Test Level 4 - asymmetrical)
Output	2 kV (Test Level 3 - asymmetrical)
Signal	2 kV (Test Level 4 - asymmetrical)
Comments	Criterion A

Surge voltage load (surge)

Standards/regulations	EN 61000-4-5

Surge voltage load (surge)

Input	2 kV (Test Level 3 - symmetrical)
	4 kV (Test Level 4 - asymmetrical)
Output	1 kV (Test Level 2 - symmetrical)
	2 kV (Test Level 3 - asymmetrical)
Signal	1 kV (Test Level 2 - asymmetrical)
Comments	Criterion A

Conducted interference

Standards/regulations	EN 61000-4-6

Conducted interference

Input/output/signal	asymmetrical
Frequency range	0.15 MHz ... 80 MHz
Comments	Criterion A
Voltage	10 V (Test Level 3)

Emitted interference

Standards/regulations	EN 61000-6-3
Radio interference voltage in acc. with EN 55011	EN 55011 (EN 55022) Class B, area of application: Industry and residential
Emitted radio interference in acc. with EN 55011	EN 55011 (EN 55022) Class B, area of application: Industry and residential

Criteria

Criterion A	Normal operating behavior within the specified limits.
Criterion B	Temporary impairment to operational behavior that is corrected by the device itself.

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Data Sheet 7

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
1	Mfg: Phoenix Contact: 24VDC Power Supply Redundancy Module Model Number: 2320157 Tags/Service: PWS1232 / PLC/RTU Control Panel Specifications: • 12...24VDC nominal input voltage range • >97% output efficiency • 20A nominal redundancy output current • 10W power loss nominal load max • DIN rail mounting

QUINT-DIODE/12-24DC/2X20/1X40 - Redundancy module

2320157

https://www.phoenixcontact.com/us/products/2320157

24VDC POWER SUPPLY REDUNDANCY MODULE

TAGS: PWS1232

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DIN rail diode module 12-24 V DC/2x20 A or 1x40 A. Uniform redundancy up to the consumer.

Product description

A safe redundant system is the result of the parallel connection of two power supply units which are decoupled from one another. To further increase system availability, QUINT DIODE provides the solution: decoupling with diode.

Your advantages

- Flexible
- Rugged design
- Consistent redundancy up to the load

Technical data

Input data

DC operation

Nominal input voltage range	12 V DC ... 24 V DC
Input voltage range	10 V DC ... 30 V DC
Voltage type of supply voltage	DC
Reverse polarity protection	yes, < 60 V
Nominal input current (I_N)	2x 20 A (-25 °C ... 60 °C)
	1x 40 A (-25 °C ... 60 °C)
Maximum current I_{max}	2x 30 A (-25 °C ... 40 °C)
	1x 60 A (-25 °C ... 40 °C)
Transient surge protection	Varistor
Voltage drop, input/output	typ. 0.5 V
Nominal input voltage range	12 V DC ... 24 V DC
Input voltage range	10 V DC ... 30 V DC
Input voltage range DC	10 V DC ... 30 V DC

Output data

Efficiency	> 97 %
Nominal output voltage	U_{In} - 0,5 V
Nominal output current (I_N)	40 A (Increasing power)
	20 A (Redundancy)
Derating	60 °C ... 70 °C (2.5 %/K)
Power loss nominal load max.	10 W (I_{OUT} = 20 A)
Connection in series	no
Derating	60 °C ... 70 °C 2.5 %/K

Connection data

Input

Connection method	Screw connection
Conductor cross-section, rigid min.	0.2 mm²
Conductor cross-section, rigid max.	6 mm²
Conductor cross-section flexible min.	0.2 mm²
Conductor cross-section flexible max.	4 mm²
Conductor cross-section AWG min.	12
Conductor cross-section AWG max.	10
Stripping length	8 mm
Screw thread	M3
Tightening torque, min	0.5 Nm
Tightening torque max	0.6 Nm

Output

Connection method	Screw connection
Conductor cross-section, rigid min.	0.5 mm²
Conductor cross-section, rigid max.	16 mm²
Conductor cross-section flexible min.	0.5 mm²
Conductor cross-section flexible max.	16 mm²
Conductor cross-section AWG min.	10
Conductor cross-section AWG max.	6
Stripping length	10 mm
Screw thread	M4
Tightening torque, min	1.2 Nm
Tightening torque max	1.5 Nm

Electrical properties

Insulation voltage input, output / housing	1000 V

Product properties

Product family	QUINT DIODE
MTBF (IEC 61709, SN 29500)	40000000 h
LED	no

Insulation characteristics

Protection class	III
Degree of pollution	2

Dimensions

Width	50 mm
Height	130 mm
Depth	125 mm
Horizontal pitch	2.8 Div.

Installation dimensions

Installation distance right/left	5 mm / 5 mm
Installation distance top/bottom	50 mm / 50 mm

Mounting

Mounting type	DIN rail mounting
Assembly note	alignable: P_N ≥50%, 5 mm horizontally, 15 mm next to active components, 50 mm vertically alignable: P_N <50%, 0 mm horizontally, 40 mm vertically top, 20 mm vertically bottom
Mounting position	horizontal DIN rail NS 35, EN 60715

Material specifications

Flammability rating according to UL 94 (housing / terminal blocks)	V0
Housing material	Metal
Housing material	Steel sheet, zinc-plated

Environmental and real-life conditions

Ambient conditions

Degree of protection	IP20
Ambient temperature (operation)	-40 °C ... 70 °C (> 60 °C Derating: 2,5 %/K)
Ambient temperature (storage/transport)	-40 °C ... 85 °C
Maximum altitude	≤ 5000 m
Climatic class	3K3 (in acc. with EN 60721)
Max. permissible relative humidity (operation)	≤ 95 % (at 25 °C, non-condensing)
Shock	18 ms, 30g, in each space direction (according to IEC 60068-2-27)
Vibration (operation)	< 15 Hz, amplitude ±2.5 mm (according to IEC 60068-2-6)
	15 Hz ... 150 Hz, 2.3g, 90 min.
Temp code	T4 (-25 ... +70 °C; > 60 °C, Derating: 2,5 %/K)

Standards and regulations

Standard – Electronic equipment for use in electrical power installations and their assembly into electrical power installations	EN 50178/VDE 0160 (PELV)
Standard - Electrical safety	EN 60950-1/VDE 0805 (SELV)
Standard – Protection against shock currents, basic requirements for protective separation in electrical equipment	EN 50178
Standard – Safety extra-low voltage	IEC 60950-1 (SELV) and EN 60204-1 (PELV)
Standard - Safe isolation	DIN VDE 0100-410

Approvals

UL approvals	UL/C-UL listed UL 508
	UL/C-UL Recognized UL 60950-1
	UL ANSI/ISA-12.12.01 Class I, Division 2, Groups A, B, C, D T3C ... T4 (Hazardous Location)

Conformity/Approvals

ATEX	Ⓔx II 3 G Ex ec IIC T4 Gc
	DEKRA 20ATEX0041 X
IECEx	Ex ec IIC T4 Gc
	IECEx DEK 20.0022X

EMC data

Electromagnetic compatibility	Conformance with EMC Directive 2014/30/EU
Low Voltage Directive	Conformance with Low Voltage Directive 2014/35/EC
EMC requirements for noise emission	EN 61000-6-3
	EN 61000-6-4
EMC requirements for noise immunity	EN 61000-6-1
	EN 61000-6-2

Surge voltage load (surge)

Standards/regulations	EN 61000-4-5

Surge voltage load (surge)

Input	2 kV (level 3 - asymmetrical: conductor to ground)
	1 kV (Level 2 - symmetrical: Conductor to conductor)
Output	2 kV (level 3 - asymmetrical: conductor to ground)
	1 kV (Level 2 - symmetrical: Conductor to conductor)
Comments	Criterion A

Surge current load (surge)

Standards/regulations	EN 61000-4-5

Surge current load (surge)

Input	2 kV (level 3 - asymmetrical: conductor to ground)
	1 kV (Level 2 - symmetrical: Conductor to conductor)
Output	2 kV (level 3 - asymmetrical: conductor to ground)
	1 kV (Level 2 - symmetrical: Conductor to conductor)

Emitted interference

Standards/regulations	EN 61000-6-3
Radio interference voltage in acc. with EN 55011	EN 55011 (EN 55022) Class B, area of application: Industry and residential
Emitted radio interference in acc. with EN 55011	EN 55011 (EN 55022) Class B, area of application: Industry and residential

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Data Sheet 8

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
1	Mfg: Weidmuller: Circuit Breaker, 120VAC, 20A Model Number: 7940102584 Tags/Service: CB1202 / PLC/RTU Control Panel Specifications: • 240VAC operating voltage • −40...85°C operating temperature range • 10kA ampere interrupting capacity • 20A current rating • Medium time delay
2	Mfg: Weidmuller: Circuit Breaker, 120VAC, 1A Model Number: 7940102579 Tags/Service: CB1211, CB1213 / PLC/RTU Control Panel Specifications: • 240VAC operating voltage • −40...85°C operating temperature range • 10kA ampere interrupting capacity • 1A current rating • Medium time delay

4	Mfg: Weidmuller: Circuit Breaker, 120VAC, 15A Model Number: 7940102582 Tags/Service: CB1205, CB1208, CB1220, CB1225 / PLC/RTU Control Panel Specifications: • 240VAC operating voltage • −40...85°C operating temperature range • 10kA ampere interrupting capacity • 15A current rating • Medium time delay
1	Mfg: Weidmuller: Circuit Breaker, 120VAC, 2A Model Number: 7940102572 Tags/Service: CB1218 / PLC/RTU Control Panel Specifications: • 240VAC operating voltage • −40...85°C operating temperature range • 10kA ampere interrupting capacity • 2A current rating • Medium time delay
2	Mfg: Weidmuller: Circuit Breaker, 24VDC, 1A Model Number: 7940102624 Tags/Service: CB1421, CB1437 / PLC/RTU Control Panel Specifications:

	• 80VDC operating voltage • −40...85°C operating temperature range • 10kA ampere interrupting capacity • 1A current rating • Medium time delay

CIRCUIT BREAKER, 120VAC

TAGS: CB1202, CB1205, CB1208, CB1220, CB1225, CB1218, CB1211, CB1213

Datasheet

QL Series Hydraulic-Magnetic Miniature Circuit Breakers



Dual mount - 1 pole



DIN mount - 2 pole



Dual mount + auxiliary

Features

- AC Circuit Breaker
- Hydraulic-Magnetic technology
- 100% rating capability, independent of ambient temperature
- One, two and three pole (UL). One, two, three pole and 1+N, 3+N (VDE).
- VDE and CCC approved, CE certified
- UL 489 Listed, CSA 22.2 certified
- Ratings 0.5A to 25A
- Optional Auxiliary Switch / Trip Alarm - factory fitted (6.5 mm wide)
- Wide range of time delays and operating currents
- Precision tripping characteristics
- Ultra compact – 13 mm wide module
- Trip indication with mid-trip handle
- Reset immediately after overload
- DIN mount product in grey shells
- Dual mount product in black shells
- Ring terminal ready (terminal screw is removable)

Applications:

- AC branch circuit protection (UL 489, IEC / EN 60847-2)
- Telecom / datacom equipment
- Lighting control
- UPS equipment
- Alternative energy equipment
- Mobile power generation equipment
- Power conditioning equipment

Auxiliary Switch, Trip Alarm & Combo Features

- Factory fitted
- Attached to right hand side of Circuit Breaker
- Compact 6.5 mm width
- AC or DC voltages
- UL 489 listed (Auxiliary Switch: 6 A 240VAC, 0.5 A 110VDC)
- IEC 60947-5-1 approved (Auxiliary Switch: 6 A 240 V AC, 0.5 A 110 V DC; Trip Alarm: 6 A 240 V AC, 0.5 A 110 V DC)

Optional Accessories

- Auxiliary Switch (DIN and Dual Mount)
- Auxiliary Switch + Trip Alarm (Dual Mount only)
- Trip Alarm (Dual Mount only)
- Handle lock
- Surface mounting clips
- Busbar
- 57 mm Escutcheon blank (Dual Mount only)
- 57 mm Safety blank (Dual Mount only)

Approvals

Weidmuller, United States
821 Southlake Blvd.
Richmond, Virginia 23236
Telephone: (800) 849-9343
Website: www.weidmuller.com
Email: customerservice@weidmuller.com

Technical data

Approvals	IEC / EN 60947-2, VDE				
Number of poles	1	2 (1+N)	2	3	4(3+N)
Operating voltages	240 V AC	240 V AC	415 V AC	415 V AC	415 V AC
Minimum current rating	0.5 A				
Maximum current rating	25 A				
Interrupting capacity	6 kA				
Operating temperature range	-40°C to +85°C				
Mounting options	DIN Rail, Dual mounting (DIN & Mini), surface mounting clip				
Time delay curves	1, 3, 9, KM, OP				

Approvals	UL 489			
Number of poles	1	1	2	3
Operating voltages	120 V AC	240 V AC	120 / 240 V AC	240 V AC
Minimum current rating	0.5 A			
Maximum current rating	25 A	25 A	25 A	25 A
Ampere interrupting capacity (AIC)	10 kA	5 kA	10 kA	5 kA
Operating temperature range	-40°C to +85°C			
Mounting options	DIN Rail, Dual mounting (DIN & Mini), surface mounting clip			
Time delay curves	1, 3, 9, KM, OP			

Insulation coordination and general data	
Endurance	10000 operations; 1500 electrical at rated current, 8500 without current (IEC 60947-2 Clause 7.2.4.2)* 6000 electrical operations (UL 489 Clause 7.1.9)*
Dielectric strength	1000-2000 Vac for one minute (IEC 60947-2 Clause 8.3.3.3)* 1000 Vac plus twice the rated voltage for one minute (UL 489 Clause 7.1.9)*
Rated impulsive withstand voltage	4 kV (IEC 60947-2)
Weight	102 g per pole, 160 g with auxiliary (unpacked)
Altitude	Certification tests done at altitude = 2000 metres. Will operate at higher altitudes.
Shock	8 G (IEC 60068-2-6) (sinusoidal wave)
Vibration	0.5 G (IEC 60068-2-6) (sinusoidal wave)
Flammability	I3 - Ignition does not persist at 850°C after glow wire is withdrawn with an oxygem index of ≥ 28
Toxicity	F1 - Smoke index of ≤ 20 which determines the fumes class
Pollution degree	PD2 - Normally only non-conductive pollution occurs. Temporary conductivity caused by condensation is to be expected.

Breaker: QL	Wire Size mm² (IEC)	Wire Gauge (UL)	Torque (IEC)	Torque (UL)	Comments
Main terminals	0.75 - 35 mm²	18 – 3 – AWG	2.5 Nm	20 in-lb	Pozidriv #2 Combi head

Long code

Example code: QL–AT-3-13-D-KM-15––R

Group	1	2	3	4	5	6	7	8	9	10
Requirement	QL Frame	Switch / Neutral	Auxiliary + Trip Alarm	Triple pole	13 mm module width	DIN Mounting	Medium delay curve KM	Current Rating 15 A	No Shunt Trip	Ring Terminal
Long Code	**QL**	**–**	**AT**	**3**	**13**	**DM**	**KM**	**15.0**	**–**	**R**

Group 1: Frame Type

Code	Description	Comments
QL	13 mm wide Miniature Circuit Breaker	UL 489, IEC / EN 60947-2, VDE, CE, CCC

Group 2: Switch/Neutral

Code	Description	Comments
–	Not applicable	Overload poles do not have any further coding
~~N~~	~~Neutral~~	~~Green Handle~~

Group 3: Auxiliary

Code	Description	Comments
–	Not applicable	Use this code if no Auxiliary used
~~A~~	~~Auxiliary Switch (1 x Aux in 1 module)~~	~~6.5 mm module fitted on right-hand side (DIN & Dual Mount)~~
~~T~~	~~Trip Alarm (1 x Trip Alarm in 1 module)~~	~~6.5 mm module fitted on right-hand side (Dual Mount only)~~
~~AT~~	~~Auxiliary Switch + Trip Alarm combo (combined in 1 module)~~	~~6.5 mm module fitted on right-hand side (Dual Mount only)~~

Group 4: No of Poles

Code	Description	Comments
1	Single pole	
~~2~~	~~Double pole~~	~~2 pole or1+N~~
~~3~~	~~Triple pole~~	
~~4~~	~~Four pole~~	~~3+N only~~

Group 5: Module Width

Code	Description	Comments
(13)	13 mm module width	13 mm per pole

Group 6: Mounting

Code	Description	Comments
~~D~~	~~DIN rail mount~~	~~DIN mount supplied in grey only – 45 mm Escutcheon, grey body~~
DM	Dual mount	Dual mount supplied in black only – 57 mm Escutcheon, black body

Group 7: Time Delays

Code	Description	Instantaneous Trip Point (x In)	Comments
~~1~~	~~Long time delay, high instantaneous trip~~	~~10 – 20~~	~~Orange handle~~
~~3~~	~~Short time delay~~	~~3 – 5~~	~~White handle~~
~~9~~	~~Long time delay~~	~~7 – 12~~	~~White handle~~
KM	Medium time delay	6 – 12	White handle
~~OP~~	~~Instantaneous~~	~~None~~	~~White handle~~

Group 8: Current Ratings

Code	Description	Comments
	~~0.5~~, 1, 2, ~~3, 4, 5, 6, 7, 8 ,9, 10, 12,~~ 15, ~~16,~~ 20, ~~25~~ A	Ratings available vary depending on certification * Other ratings are available as special orders. Check availability.

Group 9: Shunt Trip (not certified, only offered as special order)

Code	Description	Comments
–	Not applicable	If no shunt trip is required
U0	12 Vdc Shunt trip (Flying Leads)	Flying leads (+/- 60 mm long) Recommended pulse duration 100 ms to 60 s Note: Shunt trip carries no approvals - available as a special offer only
U1	24 Vdc Shunt trip (Flying Leads)	
U2	48 Vdc Shunt trip (Flying Leads)	
U3	60 Vdc Shunt trip (Flying Leads)	
U4	110 Vdc & 110 - 230 Vac Shunt Trip (Fly Leads)	

Group 10: Special Termination

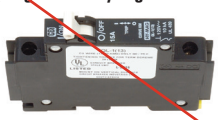
Code	Description	Comments
R	Ring Terminal ready	Terminal Screw is removable. Ring lug (5 mm)

QL Series Single Pole



Type: Single Pole (up to 240 VAC)		
Current Ratings (amps)	Description	Order Number
0.5	QL-1-13-DM-KM-0.5	7940102571
1	QL-1-13-DM-KM-1.0	7940125379
2	QL-1-13-DM-KM-2.0	7940102572
3	QL-1-13-DM-KM-3.0	7940102573
4	QL-1-13-DM-KM-4.0	7940102574
5	QL-1-13-DM-KM-5.0	7940102575
6	QL-1-13-DM-KM-6.0	7940102576
7	QL-1-13-DM-KM-7.0	7940102577
8	QL-1-13-DM-KM-8.0	7940102578
9	QL-1-13-DM-KM-9.0	7940102579
10	QL-1-13-DM-KM-10.0	7940102580
13	QL-1-13-DM-KM-13.0	7940102581
15	QL-1-13-DM-KM-15.0	7940102582
16	QL-1-13-DM-KM-16.0	7940102583
20	QL-1-13-DM-KM-20.0	7940102584
25	QL-1-13-DM-KM-25.0	7940102585

QL Series Single Pole (long time delay/high instant trip)



Type: Single Pole (up to 240 VAC) (long time delay/high instant trip)		
Current Ratings (amps)	Description	Order Number
1	QL-1-13-DM-1-1.0	7940115210
2	QL-1-13-DM-1-2.0	7940112653
3	QL-1-13-DM-1-3.0	7940115283
4	QL-1-13-DM-1-4.0	7940115284
5	QL-1-13-DM-1-5.0	7940113986
6	QL-1-13-DM-1-6.0	7940115211
7	QL-1-13-DM-1-7.0	7940114950
8	QL-1-13-DM-1-8.0	7940115282
10	QL-1-13-DM-1-10.0	7940102587
15	QL-1-13-DM-1-15.0	7940113987
16	QL-1-13-DM-1-16.0	7940114409
20	QL-1-13-DM-1-20.0	7940102588
25	QL-1-13-DM-1-25.0	7940112654

QL Series Double Pole



Type: Double Pole (up to 240 VAC)		
Current Ratings (amps)	Description	Order Number
0.1	QL-2-13-DM-KM-0.1	7940115220
0.5	QL-2-13-DM-KM-0.5	9926252000
1	QL-2-13-DM-KM-1.0	7940102643
2	QL-2-13-DM-KM-2.0	7940102644
3	QL-2-13-DM-KM-3.0	7940102645
4	QL-2-13-DM-KM-4.0	7940102646
5	QL-2-13-DM-KM-5.0	7940102647
6	QL-2-13-DM-KM-6.0	9926252006
7	QL-2-13-DM-KM-7.0	7940102649
8	QL-2-13-DM-KM-8.0	7940102650
10	QL-2-13-DM-KM-10.0	9926252010
12	QL-2-13-DM-KM-12.0	7940115938
13	QL-2-13-DM-KM-13.0	7940102652
15	QL-2-13-DM-KM-15.0	9926252015
16	QL-2-13-DM-KM-16.0	7940102654
20	QL-2-13-DM-KM-20.0	7940126749
25	QL-2-13-DM-KM-25.0	7940102656

QL Series Double Pole (long time delay/high instant trip)



Type: Double Pole (up to 240 VAC) (long time delay/high instant trip)		
Current Ratings (amps)	Description	Order Number
2	QL-2-13-DM-1-2.0	7940102657
3	QL-2-13-DM-1-3.0	7940114949
4	QL-2-13-DM-1-4.0	7940113820
5	QL-2-13-DM-1-5.0	7940102658
6	QL-2-13-DM-1-6.0	7940102659
7	QL-2-13-DM-1-7.0	7940114948
10	QL-2-13-DM-1-10.0	7940102660
13	QL-2-13-DM-1-13.0	7940113821
15	QL-2-13-DM-1-15.0	7940102661
20	QL-2-13-DM-1-20.0	7940113822
25	QL-2-13-DM-1-25.0	7940113823

QL Series Triple Pole



Type: Triple Pole (up to 240 VAC)		
Current Ratings (amps)	Description	Order Number
1	QL-3(13)-DM-KM-1.0	7940102691
2	QL-3(13)-DM-KM-2.0	7940102692
5	QL-3-13-DM-KM-5.0	7940102693
10	QL-3-13-DM-KM-10.0	7940104412
15	QL-3-13-DM-KM-15.0	7940102694
20	QL-3-13-DM-KM-20.0	7940102695
25	QL-3-13-DM-KM-25.0	7940102696

QL Series Single Pole w/ Auxiliary Contact



Type: Single Pole (up to 120 VAC)		
Current Ratings (amps)	Description	Order Number
1	QL-A-1-13-DM-KM-1.0	7940102708
2	QL-A-1-13-DM-KM-2.0	7940102709
3	QL-A-1-13-DM-KM-3.0	7940102710
5	QL-A-1-13-DM-KM-5.0	7940102711
10	QL-A-1-13-DM-KM-10.0	7940102712
15	QL-A-1-13-DM-KM-15.0	7940102713
20	QL-A-1-13-DM-KM-20.0	7940102714
25	QL-A-1-13-DM-KM-25.0	7940102715

QL Series Double Pole w/ Auxiliary Contact



Type: Double Pole (up to 240 VAC)		
Current Ratings (amps)	Description	Order Number
1	QL-A-2-13-DM-KM-1.0	7940114041
2	QL-A-2-13-DM-KM-2.0	7940119828
5	QL-A-2-13-DM-KM-5.0	7940119829
15	QL-A-2-13-DM-KM-15.0	7940119830
20	QL-A-2-13-DM-KM-20.0	7940119831

QL Series Single Pole (Alarm Contact)



Type: Single Pole (up to 120 VAC)		
Current Ratings (amps)	Description	Order Number
1	QL-T-1-13-DM-KM-1.0	7940114042
2	QL-T-1-13-DM-KM-2.0	7940114043
5	QL-T-1-13-DM-KM-5.0	7940114044
10	QL-T-1-13-DM-KM-10.0	7940102724
15	QL-T-1-13-DM-KM-15.0	7940114045
20	QL-T-1-13-DM-KM-20.0	7940114046
25	QL-T-1-13-DM-KM-25.0	7940114047

QL Series Double Pole (Alarm Contact)



Type: Double Pole (up to 240 VAC)		
Current Ratings (amps)	Description	Order Number
1	QL-T-2-13-DM-KM-1.0	7940114048
2	QL-T-2-13-DM-KM-2.0	7940114049
5	QL-T-2-13-DM-KM-5.0	7940114050
10	QL-T-2-13-DM-KM-10.0	7940114051
15	QL-T-2-13-DM-KM-15.0	7940114052
20	QL-T-2-13-DM-KM-20.0	7940114053
25	QL-T-2-13-DM-KM-25.0	7940114054

QL Series Triple Pole (Alarm Contact)



Type: Triple Pole (up to 240 VAC)		
Current Ratings (amps)	Description	Order Number
1	QL-T-3(13)-DM-KM-1.0	7940115448
2	QL-T-3(13)-DM-KM-2.0	7940115449
5	QL-T-3(13)-DM-KM-5.0	7940115450
10	QL-T-3(13)-DM-KM-10.0	7940115451
15	QL-T-3(13)-DM-KM-15.0	7940115452
20	QL-T-3(13)-DM-KM-20.0	7940115453
25	QL-T-3(13)-DM-KM-25.0	7940115454

QL Series Single Pole Combination (Auxiliary & Alarm Contact)



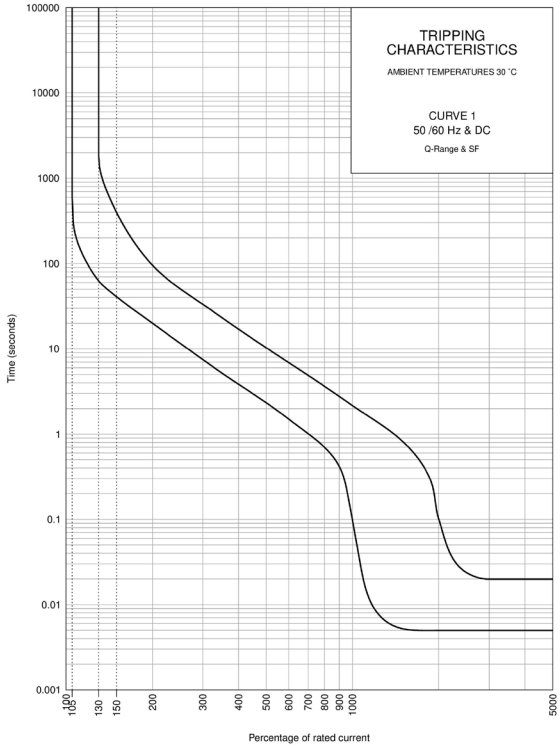
Type: Single Pole (up to 120 VAC)		
Current Ratings (amps)	Description	Order Number
1	QL-AT-1-13-DM-KM-1.0	7940114055
5	QL-AT-1-13-DM-KM-5.0	7940102725
10	QL-AT-1-13-DM-KM-10.0	7940102726
15	QL-AT-1-13-DM-KM-15.0	2853970000
20	QL-AT-1-13-DM-KM-20.0	7940114056
25	QL-AT-1-13-DM-KM-25.0	7940114057

QL Series Double Pole Combination (Auxiliary & Alarm Contact)

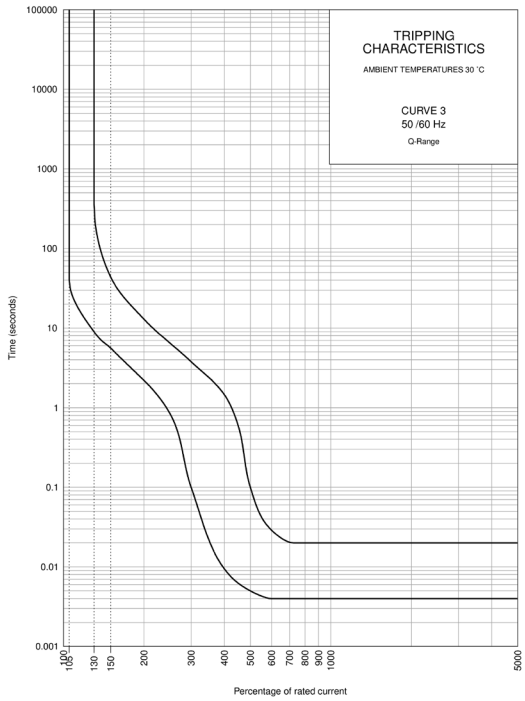


Type: Double Pole (up to 240 VAC)		
Current Ratings (amps)	Description	Order Number
1	QL-AT-2-13-DM-KM-1.0	7940114058
2	QL-AT-2-13-DM-KM-2.0	7940114059
10	QL-AT-2-13-DM-KM-10.0	7940126754
15	QL-AT-2-13-DM-KM-15.0	7940102727
20	QL-AT-2-13-DM-KM-20.0	7940114060
25	QL-AT-2-13-DM-KM-25.0	7940114061

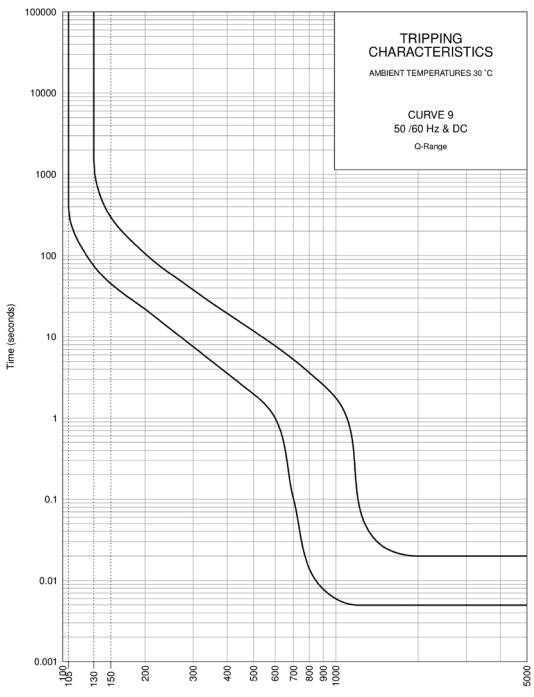
Time Delay Curves



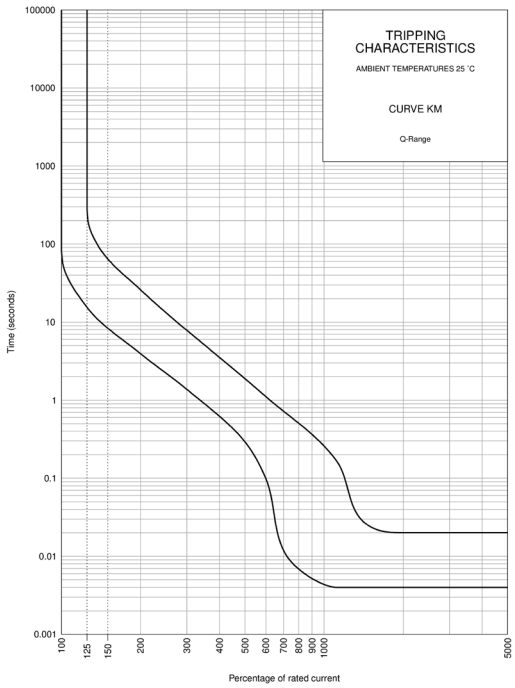
PERCENTAGE OF RATED CURRENT	105%	130%	145%	150%	200%	300%	400%	500%	600%	700%	800%	900%	1000%	2000%
MINIMUM TRIP TIME IN SECONDS	NO TRIP	63	45	41	20	7.5	3.8	2.3	1.5	1.01	0.7	0.42	0.1	0.005
MAXIMUM TRIP TIME IN SECONDS	NO TRIP	1800	503	400	96	34	17	10.3	6.8	4.9	3.6	2.8	2.2	0.1



PERCENTAGE OF RATED CURRENT	105%	130%	145%	150%	200%	300%	400%	500%	600%	700%	800%	900%	1000%	2000%
MINIMUM TRIP TIME IN SECONDS	NO TRIP	9	6	5.6	2.2	0.1	0.009	0.005	0.004	0.004	0.004	0.004	0.004	0.004
MAXIMUM TRIP TIME IN SECONDS	NO TRIP	365	57	44	13	3.8	1.45	0.1	0.028	0.02	0.02	0.02	0.02	0.02

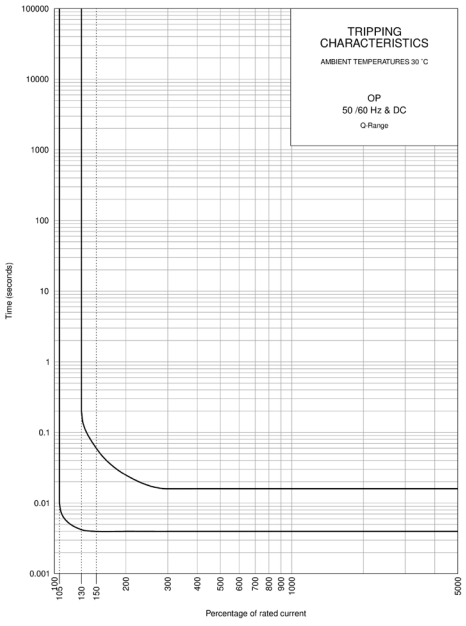


PERCENTAGE OF RATED CURRENT	105%	130%	145%	150%	200%	300%	400%	500%	600%	700%	800%	900%	1200%	2000%
MINIMUM TRIP TIME IN SECONDS	NO TRIP	75	50	45	22	7.6	3.6	2	1	0.1	0.01	0.008	0.005	0.005
MAXIMUM TRIP TIME IN SECONDS	NO TRIP	1500	360	295	100	38	19	11.8	7.7	5.3	3.6	2.6	0.06	0.02



PERCENTAGE OF RATED CURRRENT	100%	125%	130%	145%	200%	300%	400%	500%	600%	700%	800%	900%	1000%	2000%
MINMUM TRIP TIME IN SECONDS	NO TRIP	16	13.2	9.2	26	1.4	0.6	0.3	0.1	0.01	0.006	0.005	0.004	0.004
MAXIMUM TRIP TIME IN SECONDS	NO TRIP	279	140	74	4	7.8	3.5	1.9	1.1	0.7	0.5	0.36	0.25	0.02

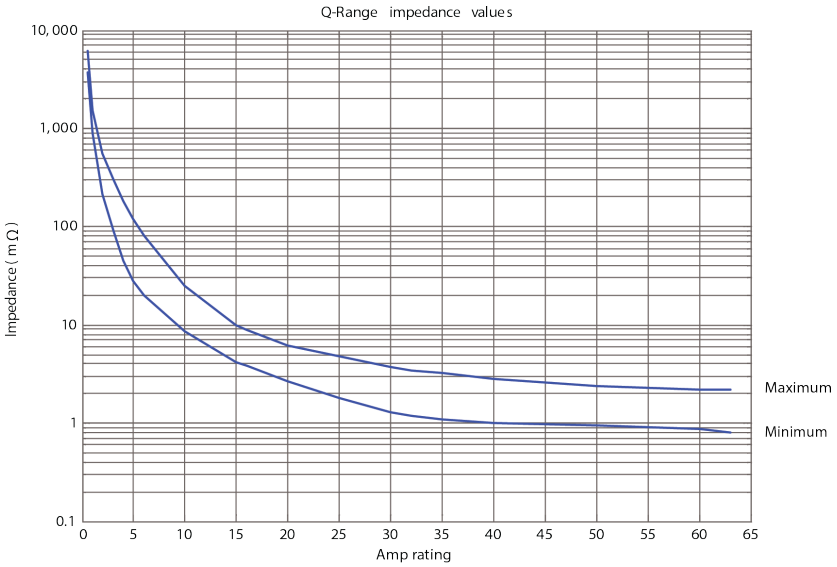
Time Delay Curves



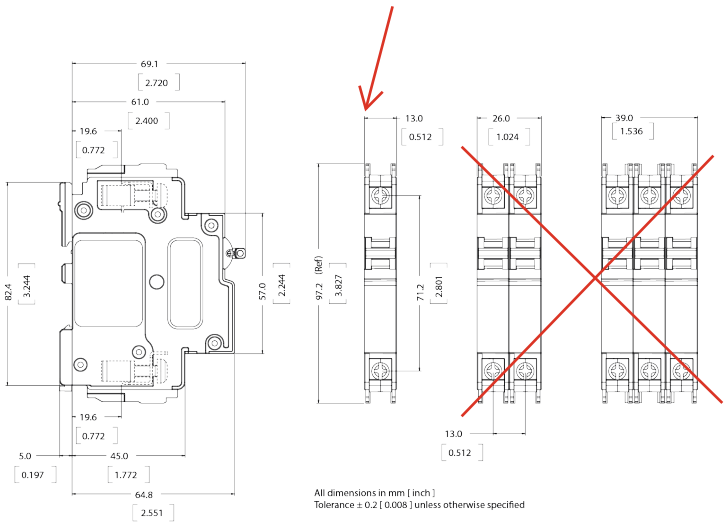
PERCENTAGE OF RATED CURRENT	105%	130%	145%	150%	200%	300%	400%	500%	600%	700%	800%	900%	1000%	2000%
MINIMUM TRIP TIME IN SECONDS	NO TRIP	0.0042	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.004
MAXIMUM TRIP TIME IN SECONDS	NO TRIP	0.2	0.07	0.06	0.025	0.016	0.016	0.016	0.016	0.016	0.016	0.016	0.016	0.016

* The published time delay curves are generated at 30°C ambient temperature with the Circuit Breaker mounted in the up-right position. The “must hold”, “must trip” and “instantaneous trip” current values are not affected by temperature, although delay time for the other operating current values may have to be adjusted using the temperature compensation curve which is available on request.

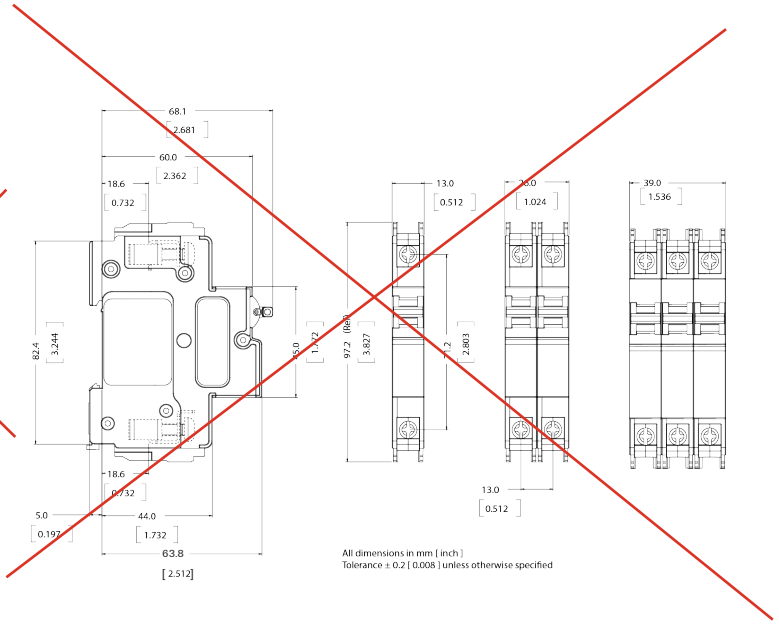
Internal Impedance vs Current Rating



Outline Dimensions: Dual mount



Outline Dimensions: DIN mount



CIRCUIT BREAKER, 24VDC

Datasheet

QY Series Hydraulic-Magnetic Miniature Circuit Breakers

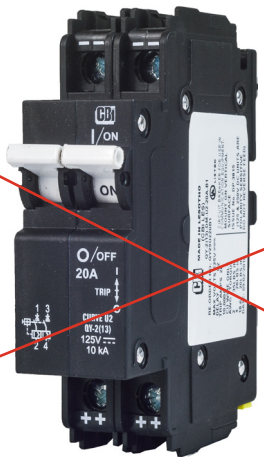
TAGS: CB1421, CB1437



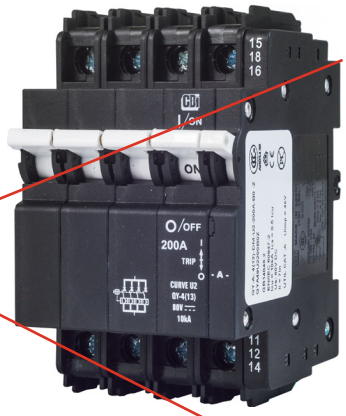
Dual mount 1 Pole



DIN mount 1 Pole



DIN mount 2 Pole



DIN mount 4 Pole + Auxiliary

Features

- DC Circuit Breaker
- Hydraulic-Magnetic Technology
- 100% Rating Capability Independent of Ambient Temperature
- Up to Two Poles (Four Poles Parallel)
- VDE, CE and CCC approved
- UL Listed (UL 489A)
- Ratings up to 200A
- Optional Auxiliary / Trip Alarm Switch - Factory Fitted
- Optional Shunt Trip
- Wide Range of Time Delays and Operating Currents
- Precision Tripping
- Ultra Compact – 13mm Width Module
- Trip Indication with Mid-Trip Handle
- Reset Immediately After Overload

Applications

- DC branch circuit protection
- Telecommunication equipment
- UPS equipment
- Alternative energy equipment
- Battery protection & switching
- Telecommunication DC power distribution

Approvals

Weidmüller

Technical Data

Approval	IEC / EN 60947-2						
Number of Poles	1	2	2 parallel	2 parallel	3 parallel	4 parallel	2 series
Operating Voltages	80 Vdc, 125 Vdc		80 Vdc	125 Vdc	80 Vdc		250 Vdc
Minimum Current Rating	0.1A	0.1A	30A	30A	110A	200A	0.1A
Maximum Current Rating	63A	50A	100A	100A	150A	200A	50A
Ultimate S/C Breaking Capacity (Icu)	10 kA						

Approval	UL 489A							
Number of Poles	1	2	2 parallel	2 parallel	3 parallel	4 parallel	2 series	4 series
Operating Voltages	80 Vdc, 125 Vdc		80 Vdc	125 Vdc	80/125 Vdc	80 Vdc	250 Vdc	600 Vdc
Minimum Current Rating	0.1A	0.1A	30A	20A	110A	200A	0.1A	0.1A
Maximum Current Rating	63A	50A	100A	100A	150A	200A	50A	20A
Ampere Interrupting Capacity (AIC)	10 kA							

Product Type	QY
Ambient Operating Temperature	-40°C to +85°C
Mounting Options	Dual mounting (DIN & mini), Surface mounting clip
Time Delay Curves	1,9, U2, U3, OP
Endurance	10000 operations - 1500 with current, 8500 without current (IEC 60947-2 Clause 7.2.4.2)* Minimum of 1000 operations with current (UL 489A Clause 12)*
Dielectric Strength	1000 - 2000 Vac for one minute (IEC 60947-2 Clause 8.3.3.3)* 1000 Vac plus twice the rated voltage for one minute (UL 489A Clause 8)*
Rated Impulsive Withstand Voltage	4 kV (IEC 6097-2 Clause 8.3.3.2)*
Weight	102 g per pole, 160 g with auxiliary (unpacked)
Altitude	Certification tests done at altitude = 2000 meters. Will operate at higher altitudes
Shock	16 G (IEC 60068-2-27)
Vibration	2 G (IEC 60068-2-6) (sinusoidal wave)
Flammability	I3 - ignition does not persist at 850°C after glow wire is withdrawn with an oxygen index of ≥ 28
Toxicity	F1 - Smoke index of ≤ 20 which determines the fume class
Pollution Degree	PD2 - Normally only non-conductive pollution occurs. Temporary conductivity caused by condensation is to be expected.

Circuit Breaker	Wire Size mm² (IEC)	Wire Gauge (UL)	Torque (IEC)	Torque (UL)	Comments
1 Pole & 2 Pole	0.75 - 25 mm²	18 -2 - AWG	2.5 Nm	20 in-lb	Pozidriv #2 Combi head
2 Pole Parallel	50 mm²	14 - 1/0 - AWG	3.2 Nm	28 in-lb	Bridge terminal
3 Pole Parallel	95 mm²	14 - 5/0 - AWG	5.6 Nm	50 in-lb	Bridge terminal

Long Code

Example Code: QY-3-13-D-U2-150.0-B0

Group	1	2	3	4	5	6	7	8
Requirement	QY Frame	Auxiliary	Triple pole	13 mm module width	DIN Rail	Medium delay curve U2	Current rating 150 A	Voltage 80 Vdc
Long Code	**QY**	**A**	**3**	**13**	**D**	**U2**	**150.0**	**B0**

Group 1: Frame Type

Code	Description	Comments
QY	Miniature Circuit Breaker	UL 489A, IEC / EN 60947-2, GB 14048.2

Group 2: Auxiliary

Code	Description	Comments
–	Not applicable	if auxiliary is not required
~~A~~	~~Auxiliary switch 1 x Aux in 1 module)~~	~~6.5 mm module fitted on right hand side (DIN & Dual mount)~~
~~T~~	~~Trip alarm (1 xx Trip alarm in 1 module)~~	~~6.5 mm module fitted on right hand side (Dual mount)~~
~~AT~~	~~Auxiliary switch + trip alarm combo (combined in 1 module)~~	~~6.5 mm module fitted on right hand side (Dual mount)~~

Group 3: No. of Poles

Code	Description	Comments
1	Single pole	
~~2~~	~~Double pole~~	
~~3~~	~~Triple pole~~	

Group 4: Module Width

Code	Description	Comments
13	13 mm module width	13 mm

Group 5: Mounting

Code	Description	Comments
~~D~~	~~DIN rail mount~~	~~DIN mount supplied in grey only - 45 mm escutcheon, grey body~~
DM	Dual mount	Dual mount supplied in black only - 57 mm escutcheon, black body

Group 6: Time Delays

Code	Description	Instantaneous Trip Point (x In)	Comments
~~1~~	~~Long time relay, high instantaneous trip~~	~~10 - 20~~	~~Orange handle~~
~~9~~	~~Long time delay~~	~~7 - 12~~	~~White handle~~
U2	Medium time delay	5 - 10	White handle
~~U3~~	~~Short time delay~~	~~3 - 5~~	~~White handle~~
~~OP~~	~~Instantaneous~~	~~None~~	~~White handle~~

Group 7: Current Ratings

Code / Description	Comments
~~0.1, 0.2, 0.3, 0.5~~, 1, ~~2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 15, 16, 20, 25, 30, 32, 35, 40, 45, 50, 60, 63, 70, 80, 90, 100, 120, 125, 150, 200 A~~	Ratings available vary depending on certification, bridging configuration and voltage. *Other ratings are available as special orders.

Group 8: Voltage (see diagram on pg 7)

Code	Voltage	Description	Comments
B0	80 Vdc	Not polarity sensitive	
B1	125 Vdc	Polarity sensitive. Positive bottom.	
B2	250 Vdc	Polarized. Positive bottom	Two poles intended to be bridged in series from the Top of (Pole 1) to the bottom of (Pole 2), bridging to be done by customer
B3	250 Vdc	Polarized. Positive bottom / Positive top	Factory bridged at the top
B4	600 Vdc	Polarized. Positive bottom / Positive top / Positive bottom / Positive top	Factory bridged top / bottom / top
T1	125 Vdc	Polarized. Positive top	
T2	250 Vdc	Polarized. Positive top	Two poles intended to be bridged in series from the bottom of (Pole 1) to top of (Pole 2) - bridging to be done by customer
T3	250 Vdc	Polarized. Positive top / Positive bottom	Factory bridged at the bottom
T4	600 Vdc	Polarized. Positive top / Positive bottom / Positive top / Positive bottom	Factory bridged bottom / top / bottom
B1 - T1	125 Vdc	Polarity sensitive	1 Pole Positive bottom / 2 Pole positive top
T3 - B3	250 Vdc	Polarity sensitive	1 & 2 Pole Bottom bridged / 3 & 4 Pole top bridged in series

QY Series Single Pole (up to 80 VDC)



Type: Single Pole (up to 80 VDC)		
Current Ratings (amps)	Description	Order Number
0.5	QY-1-13-DM-U2-0.5	7940102623
1	QY-1-13-DM-U2-1.0	7940102624
2	QY-1-13-DM-U2-2.0	7940102625
3	QY-1-13-DM-U2-3.0	7940102626
4	QY-1-13-DM-U2-4.0	7940102627
5	QY-1-13-DM-U2-5.0	7940102628
6	QY-1-13-DM-U2-6.0	7940102629
7	QY-1-13-DM-U2-7.0	7940102630
8	QY-1-13-DM-U2-8.0	7940102631
10	QY-1-13-DM-U2-10.0	9926251810
15	QY-1-13-DM-U2-15.0	9926251815
20	QY-1-13-DM-U2-20.0	9926251820
25	QY-1-13-DM-U2-25.0	7940102635
30	QY-1-13-DM-U2-30.0	7940102636
35	QY-1-13-DM-U2-35.0	7940102637
40	QY-1-13-DM-U2-40.0	7940102638
45	QY-1-13-DM-U2-45.0	7940102639
50	QY-1-13-DM-U2-50.0	7940102640
60	QY-1-13-DM-U2-60.0	9926251960

QY Single Pole Positive Bottom (up to 125 VDC)



Type: Single Pole (up to 125 VDC)		
Current Ratings (amps)	Description	Order Number
1	QY-1-13-DM-U2-1.0-B1	7940102609
2	QY-1-13-DM-U2-2.0-B1	9926251802
3	QY-1-13-DM-U2-3.0-B1	7940102611
4	QY-1-13-DM-U2-4.0-B1	7940102612
5	QY-1-13-DM-U2-5.0-B1	7940102613
6	QY-1-13-DM-U2-6.0-B1	7940116300
10	QY-1-13-DM-U2-10.0-B1	7940102615
15	QY-1-13-DM-U2-15.0-B1	9926251815
20	QY-1-13-DM-U2-20.0-B1	7940102617
25	QY-1-13-DM-U2-25.0-B1	7940102618
30	QY-1-13-DM-U2-30.0-B1	7940102619
35	QY-1-13-DM-U2-35.0-B1	7940112672
40	QY-1-13-DM-U2-40.0-B1	7940102620
50	QY-1-13-DM-U2-50.0-B1	7940102621
60	QY-1-13-DM-U2-60.0-B1	7940102622

QY Series Double Pole (125 VDC)



Type: Double Pole (125 VDC)		
Current Ratings (amps)	Description	Order Number
0.5	QY-2-13-DM-U2-0.5	7940109143
1	QY-2-13-DM-U2-1.0	7940109144
2	QY-2-13-DM-U2-2.0	7940109145
3	QY-2-13-DM-U2-3.0	7940109146
5	QY-2-13-DM-U2-4.0	7940109147
5	QY-2-13-DM-U2-5.0	7940102685
10	QY-2-13-DM-U2-10.0	7940102686
15	QY-2-13-DM-U2-15.0	7940126751
20	QY-2-13-DM-U2-20.0	7940126752
25	QY-2-13-DM-U2-25.0	7940109148
30	QY-2-13-DM-U2-30.0	7940126753
35	QY-2-13-DM-U2-35.0	7940109149
40	QY-2-13-DM-U2-40.0	7940115443
45	QY-2-13-DM-U2-45.0	7940109150
50	QY-2-13-DM-U2-50.0	7940109151
60	QY-2-13-DM-U2-60.0	7940109152

QY Series Single Pole Long Time Relay (up to 80 VDC)



Type: Single Pole (up to 80 VDC)		
Current Ratings (amps)	Description	Order Number
1	QY-1-13-DM-1-1.0	7940112648
2	QY-1-13-DM-1-2.0	7940112649
3	QY-1-13-DM-1-3.0	7940115098
4	QY-1-13-DM-1-4.0	7940112650
5	QY-1-13-DM-1-5.0	7940112651
7	QY-1-13-DM-1-7.0	7940115097
25	QY-1-13-DM-1-25.0	7940112652

QY Series Single Pole w/ Auxiliary Contact (up to 80 VDC)



Type: Single Pole (up to 80 VDC)		
Current Ratings (amps)	Description	Order Number
0.5	QY-A-1-13-DM-U2-0.5	7940102717
1	QY-A-1-13-DM-U2-1.0	7940102718
2	QY-A-1-13-DM-U2-2.0	7940102719
3	QY-A-1-13-DM-U2-3.0	7940102720
4	QY-A-1-13-DM-U2-4.0	7940102721
5	QY-A-1-13-DM-U2-5.0	7940102722
6	QY-A-1-13-DM-U2-6.0	7940115105
8	QY-A-1-13-DM-U2-8.0	7940104413
9	QY-A-1-13-DM-U2-9.0	7940115104

QY Series Single Pole Instant Trip (up to 80 VDC)



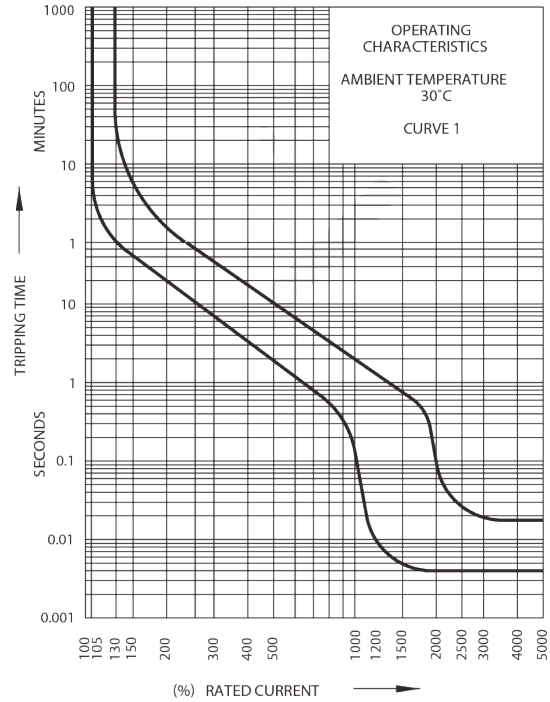
Type: Single Pole (up to 80 VDC)		
Current Ratings (amps)	Description	Order Number
1	QY-1-13-DM-OP-1.0	7940114110
2	QY-1-13-DM-OP-2.0	7940102606
3	QY-1-13-DM-OP-3.0	7940112396
5	QY-1-13-DM-OP-5.0	7940112397
6	QY-1-13-DM-OP-6.0	7940102607
10	QY-1-13-DM-OP-10.0	7940114112
20	QY-1-13-DM-OP-20.0	7940102608

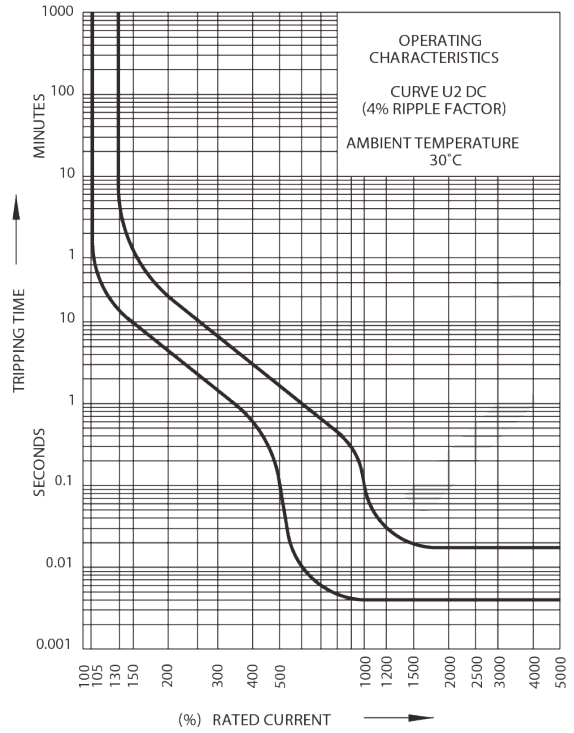
QY Series Double Pole Positive Bottom (125 VDC)



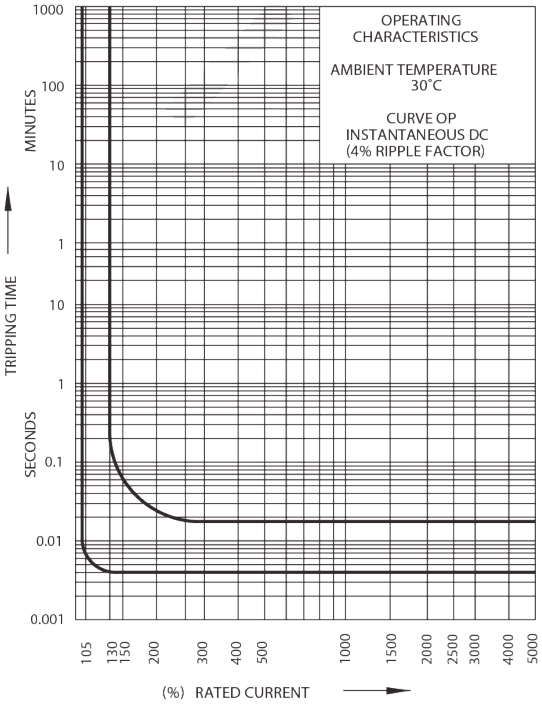
Type: Double Pole (125 VDC)		
Current Ratings (amps)	Description	Order Number
5	QY-2-13-DM-U2-5.0-B1	7940102677
10	QY-2-13-DM-U2-10.0-B1	7940102678
15	QY-2-13-DM-U2-15.0-B1	7940102679
20	QY-2-13-DM-U2-20.0-B1	7940126750
30	QY-2-13-DM-U2-30.0-B1	7940102681
32	QY-2-13-DM-U2-32.0-B1	7940102682

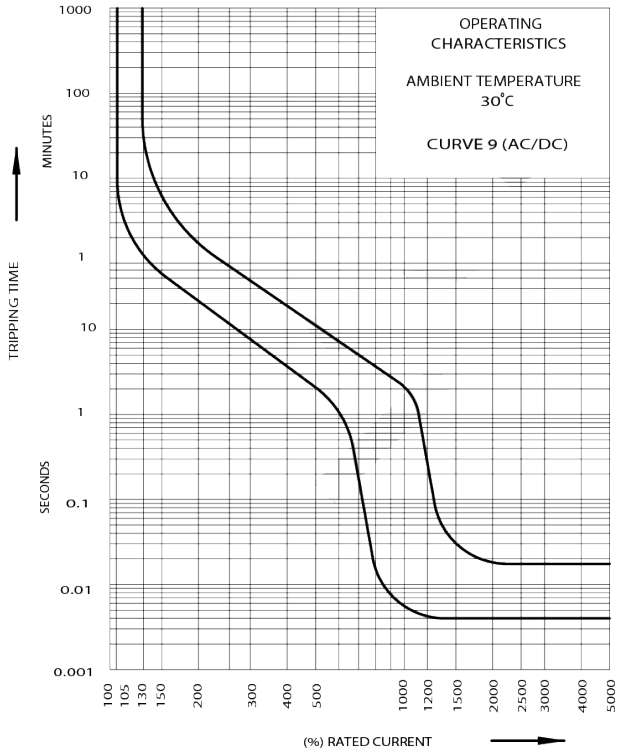
Standard Time Delay Curves



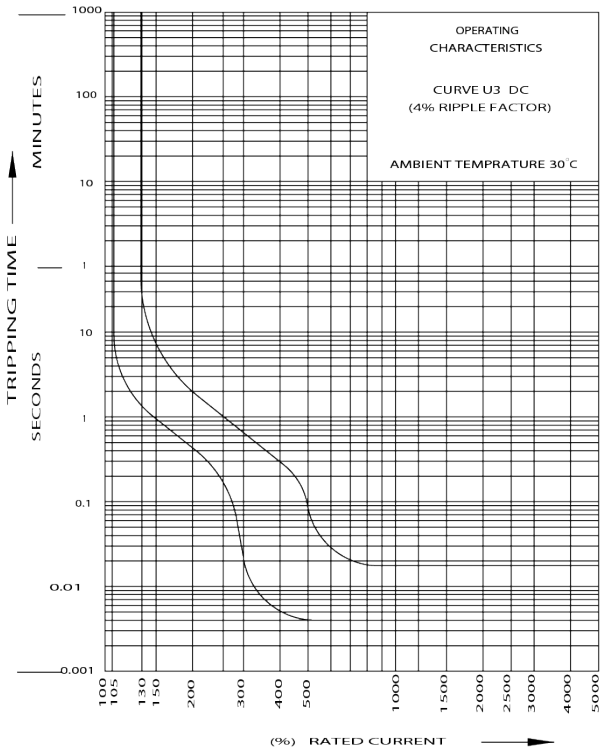


Standard Time Delay Curves

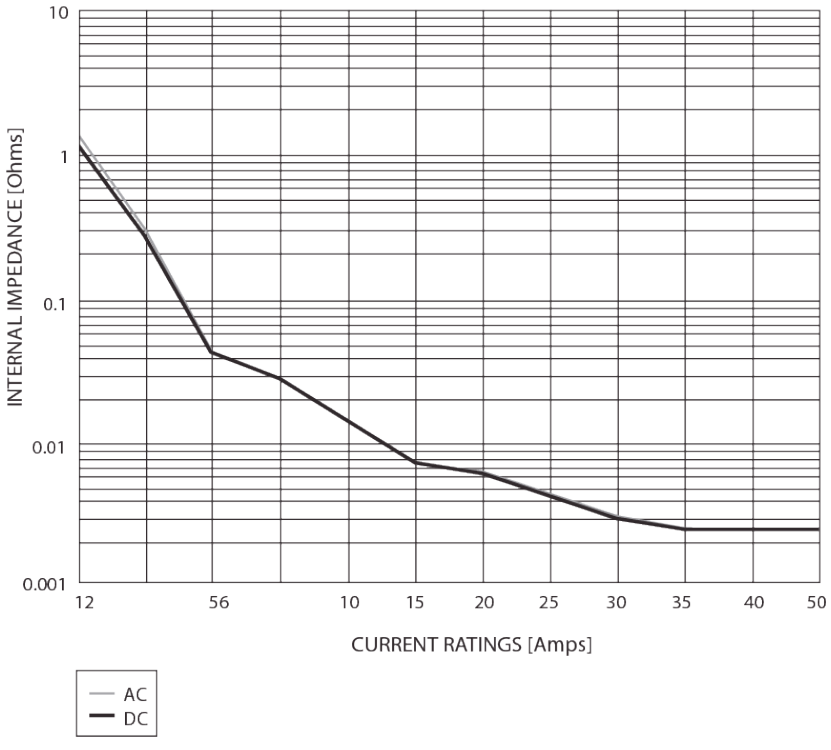




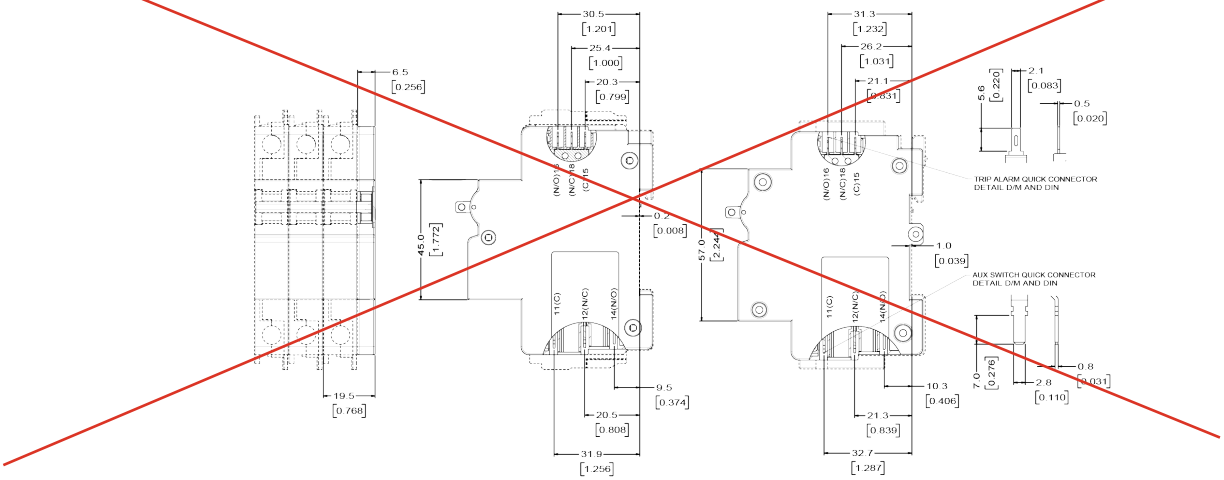
Standard Time Delay Curves



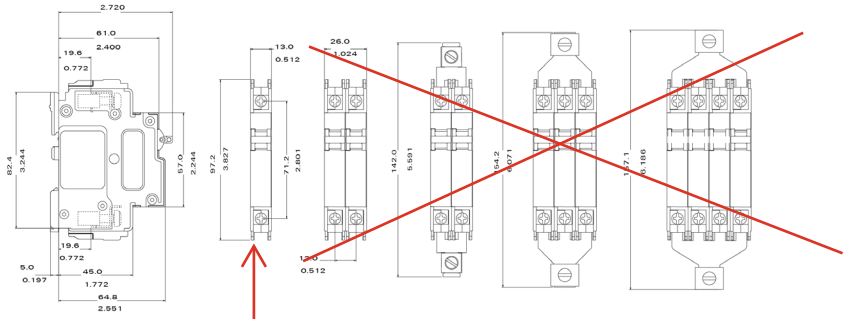
Internal Impedance vs Current Rating



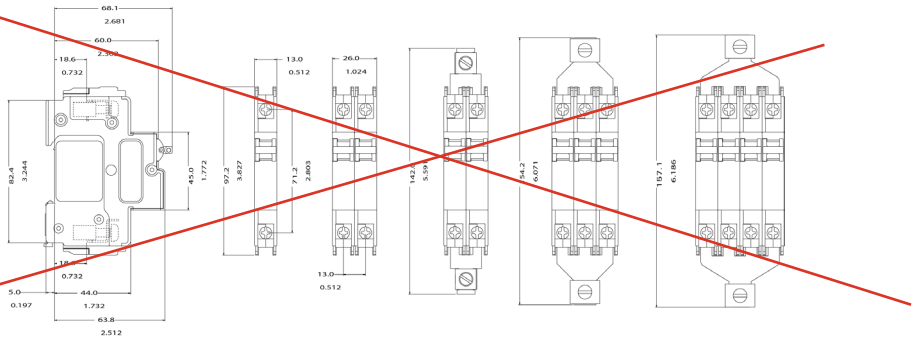
Auxiliary Outline Dimensions: DIN and Dual Mount



Outline Dimensions: Dual mount



Outline Dimensions: DIN mount



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Data Sheet 9

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
1	Mfg: Hubbel: Duplex GFCI Receptacle, 15A Model Number: GFRST15W Tags/Service: RECP1208 / PLC/RTU Control Panel Specifications: • 125VAC nominal supply voltage • 15A nominal current rating • 2-pole, 3-wire grounding • 5-15R outlet • White receptacle color

DUPLEX GFCI RECEPTACLE, 15A

HUBBELL
Wiring Device-Kellems

TAGS: RECP1208

Replaced By GFR15W - Heavy Duty Commercial AUTOGUARD® Self-Test GFCI Receptacle, 15A, White



By Hubbell Wiring Device-Kellems
Catalog # GFRST15W

Power Protection Devices, Receptacle, Self Test, GFCI, 15A 125V, 2-Pole 3-Wire Grounding, 5-15R, White

General

Catalog Number	GFRST15W
Color	White
EU RoHS Indicator	Contact Manufacturer
Series	Heavy Duty
Style	Standard
Type	Standard
	Standard
UPC	883778122071

Electrical Ratings

Amperage Rating	15 A
Current / Amperage Rating	15 A
Voltage Rating Description	125 V

Certifications And Compliance

Industry Standard(s)	Meets All NEC® Requirements
Nema Rating	5-15R

Logistics

Carton Quantity	10
Pallet Quantity	2240

Product Assets

Catalogs - WDK Catalog Full 2024
Literature - Hubbell SystemOne Recessed Fire-Rated Poke-Throughs
Warranty - Industrial & Commercial Products

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Data Sheet 10

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
10	Mfg: Schneider Electric: Miniature Relay, 120VAC, 8 Blade Model Number: RXM2AB2F7 Tags/Service: CR1203, CR1206, CR1702, CR1703, CR1704, CR1705, CR1706, CR1707, CR1708, CR1709 / PLC/RTU Control Panel Specifications: • 120VAC control circuit voltage • 10A continuous output current • 12A rated operational current • Miniature relay type with status LED • Lockable test button control type • 2 C/O contact composition • 100,000 cycles resistive electrical durability • 1.2VA average coil consumption
10	Mfg: Schneider Electric: Miniature Relay Socket, 120VAC, DIN Model Number: RXZE2M114 Tags/Service: CR1203, CR1206, CR1702, CR1703, CR1704, CR1705, CR1706, CR1707, CR1708, CR1709 / PLC/RTU Control Panel Specifications: • For RXM 2 C/O series plug-in relay • 10A conventional free air thermal current

	• < 300V UL rated operational voltage

Product data sheet

MINIATURE RELAY, 120VAC, 8 BLADE

Specifications

TAGS: CR1203, CR1206, CR1702, CR1703, CR1704, CR1705, CR1706, CR1707, CR1708, CR1709



miniature plug in relay, Harmony Electromechanical Relays, 12A, 2CO, with LED, lockable test but to n, 120V AC

RXM2AB2F7

Product availability: Stock - Normally stocked in distribution facility

Main

Range of Product	Harmony Electromechanical Relays
Series name	RXM series
Product or Component Type	Plug-in relay
Relay Type	Miniature relay
Contacts type and composition	2 C/O
Status LED	With
Control Type	Lockable test button
[Uc] control circuit voltage	120 V AC 50/60 Hz
[Ithe] conventional enclosed thermal current	12 A
Continuous output current	10 A

Complementary

[Uimp] rated impulse withstand voltage	4 kV 1.2/50 µs
[Ie] rated operational current	~~12 A 28 V DC) NO IEC~~ ~~12 A 250 V AC) NO IEC~~ ~~6 A 28 V DC) NC IEC~~ ~~6 A 250 V AC) NC IEC~~ ~~12 A 28 V DC) UL~~ 12 A 277 V AC) UL
Minimum switching capacity	170 mW 10 mA, 17 V
Electrical durability	100000 cycles resistive
Average coil consumption in VA	1.2 60 Hz
Rated operational voltage limits	96...132 V AC
[Ui] rated insulation voltage	~~250 V IEC~~ ~~300 V CSA~~ 300 V UL
Average consumption	1.2 VA 60 Hz
Maximum switching voltage	250 V IEC
Drop-out voltage threshold	>= 0.15 Uc
Load current	12 A 250 V AC ~~12 A 28 V DC~~
Operating time	20 ms
Maximum switching capacity	3000 VA/336 W
Average resistance	4430 Ohm 20 °C +/- 15 %

Price is “List Price” and may be subject to a trade discount – check with your local distributor or retailer for actual price.

Disclaimer: This documentation is not intended as a substitute for and is not to be used for determining suitability or reliability of these products for specific user applications

Mechanical durability	10000000 cycles
Safety reliability data	B10d = 100000
Operating rate	<= 1200 cycles/hour under load <= 18000 cycles/hour no-load
Utilisation coefficient	20 %
CAD overall height	3.1 in (79 mm)
CAD overall depth	3.09 in (78.45 mm)
reset time	20 ms
Dielectric strength	1300 V AC between contacts with micro disconnection 2000 V AC between coil and contact with basic insulation 2000 V AC between poles with basic insulation
Compatibility code	RXM
Protection category	RT I
Pollution degree	3
Operating position	Any position
Test levels	Level A group mounting
Device presentation	Complete product
Contacts material	AgNi
Shape of pin	Flat (faston type)
Net Weight	0.082 lb(US) (0.037 kg)

Environment

Ambient air temperature for operation	-40…131 °F (-40…55 °C)
IP degree of protection	IP40 conforming to IEC 60529
Standards	CSA C22.2 No 14 IEC 61810-1 UL 508
Product Certifications	UL Lloyd's CE CSA GOST IECEE CB Scheme
Ambient Air Temperature for Storage	-40…185 °F (-40…85 °C)
Vibration resistance	3 gn +/- 1 mm 10…150 Hz)5 cycles in operation 5 gn +/- 1 mm 10…150 Hz)5 cycles not operating
Shock resistance	10 gnin operation 30 gnnot operating

Ordering and shipping details

Category	US10CP221127
Discount Schedule	0CP2
GTIN	3389119403504
Returnability	Yes
Country of origin	CN

Packing Units

Unit Type of Package 1	PCE
Nbr. of units in pkg.	1
Package 1 Height	0.827 in (2.100 cm)
Package 1 Width	1.063 in (2.700 cm)
Package 1 Length	1.811 in (4.600 cm)
Package weight(Lbs)	1.305 oz (37.000 g)
Unit Type of Package 2	BB1
Number of Units in Package 2	10
Package 2 Height	1.181 in (3.000 cm)
Package 2 Width	4.016 in (10.200 cm)
Package 2 Length	4.921 in (12.500 cm)
Package 2 Weight	13.686 oz (388.000 g)
Unit Type of Package 3	S02
Number of Units in Package 3	240
Package 3 Height	5.906 in (15.000 cm)
Package 3 Width	11.811 in (30.000 cm)
Package 3 Length	15.748 in (40.000 cm)
Package 3 Weight	21.625 lb(US) (9.809 kg)

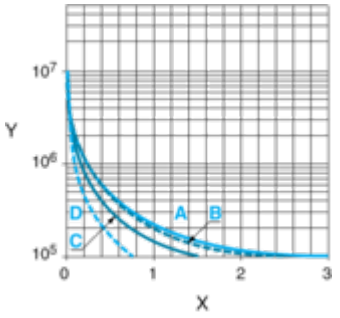
Contractual warranty

Warranty (in months)	18

Performance Curves

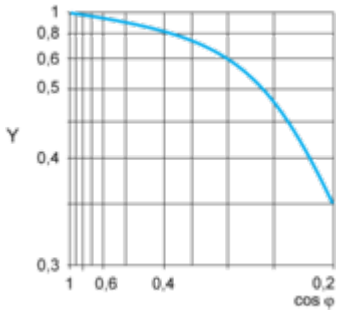
Electrical Durability of Contacts

Durability (inductive load) = durability (resistive load) x reduction coefficient.
Resistive AC load



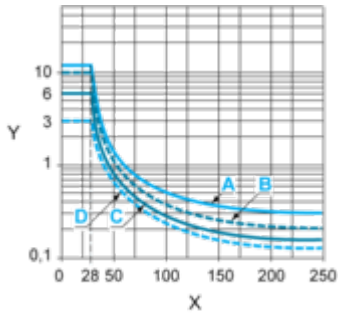
X Switching capacity (kVA)
Y Durability (Number of operating cycles)
A RXM2AB•••
~~**B** RXM3AB•••~~
~~**C** RXM4AB•••~~
~~**D** RXM4GB•••~~

Reduction coefficient for inductive AC load (depending on power factor cos ϕ)



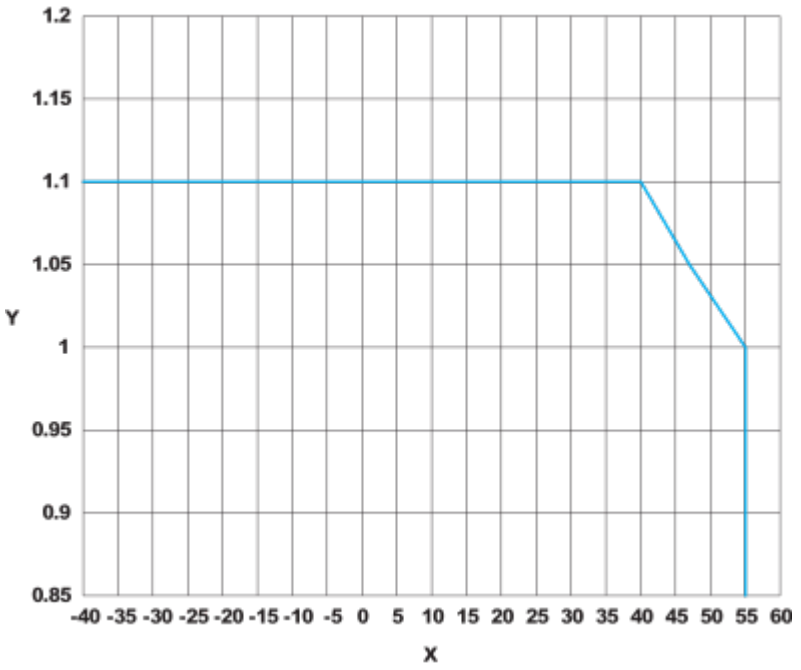
Y Reduction coefficient (A)

Maximum switching capacity on resistive DC load



X Voltage DC
Y Current DC
A RXM2AB•••
~~**B** RXM3AB•••~~
~~**C** RXM4AB•••~~
~~**D** RXM4GB•••~~
Note : These are typical curves, actual durability depends on load, environment, duty cycle, etc.
For inductive load, to increase relay life cycles, please add a proper load protection circuit (eg: RC protection/Varistor/free Wheeling diode -DC load only-).
For low level loads (below 10mA), we recommend to use RXM*GB series with bifurcated contacts relays instead.

AC Coil Voltage and Operating Temperature under continuous duty



X : Operating temperature (°C)
Y : AC coil voltage (UC)

Product data sheet

MINIATURE RELAY SOCKET, 120VAC, DIN

Specifications

TAGS: CR1203, CR1206, CR1702, CR1703, CR1704, CR1705, CR1706, CR1707, CR1708, CR1709



Harmony, plugin relay socket, mixed contact, 10 A, 250 V, screw clamp, for RXM2 or RXM4 relays

RXZE2M114

Product availability: Stock - Normally stocked in distribution facility

Main

Range of Product	Harmony Electromechanical Relays Harmony Timer Relays
Product compatibility	Plug-in relay RXM 2 C/O ~~Plug-in relay RXM 4 C/O~~ ~~Plug-in relay REXL 2 C/O~~ ~~Plug-in relay REXL 4 C/O~~ ~~Plug-in relay 782H 4 C/O~~
Device short name	RXZ
[Ith] conventional free air thermal current	10 A
Product or Component Type	Socket
Contacts type and composition	4 C/O

Complementary

[Ue] rated operational voltage	< 250 V IEC < 300 V UL
Connections - terminals	Screw clamp terminals, 1 x 0.34...1 x 1 mm² AWG 22...AWG 17) flexible with cable end Screw clamp terminals, 2 x 0.34...2 x 1 mm² AWG 22...AWG 17) flexible with cable end Screw clamp terminals, 1 x 0.5...1 x 1.5 mm² AWG 20...AWG 16) solid without cable end Screw clamp terminals, 2 x 0.5...2 x 1.5 mm² AWG 20...AWG 16) solid without cable end
Shape of pin	Flat
Tightening torque	< 1 N.m M3
Fixing mode	By screw (panel) Clip-on (35 mm symmetrical DIN rail)
Wire stripping length	0.3 in (8 mm)
Contact terminal arrangement	Mixed
[Uimp] rated impulse withstand voltage	4 kV
Overvoltage category	III
Pollution degree	3
Sale per indivisible quantity	10
Width	1.2 in (30 mm)
Net Weight	0.112 lb(US) (0.051 kg)

Environment

IP degree of protection	IP20 conforming to IEC 60529

Price is “List Price” and may be subject to a trade discount – check with your local distributor or retailer for actual price.

Disclaimer: This documentation is not intended as a substitute for and is not to be used for determining suitability or reliability of these products for specific user applications

Dielectric strength	2500 V
Standards	IEC 61984
Product Certifications	UL CSA Lloyd's
Marking	CE
Ambient Air Temperature for Storage	-40…185 °F (-40…85 °C)
Ambient Air Temperature for Operation	-40…131 °F (-40…55 °C)

Ordering and shipping details

Category	US10CP221128
Discount Schedule	0CP2
GTIN	3389119404242
Returnability	Yes
Country of origin	CN

Packing Units

Unit Type of Package 1	PCE
Nbr. of units in pkg.	1
Package 1 Height	1.024 in (2.600 cm)
Package 1 Width	1.220 in (3.100 cm)
Package 1 Length	2.756 in (7.000 cm)
Package weight(Lbs)	1.975 oz (56.000 g)
Unit Type of Package 2	BB1
Number of Units in Package 2	10
Package 2 Height	2.992 in (7.600 cm)
Package 2 Width	3.504 in (8.900 cm)
Package 2 Length	6.102 in (15.500 cm)
Package 2 Weight	21.164 oz (600.000 g)
Unit Type of Package 3	S03
Number of Units in Package 3	200
Package 3 Height	11.811 in (30.000 cm)
Package 3 Width	11.811 in (30.000 cm)
Package 3 Length	15.748 in (40.000 cm)
Package 3 Weight	27.445 lb(US) (12.449 kg)

Contractual warranty

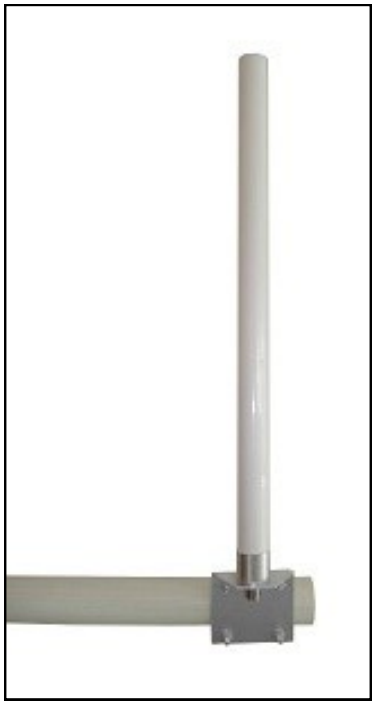
Warranty (in months)	18

Data Sheet 11

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
1	Mfg: Ventev: Cellular Antenna Model Number: T09060010006HM Tags/Service: ANTENNA / PLC/RTU Control Panel Specifications: • 824 960 MHz frequency range • 136MHz bandwidth • 6dBi gain • 360° horizontal beamwidth, 30° vertical beamwidth • 50Ω nominal impedance • 100W maximum power rating • Vertical polarization • N-Style female jack connector • 60m/s rated wind velocity • Fiberglass radome • −22°F 158°F operating temperature range

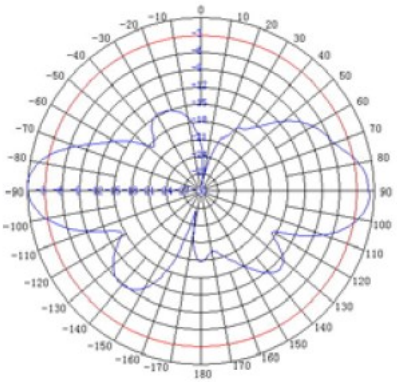
824-960 MHz 6 dBi Fiberglass Omnidirectional Antenna with N-style Jack Connector, Horizontal Mount

Ventev's Terrawave 6 dBi Fiberglass Omni Antenna is designed for applications operating in 824 - 960 MHz frequency, including radio frequency identification (RFID), cellular systems, global system for mobile (GSM) communications, code-division multiple access (CDMA) systems, point-to-multipoint communications and wireless video links. It features high gain and is vertically polarized. Includes a heavy duty mounting bracket and an N-style jack connector. *Every TerraWave antenna is covered by the Company's two year TerraNet warranty program. Visit www.terrawave.com for additional information.*

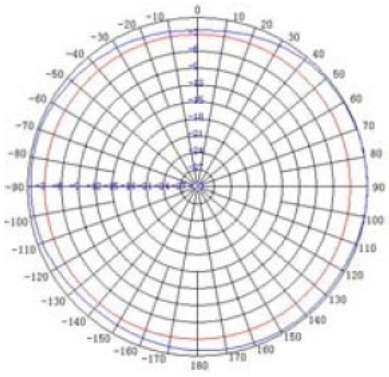


Specifications	
Model	T09060O10006HM
Frequency Range	824 ~ 960 MHz
Bandwidth	136 MHz
Gain	6 dBi
Horizontal Beamwidth	360°
Vertical Beamwidth	30°
VSWR	≤ 1.5
Nominal Impedance	50 Ohms
Maximum Power Rating	100 Watts
Polarization	Vertical
Connector	N-Style Jack (F)
Dimensions	22.05"x 1.42"
Rated Wind Velocity	60 m/s
Radome	Fiberglass
Pole Diameter	1.38" ~ 1.97"
Operating Temperature Range	-22°F ~ 158°F

E Plane Pattern



H Plane Pattern



Data Sheet 12

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
1	Mfg: Moxa: Unmanaged Ethernet Switch, 5 Port Model Number: EDS-205 Tags/Service: ETH1403 / PLC/RTU Control Panel Specifications: • 5 10/100BaseT(X) RJ45 ports • 24VDC input operating voltage • 0.77W maximum power consumption • Plastic, IP30 rated housing • 14 to 140°F operating temperature range

EDS-205 Series

UNMANAGED ETHERNET SWITCH, 5 PORT

TAGS: ETH1403

5-port entry-level unmanaged Ethernet switches



Features and Benefits

- 10/100BaseT(X) (RJ45 connector)
- IEEE802.3/802.3u/802.3x support
- Broadcast storm protection
- DIN-rail mounting ability
- -10 to 60°C operating temperature range

Certifications

Introduction

The EDS-205 Series supports IEEE 802.3/802.3u/802.3x with 10/100M, full/half-duplex, MDI/MDIX auto-sensing RJ45 ports. The EDS-205 Series is rated to operate at temperatures ranging from -10 to 60°C, and is rugged enough for any harsh industrial environment. The switches can be easily installed on a DIN rail as well as in distribution boxes. The DIN-rail mounting capability, wide operating temperature, and the IP30 housing with LED indicators make the plug-and-play EDS-205 switches reliable and easy to use.

Specifications

Ethernet Interface	
Standards	IEEE 802.3 for 10BaseT IEEE 802.3u for 100BaseT(X) IEEE 802.3x for flow control
10/100BaseT(X) Ports (RJ45 connector)	5 Full/Half duplex mode Auto MDI/MDI-X connection Auto negotiation speed

Switch Properties	
Processing Type	Store and Forward
MAC Table Size	2 K
Packet Buffer Size	768 kbits

Power Parameters	
Input Voltage	12/24/48 VDC, 24 VAC[1]
Operating Voltage	9.6 to 60 VDC, 18 to 30 VAC
Input Current	12 to 48 VDC, 0.06 to 0.02 A or 24 VAC, 50-60Hz, 0.07 A (max.)
Power Consumption	0.77 W (max.)
Connector	1 removable 3-contact terminal block(s)

1. 24 VAC only for models with hardware version 3.0.0 or later

Overload Current Protection	Supported
Reverse Polarity Protection	Supported

Physical Characteristics

Housing	Plastic
IP Rating	IP30
Dimensions	24.9 x 100 x 74 mm (0.98 x 3.94 x 2.91 in)
Weight	111 g (0.24 lb)
Installation	DIN-rail mounting

Environmental Limits

Operating Temperature	-10 to 60°C (14 to 140°F)
Storage Temperature (package included)	-40 to 85°C (-40 to 185°F)
Ambient Relative Humidity	5 to 95% (non-condensing)

Standards and Certifications

Safety	UL 61010-2-201 EN 62368-1
EMC	EN 55032/35
EMI	FCC Part 15B Class A
EMS	IEC 61000-4-2 ESD: Contact: 4 kV; Air: 8 kV IEC 61000-4-3 RS: 80 MHz to 1 GHz: 10 V/m IEC 61000-4-4 EFT: Power: 2 kV; Signal: 1 kV IEC 61000-4-5 Surge: Power: 2 kV; Signal: 1 kV IEC 61000-4-6 CS: 10 V IEC 61000-4-8 PFMF
Shock	IEC 60068-2-27
Vibration	IEC 60068-2-6
Freefall	IEC 60068-2-32

MTBF

Time	18,921,624 hrs
Standards	Telcordia (Bellcore), GB

Warranty

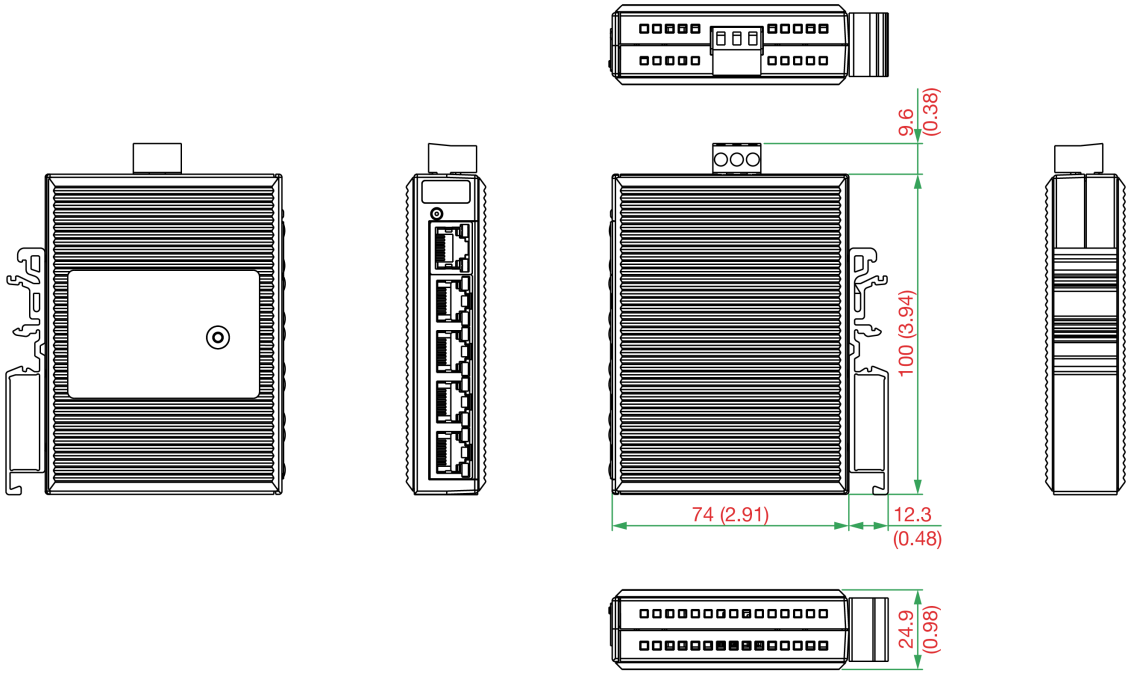
Warranty Period	5 years
Details	See www.moxa.com/warranty

Package Contents

Device	1 x EDS-205 Series switch
Documentation	1 x quick installation guide 1 x warranty card

Dimensions

Unit: mm (inch)



Ordering Information

Model Name	10/100BaseT(X) Ports RJ45 Connector	Housing Material	Operating Voltage	Operating Temp.
EDS-205	5	Plastic	12-48 VDC, 24 VAC	-10 to 60°C

Accessories (sold separately)

Power Supplies

MDR-40-24	DIN-rail 24 VDC power supply with 40W/1.7A, 85 to 264 VAC, or 120 to 370 VDC input, -20 to 70°C operating temperature
MDR-60-24	DIN-rail 24 VDC power supply with 60W/2.5A, 85 to 264 VAC, or 120 to 370 VDC input, -20 to 70°C operating temperature

Rack-Mounting Kits

RK-4U	19-inch rack-mounting kit

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Data Sheet 13

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
2	Mfg: Mersen: Miniature Fuse, Fast Acting, 2A Model Number: MI5HF25V2/100 Tags/Service: FU1403, FU1407 / PLC/RTU Control Panel Specifications: • 2A current rating • 5 mm x 20 mm size • 1500A breaking capacity at 250VAC • 1.17W loss at 100% In • 0.17V maximum voltage drop
4	Mfg: Mersen: Miniature Fuse, Fast Acting, 0.5A Model Number: MI5HF25V0.5/100 Tags/Service: FU1501, FU1520, FU1601, FU1920 / PLC/RTU Control Panel Specifications: • 0.5A current rating • 5 mm x 20 mm size • 1500A breaking capacity at 250VAC • 0.2W loss at 100% In • 0.01V maximum voltage drop

8	Mfg: Mersen: Fast Blow Fuse, 3A Model Number: GGM3 Tags/Service: FU1758, FU1759, FU1760, FU1761, FU1762, FU1763, FU1764, FU1765 / PLC/RTU Control Panel Specifications: • Glass body • 5 mm x 20 mm size • 10kA I.R. at 125VAC • 3A current rating
14	Mfg: Phoenix Contact: Fuse Terminal Block, 120VAC Model Number: 3046100 Tags/Service: FU1758, FU1759, FU1760, FU1761, FU1762, FU1763, FU1764, FU1765, FU1501, FU1520, FU1601, FU1403, FU1407, FU1920 / PLC/RTU Control Panel Specifications: • Category III overvoltage protection • 4kV rated surge voltage • 1.02W maximum power dissipation for nominal condition • For G / 5x20 fuses • 110VAC…250VAC LED voltage range

MINIATURE FUSE, FAST ACTING

TAGS: FU1501, FU1520, FU1601, FU1403, FU1407,FU1920

Fuses 5x20 - Quick acting - 5HF

IEC MINIATURE FUSES

5X20



FEATURES & BENEFITS

- 5x20
- Fast action
- Safe protection system

APPLICATIONS

- Electrical equipment
- Industrial power devices
- Circuit boards

STANDARDS

- IEC 60127-2 sheet 1

Mersen's 5HF miniature fuses cover a range including multiple amperages for 250VAC, for the protection of electrical distribution circuits and other industrial applications. These fuses are available in the size 5X20. 5HF miniature fuses are fast-acting fuses that protect your electrical equipment from the slightest high current. All our technology and processes are designed to ensure the best technical performance.

TECHNICAL DATA OVERVIEW

Voltage AC	250 VAC
Ratings (A)	0,1 A to 20 A
Speed/Characteristic	Quick acting
I.R. AC (IEC/UL)	1500 A
Body Material	Ceramic
Construction	50 mA - 500 mA : without extinguishing agent 630 mA - 16 A : with extinguishing agent
Contact caps	Brass, nickel plated

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PRODUCT RANGE

5X20 Fast Acting 5HF - Set of 100

Catalog number	Item number	Size	Rated voltage	Rated current I_n	Pre-arcing I^2t	Breaking capacity @ Nominal current	Watt Loss @ 100% In	Max. voltage drop
MI5HF25V0.1/100	G095548Q	5x20	250 V	0,1 A [1]	0.0035 A²s	1500 A	0.44 W	2.29 V
MI5HF25V0.125/100	H095549Q	5x20	250 V	0,125 A [1]	0.0059 A²s	1500 A	0.57 W	1.79 V
MI5HF25V0.16/100	J095550Q	5x20	250 V	0,16 A [1]	0.0055 A²s	1500 A	0.55 W	1.7 V
MI5HF25V0.2/100	Q095556Q	5x20	250 V	0,2 A [1]	0.016 A²s	1500 A	0.62 W	1.5 V
MI5HF25V0.25/100	R095557Q	5x20	250 V	0,25 A [1]	0.046 A²s	1500 A	0.73 W	1.28 V
MI5HF25V0.315/100	S095558Q	5x20	250 V	0,315 A [1]	0.072 A²s	1500 A	0.83 W	1.11 V
MI5HF25V0.4/100	T095559Q	5x20	250 V	0,4 A [1]	0.25 A²s	1500 A	0.88 W	1.13 V
MI5HF25V0.5/100	V095560Q	5x20	250 V	0,5 A [1]	0.17 A²s	1500 A	0.2 W	0.01 V
MI5HF25V0.63/100	W095561Q	5x20	250 V	0,63 A [1]	0.11 A²s	1500 A	1.65 W	1.04 V
MI5HF25V0.8/100	X095562Q	5x20	250 V	0,8 A [1]	0.18 A²s	1500 A	1.52 W	0.76 V
MI5HF25V1/100	Y095563Q	5x20	250 V	1 A [1]	0.17 A²s	1500 A	0.99 W	0.3 V
MI5HF25V1.25/100	C095567Q	5x20	250 V	1,25 A [1]	0.58 A²s	1500 A	0.72 W	0.2 V
MI5HF25V1.6/100	D095568Q	5x20	250 V	1,6 A	0.89 A²s	1500 A	1.19 W	0.23 V
MI5HF25V2/100	E095569Q	5x20	250 V	2 A	1.91 A²s	1500 A	1.17 W	0.17 V
MI5HF25V2.5/100	F095570Q	5x20	250 V	2,5 A	2.61 A²s	1500 A	1.55 W	0.22 V
MI5HF25V3.15/100	G095571Q	5x20	250 V	3,15 A	2.68 A²s	1500 A	1.89 W	0.19 V
MI5HF25V4/100	H095572Q	5x20	250 V	4 A	12.9 A²s	1500 A	1.94 W	0.14 V
MI5HF25V5/100	J095573Q	5x20	250 V	5 A	31.2 A²s	1500 A	2.16 W	0.13 V
MI5HF25V6.3/100	K095574Q	5x20	250 V	6,3 A	79.3 A²s	1500 A	1.81 W	0.1 V
MI5HF25V8/100	J085407Q	5x20	250 V	8 A [2]	86.7 A²s	1500 A	3.08 W	0.13 V
MI5HF25V10/100	K085408Q	5x20	250 V	10 A [2]	204.4 A²s	1500 A	3.06 W	0.1 V
MI5HF25V12.5	G210341	5x20	250 V	12,5 A [2]	433.3 A²s	1500 A	3.22 W	0.87 V
MI5HF25V16	H210342	5x20	250 V	16 A [2]	930.6 A²s	1500 A	2.9 W	0.68 V
MI5HF25V20	J210343	5x20	250 V	20 A [2]	-	1500 A	-	-

[1] No Semko Approval
[2] Set of 50

5X20 Fast Acting 5HF - Set of 1000

Catalog number	Item number	Size	Rated voltage	Rated current I_n	Pre-arcing I^2t	Breaking capacity @ Nominal current	Watt Loss @ 100% In	Max. voltage drop
MI5HF25V0.16/1000	J095550T	5x20	250 V	0,16 A	0.0055 A²s	1500 A	0.55 W	1.7 V
MI5HF25V0.25/1000	R095557T	5x20	250 V	0,25 A	0.046 A²s	1500 A	0.732 W	1.28 V
MI5HF25V0.315/1000	R095557T	5x20	250 V	0,315 A	0.072 A²s	1500 A	0.829 W	1.11 V

5X20 Fast Action 5HF - Assortment Box

Catalog number	Item number	Size	Rated voltage	Rated current I_n
MIBOX5HF	B229863	5x20	250 V	0,1 to 6,3 A

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PRODUCT RANGE

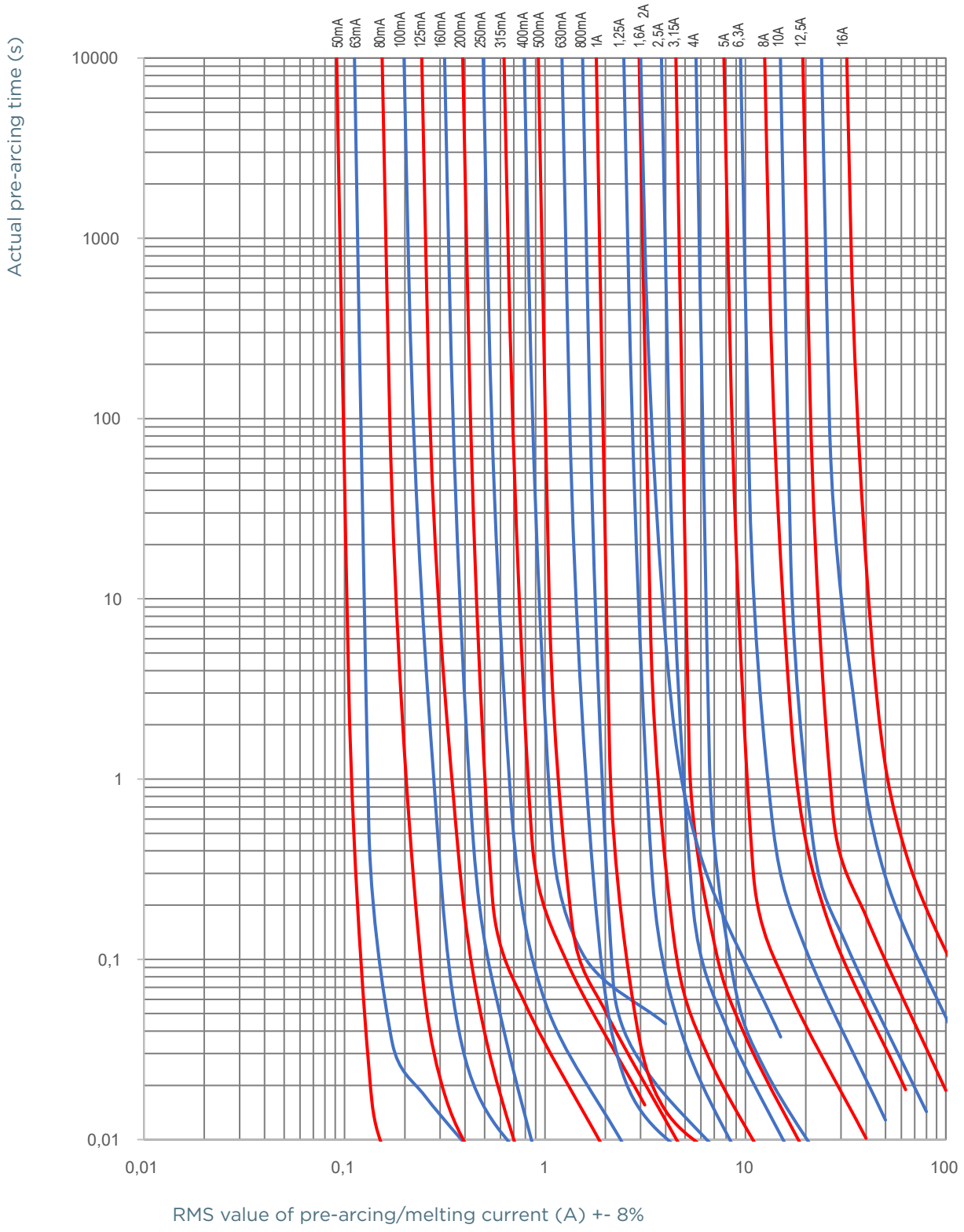
Composition of the box

Catalog number	Item number	Size	Rated voltage	Rated current I_n	Pre-arcing I^2t	Breaking capacity @ Nominal current	Watt Loss @ 100% In	Max. voltage drop
MI5HF25V0.1	G095548	5x20	250 V	0,1 A	0.0035 A²s	1500 A	0.44 W	2.29 V
MI5HF25V0.125	H095549	5x20	250 V	0,125 A	0.0059 A²s	1500 A	0.57 W	1.79 V
MI5HF25V0.16	J095550	5x20	250 V	0,16 A	0.0055 A²s	1500 A	0.55 W	1.7 V
MI5HF25V0.2	Q095556	5x20	250 V	0,2 A	0.016 A²s	1500 A	0.62 W	1.5 V
MI5HF25V0.25	R095557	5x20	250 V	0,25 A	0.046 A²s	1500 A	0.73 W	1.28 V
MI5HF25V0.315	S095558	5x20	250 V	0,315 A	0.072 A²s	1500 A	0.83 W	1.11 V
MI5HF25V0.4	T095559	5x20	250 V	0,4 A	0.25 A²s	1500 A	0.88 W	1.13 V
MI5HF25V0.5	V095560	5x20	250 V	0,5 A	0.17 A²s	1500 A	0.2 W	0.01 V
MI5HF25V0.63	W095561	5x20	250 V	0,63 A	0.11 A²s	1500 A	1.65 W	1.04 V
MI5HF25V0.8	X095562	5x20	250 V	0,8 A	0.18 A²s	1500 A	1.52 W	0.76 V
MI5HF25V1	Y095563	5x20	250 V	1 A	0.17 A²s	1500 A	0.99 W	0.3 V
MI5HF25V1.25	C095567	5x20	250 V	1,25 A	0.58 A²s	1500 A	0.72 W	0.2 V
MI5HF25V1.6	D095568	5x20	250 V	1,6 A	0.89 A²s	1500 A	1.13 W	0.26 V
MI5HF25V2	E095569	5x20	250 V	2 A	1.91 A²s	1500 A	1.17 W	0.17 V
MI5HF25V2.5	F095570	5x20	250 V	2,5 A	2.61 A²s	1500 A	1.55 W	0.22 V
MI5HF25V3.15	G095571	5x20	250 V	3,15 A	2.68 A²s	1500 A	1.89 W	0.19 V
MI5HF25V4	H095572	5x20	250 V	4 A	12.9 A²s	1500 A	1.94 W	0.14 V
MI5HF25V6.3	K095574	5x20	250 V	6,3 A	79.3 A²s	1500 A	1.81 W	0.1 V

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Fuses 5x20 - Quick acting - 5HF

TIME CURRENT CHARACTERISTIC CURVES



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TECHNICAL CHARACTERISTICS

In - t characteristics

Rated current factor		Melting time :		
		50 mA - 3,15 A	4A - 10A	12,5 A - 16 A
1,5 - In	tmin	60 min	60 min	60 min
	tmax	-	-	-
2,1 - In	tmin	0	0	0
	tmax	30 min	30 min	30 min
2,75 - In	tmin	10 ms	10 ms	40 ms
	tmax	2 s	3 s	20 s
4 - In	tmin	3 ms	3 ms	10 ms
	tmax	300 ms	300 ms	1 s
10 - In	tmin	0	0	0
	tmax	20 ms	20 ms	30 ms

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FAST BLOW FUSE, 3A

TAGS: FU1758-65

GGC/GGC-V

- Glass body
- Fast-acting
- 1/4" x 1-1/4"
- 1-1/2"Axial leads optional



1/10A through 10A, 250VAC, UL Listed and CSA Certified

12A and 15A, 125VAC, UL Listed and CSA Certified

20A through 30A, 32VAC

GGM/GGM-V

- Glass body
- Fast-acting
- 5 mm x 20 mm
- 1-1/2" Axial leads optional



1/16A through 3A, 250VAC, UL Listed and CSA Certified

4A through 8A, 125VAC, UL Listed and CSA Certified

10A and 15A, 250VAC, UL Recognized

STANDARD FUSE AMPERE RATINGS

Catalog Number	Axial Lead Cat. No	Ampere Rating	Volts	I.R.
GGC1/10	GGC-V1/10	1/10A	250V	2
GGC1/8	GGC-V1/8	1/8A	250V	2
GGC15/100	GGC-V15/100	15/100A	250V	2
GGC175/1000	GGC-V175/1000	175/1000A	250V	2
GGC3/16	GGC-V3/16	3/16A	250V	2
GGC2/10	GGC-V2/10	2/10A	250V	2
GGC1/4	GGC-V1/4	1/4A	250V	2
GGC3/10	N/A	3/10A	250V	2
GGC3/8	GGC-V3/8	3/8A	250V	2
GGC4/10	GGC-V4/10	4/10A	250V	2
GGC1/2	GGC-V1/2	1/2A	250V	2
GGC6/10	GGC-V6/10	6/10A	250V	2
GGC3/4	GGC-V3/4	3/4A	250V	2
GGC8/10	GGC-V8/10	8/10A	250V	2
GGC1	GGC-V1	1A	250V	2
GGC1-1/4	GGC-V1-1/4	1-1/4A	250V	2
GGC1-1/2	GGC-V1-1/2	1-1/2A	250V	2
GGC1-3/4	N/A	1-3/4A	250V	2
GGC1-6/10	GGC-V1-6/10	1-6/10A	250V	2
GGC2	GGC-V2	2A	250V	2
GGC2-1/4	GGC-V2-1/4	2-1/4A	250V	2
GGC2-1/2	GGC-V2-1/2	2-1/2A	250V	2
GGC3	GGC-V3	3A	250V	2
GGC3-2/10	GGC-V3-2/10	3-2/10A	250V	3
GGC4	GGC-V4	4A	250V	2
GGC5	GGC-V5	5A	250V	2
GGC6	GGC-V6	6A	250V	2
GGC6-3/10	GGC-V6-3/10	6-3/10A	250V	2
GGC7	GGC-V7	7A	250V	2
GGC7-1/2	N/A	7-1/2A	250V	2
GGC8	GGC-V8	8A	250V	2
GGC10	GGC-V10	10A	250V	2
GGC12	GGC-V12	12A	125V	4
GGC15	GGC-V15	15A	125V	4
GGC20	GGC-V20	20A	32V	5
GGC25	GGC-V25	25A	32V	5
GGC30	GGC-V30	30A	32V	5

2. 250VAC @ 200A I.R./125VAC @ 10kA I.R.
3. 250VAC @ 10kA I.R.
4. 125VAC @ 10kA I.R.
5. 32VAC @ 1000A I.R.

STANDARD FUSE AMPERE RATINGS

Catalog Number	Axial Lead Cat. No	Ampere Rating	Volts	I.R.
~~GGM1/16~~	~~GGM-V1/16~~	~~1/16A~~	~~250V~~	~~2~~
~~GGM1/10~~	~~GGM-V1/10~~	~~1/10A~~	~~250V~~	~~3~~
~~GGM1/8~~	~~GGM-V1/8~~	~~1/8A~~	~~250V~~	~~3~~
~~GGM2/10~~	~~GGM-V2/10~~	~~2/10A~~	~~250V~~	~~3~~
~~GGM1/4~~	~~GGM-V1/4~~	~~1/4A~~	~~250V~~	~~3~~
~~GGM3/10~~	~~GGM-V3/10~~	~~3/10A~~	~~250V~~	~~3~~
~~GGM4/10~~	~~GGM-V4/10~~	~~4/10A~~	~~250V~~	~~1~~
~~GGM1/2~~	~~GGM-V1/2~~	~~1/2A~~	~~250V~~	~~3~~
~~GGM6/10~~	~~GGM-V6/10~~	~~6/10A~~	~~250V~~	~~1~~
~~GGM7/10~~	~~GGM-V7/10~~	~~7/10A~~	~~250V~~	~~3~~
~~GGM3/4~~	~~GGM-V3/4~~	~~3/4A~~	~~250V~~	~~3~~
~~GGM8/10~~	~~GGM-V8/10~~	~~8/10A~~	~~250V~~	~~1~~
~~GGM1~~	~~GGM-V1~~	~~1A~~	~~250V~~	~~3~~
~~GGM1-2/10~~	~~N/A~~	~~1-2/10A~~	~~250V~~	~~2~~
~~GGM1-1/4~~	~~GGM-V1-1/4~~	~~1-1/4A~~	~~250V~~	~~3~~
~~GGM1-1/2~~	~~GGM-V1-1/2~~	~~1-1/2A~~	~~250V~~	~~3~~
~~GGM2~~	~~GGM-V2~~	~~2A~~	~~250V~~	~~3~~
~~GGM2-1/2~~	~~GGM-V2-1/2~~	~~2-1/2A~~	~~250V~~	~~3~~
GGM3	GGM-V3	3A	250V	3
~~GGM4~~	~~GGM-V4~~	~~4A~~	~~125V~~	~~4~~
~~GGM5~~	~~GGM-V5~~	~~5A~~	~~125V~~	~~4~~
~~GGM6~~	~~GGM-V6~~	~~6A~~	~~125V~~	~~4~~
~~GGM7~~	~~GGM-V7~~	~~7A~~	~~125V~~	~~4~~
~~GGM8~~	~~GGM-V8~~	~~8A~~	~~125V~~	~~4~~
~~GGM15~~	~~GGM-V15~~	~~15A~~	~~250V~~	~~5~~

1. 250VAC @ 35A I.R./125VAC @ 10kA I.R.
2. 250VAC @ 100A I.R./125VAC @ 10kA I.R.
3. 250VAC @ 200A I.R./125VAC @ 10kA I.R.
4. 125VAC @ 10kA I.R.
5. 250VAC @ 100A I.R.

FUSE TERMINAL BLOCK, 120VAC

UT 4-HESILA 250 (5X20) - Fuse modular terminal block

TAGS: FU1403, FU1407, FU1501, FU1520, FU1601, FU1758-1765, FU1920

3046100

https://www.phoenixcontact.com/us/products/3046100

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Fuse modular terminal block, fuse type: Glass / ceramics / ..., fuse type: G / 5 x 20, nom. voltage: 250 V, nominal current: 6.3 A, connection method: Screw connection, Rated cross section: 4 mm^2, cross section: 0.14 mm^2 - 6 mm^2, mounting type: NS 35/7,5, NS 35/15, color: black

Your advantages

- Globally recognized: Internationally proven screw connection
- Maintenance-free and vibration-resistant thanks to the patented Reakdyn principle
- Easy integration and replacement of fuses with the lever element
- Easy checking of the fuses with optical signal unit
- Long-term stable connections with the use of high-quality materials
- Long-term stable connections with the use of high-quality materials
- Maximum efficiency in the smallest space - thanks to integrated level bridging, the connections are connected across levels

Technical data

Notes

Order information:	Fuse-link not supplied as standard
General	The current is determined by the fuse used, the voltage by the selected LED. If the fuse is faulty, the downstream circuit will not be disconnected.

Product properties

Product family	UT
Area of application	Railway industry
	Machine building
	Plant engineering
Number of connections	2
Number of rows	1
Potentials	1

Insulation characteristics

Overvoltage category	III
Degree of pollution	3

Electrical properties

Fuse type	Glass / ceramics / ...
Rated surge voltage	4 kV
Maximum power dissipation for nominal condition	1.02 W
Fuse	G / 5 x 20
LED voltage range	110 V AC/DC ... 250 V AC/DC
LED current range	0.41 mA ... 0.96 mA
Maximum power dissipation	max. 1.6 W (with single arrangement of the fuse terminal block in the event of overload)
	~~max. 1.6 W (With interconnected arrangement of several fuse terminal blocks in the event of overload)~~
	max. 4 W (with single arrangement of the fuse terminal block in the event of a short-circuit)
	~~max. 2.5 W (With interconnected arrangement of several fuse terminal blocks in the event of a short-circuit)~~

Input data

LED voltage range	110 V AC/DC ... 250 V AC/DC

Connection data

Number of connections per level	2
Nominal cross section	4 mm²
Connection method	Screw connection
Screw thread	M3

Tightening torque	0.6 ... 0.8 Nm
Stripping length	9 mm
Internal cylindrical gage	A4
Connection in acc. with standard	IEC 60947-7-3
Conductor cross-section rigid	0.14 mm² ... 6 mm²
Cross section AWG	26 ... 10 (converted acc. to IEC)
Conductor cross-section flexible	0.14 mm² ... 6 mm²
Conductor cross-section, flexible [AWG]	26 ... 10 (converted acc. to IEC)
Conductor cross-section flexible ultrasound-compressed	0.34 mm² ... 6 mm²
Conductor cross-section, flexible [AWG] ultrasound-compressed	22 ... 10 (converted acc. to IEC)
Conductor cross-section flexible (ferrule without plastic sleeve)	0.25 mm² ... 4 mm²
Flexible conductor cross-section (ferrule with plastic sleeve)	0.14 mm² ... 4 mm²
2 conductors with same cross section, rigid	0.14 mm² ... 1.5 mm²
2 conductors with same cross section, flexible	0.14 mm² ... 1.5 mm²
2 conductors with same cross section, flexible, with ferrule without plastic sleeve	0.14 mm² ... 1.5 mm²
2 conductors with the same cross section, flexible, with TWIN ferrule with plastic sleeve	0.5 mm² ... 1.5 mm²
Nominal cross section	4 mm²
Nominal current	6.3 A
Maximum load current	6.3 A (the current is determined by the fuse used)
Nominal voltage	250 V

Dimensions

Width	6.2 mm
Height	57.8 mm
Depth on NS 35/7,5	75.6 mm
Depth on NS 35/15	83.1 mm

Material specifications

Color	black (RAL 9005)
Flammability rating according to UL 94	V0
Insulating material group	I
Insulating material	PA
Static insulating material application in cold	-60 °C
Temperature index of insulation material (DIN EN 60216-1 (VDE 0304-21))	125 °C
Relative insulation material temperature index (Elec., UL 746 B)	130 °C
Fire protection for rail vehicles (DIN EN 45545-2) R22	HL 1 - HL 3
Fire protection for rail vehicles (DIN EN 45545-2) R23	HL 1 - HL 3
Fire protection for rail vehicles (DIN EN 45545-2) R24	HL 1 - HL 3
Fire protection for rail vehicles (DIN EN 45545-2) R26	HL 1 - HL 3
Calorimetric heat release NFPA 130 (ASTM E 1354)	27,5 MJ/kg
Surface flammability NFPA 130 (ASTM E 162)	passed
Specific optical density of smoke NFPA 130 (ASTM E 662)	passed

Smoke gas toxicity NFPA 130 (SMP 800C)	passed

Mechanical properties

Mechanical data

Open side panel	No

Environmental and real-life conditions

Ambient conditions

Ambient temperature (operation)	-60 °C ... 110 °C (Operating temperature range incl. self-heating; for max. short-term operating temperature, see RTI Elec.)
Ambient temperature (storage/transport)	-25 °C ... 60 °C (for a short time, not exceeding 24 h, -60 °C to +70 °C)
Ambient temperature (assembly)	-5 °C ... 70 °C
Ambient temperature (actuation)	-5 °C ... 70 °C
Permissible humidity (operation)	20 % ... 90 %
Permissible humidity (storage/transport)	30 % ... 70 %

Standards and regulations

Connection in acc. with standard	IEC 60947-7-3

Mounting

Mounting type	NS 35/7,5
	NS 35/15

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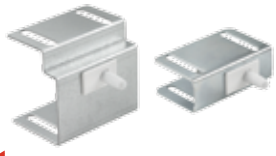
Data Sheet 14

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
2	Mfg: Hoffman: Remote Door Switch Model Number: ALFSWD Tags/Service: LS1205 / PLC/RTU Control Panel Specifications: • NEMA 4X rated door switch assembly • Mounts on enclosure frame and includes mounted hardware • 14ga. steel mounting plate with a plated finish

REMOTE DOOR SWITCH

TAGS: LS1205

REMOTE DOOR SWITCHES



INDUSTRY STANDARDS

UL508A Recognized, Type 3R, 4, 4X, 12; File E61997
cUL Recognized per CSA C22.2 No. 14-13; Type 3R, 4, 4X, 12; File E61997

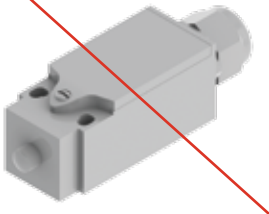
FEATURES

- The remote door switch can activate a light or other electrical component when the enclosure door is opened
- Mounts on enclosure frame and includes mounting hardware
- Mounting plate is 14 gauge steel with a plated finish
- Can be hard-wired to the PaneLite LED or Fluorescent light or connected via the PaneLite Door Switch Cable

BULLETIN: A80LT, P20

Catalog Number	Description
ALFSWD	Door switch assembly (order connection cable separately)
~~PLFSWD~~	~~Door switch assembly for ProLine and ProLine G2 (order connection cable separately)~~

TOUCH-SAFE UL LIGHT SWITCH



APPLICATION

The Touch-Safe light switch is designed to be used with Hoffman light kits (AC and DC). It provides a UL listed touch-safe switch that can be used on many enclosure types and includes hardware for most applications. The light switch maintains enclosure overall rating up to UL 508A Type 4X or 12.

FEATURES

- UL listed for a touch-safe wire connection
- Easily mounted to various enclosure types; common bracket and hardware included for many enclosure types (NEMA, Concept/ Fusion G7, ProLine, Free-Stand Type 12)
- Cable PG compression hub
- Wide operating temperature range: -49°F to +158°F (-45°C to 70°C)
- Rugged die-cast anodized zinc construction
- Connection 3 cage clamps for solid and stranded wire AWG 20-14 (0.5-2.5mm²)
- Protection class I (grounded)

BULLETIN: A80LT

Catalog Number	Max. Cable Dia.	Max. Voltage
LDSWITCH	.375 in.	250 AC / 48 DC

MOUNTING BRACKET KIT FOR LIGHT PACKAGE



Kit simplifies mounting light package in Hoffman PROLINE disconnect enclosures. Includes brackets, all mounting hardware and complete instructions.

BULLETIN: A80LT

Catalog Number	Description
PDLFBRKT	Mounting Bracket Kit

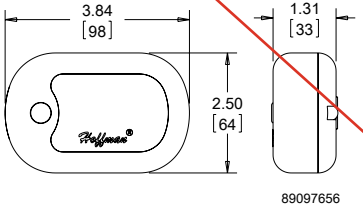
LED PUCK LIGHT



The LED Puck Light is ideal for remote and darkened applications. This versatile light provides mounting flexibility; it can be magnetically attached to flat steel surfaces or can be hung with a swivel hook. This small form factor, light-weight LED light provides superior lighting performance with minimal power consumption. It can be used as a three-LED flashlight or as a Chip on Board (COB) work light with operating temperature of 40 F to 120 F (4 C to 48 C). An on/ off switch is incorporated in the light and three AAA batteries are included.

BULLETIN: A80LT

Catalog Number	AxBxC in./mm	Product Weight
LEDPUCK	2.25 x 3.75 x 1.38 57 x 95 x 35	0.22 lbs.



Data Sheet 15

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
1	Mfg: Modicon: Power Supply Module Model Number: BMXCPS3500H Tags/Service: PWS1301 / PLC/RTU Control Panel Specifications: • Modicon X80 power supply module • 100...240VAC primary voltage • 23.4W, 24VDC secondary power to I/O module and processor • 0.12kVA apparent power • 30A inrush current at 120VAC • 1.3A current at 24VDC secondary voltage • 8.5W maximum power dissipation • Reset push-button cold restart • 1 green rack voltage OK LED • 1 green sensor voltage LED

Product data sheet

POWER SUPPLY MODULE

Specifications

TAGS: PWS1301



power supply module, Modicon X80, 100 to 240V AC, 36W, severe environment

BMXCPS3500H

Product availability: Stock - Normally stocked in distribution facility

Main

Range of Product	Modicon X80
Product or Component Type	Power supply module
backplane compatibility	Not compatible with BMEXBP..02
Product Specific Application	For severe environments
Primary Voltage	100...240 V
Supply circuit type	AC
Secondary power	~~15 W 3.3 V DC 32…140 °F (0…60 °C) I/O module logic power supply~~ ~~31.2 W 24 V DC 32…140 °F (0…60 °C) I/O module power supply and processor~~ ~~11.3 W 3.3 V DC -13…158 °F (-25…70 °C) I/O module logic power supply~~ 23.4 W 24 V DC -13…158 °F (-25…70 °C) I/O module power supply and processor ~~21.6 W 24 V DC 32…140 °F (0…60 °C) sensor power supply~~ ~~16.2 W 24 V DC -13…158 °F (-25…70 °C) sensor power supply~~

Complementary

Primary voltage limit	85...264 V
Network Frequency	50/60 Hz
Network frequency limits	47…63 Hz
Apparent power	0.12 kVA
Input current	~~0.52 A 240 V~~ 1.04 A 115 V
Inrush current	30 A 120 V ~~60 A 240 V~~
I²t on activation	1 A².s 120 V ~~3 A².s 240 V~~
It on activation	0.05 A.s 120 V ~~0.07 A.s 240 V~~
MTBF reliability	4300000 H
Protection type	Internal fuse not accessible primary circuit Overload protection secondary circuit Overvoltage protection secondary circuit Short-circuit protection secondary circuit
Current at secondary voltage	~~0.9 A 24 V DC sensor power supply~~ 1.3 A 24 V DC I/O module power supply and processor ~~4.5 A 3.3 V DC I/O module logic power supply~~
Maximum power dissipation in W	8.5 W
Status LED	1 LED (Green) rack voltage OK 1 LED (Green) sensor voltage
Control Type	RESET push-button cold restart

Price is “List Price” and may be subject to a trade discount – check with your local distributor or retailer for actual price.

Disclaimer: This documentation is not intended as a substitute for and is not to be used for determining suitability or reliability of these products for specific user applications

Electrical connection	1 connector 2 alarm relay 1 connector 5 line supply, protective earth, 24 V DC input sensor
Insulation resistance	>= 100 MOhm primary/ground >= 100 MOhm primary/secondary
Net Weight	0.79 lb(US) (0.36 kg)

Environment

Immunity to microbreaks	1 ms
Dielectric strength	~~1500 V primary/secondary I/O module logic power supply~~ 1500 V primary/secondary I/O module power supply and processor ~~2300 V primary/secondary sensor power supply~~ ~~1500 V primary/ground~~ ~~500 V 24 V sensor output/ground~~
Vibration resistance	3 gn
Shock resistance	30 gn
IP Degree of Protection	IP20
Directives	2014/35/EU - low voltage directive 2014/30/EU - electromagnetic compatibility
Product certifications	CE UL CSA RCM EAC Merchant Navy
Standards	IEC 61131-2 IEC 61010-2-201 UL 61010-2-201 CSA C22.2 No 61010-2-201 IACS E10 EN/IEC 61000-6-5, interface type 1 and type 2 EN/IEC 61850-3, location G
Environmental characteristic	Gas resistant class Gx ISA S71.04 Gas resistant class 3C4 IEC 60721-3-3 Dust resistant class 3S4 IEC 60721-3-3 Sand resistant class 3S4 IEC 60721-3-3 Salt resistant level 2 IEC 68252 Mold growth resistant class 3B2 IEC 60721-3-3 Fungal spore resistant class 3B2 IEC 60721-3-3 Hazardous location class I division 2
Ambient Air Temperature for Storage	-40…185 °F (-40…85 °C)
Ambient Air Temperature for Operation	-13…158 °F (-25…70 °C)
Relative humidity	5…95 % 131 °F (55 °C) without condensation
Protective treatment	Conformal coating
Operating altitude	0...6561.68 ft (0...2000 m) 2000...5000 m with derating factor

Ordering and shipping details

Category	US1PC3418160
Discount Schedule	PC34
GTIN	3595864029894
Returnability	Yes
Country of origin	ID

Packing Units

Unit Type of Package 1	PCE
Nbr. of units in pkg.	1
Package 1 Height	5.118 in (13.000 cm)
Package 1 Width	6.102 in (15.500 cm)
Package 1 Length	6.299 in (16.000 cm)
Package weight(Lbs)	19.048 oz (540.000 g)
Unit Type of Package 2	S04
Number of Units in Package 2	12
Package 2 Height	11.811 in (30.000 cm)
Package 2 Width	15.748 in (40.000 cm)
Package 2 Length	23.622 in (60.000 cm)
Package 2 Weight	16.700 lb(US) (7.575 kg)

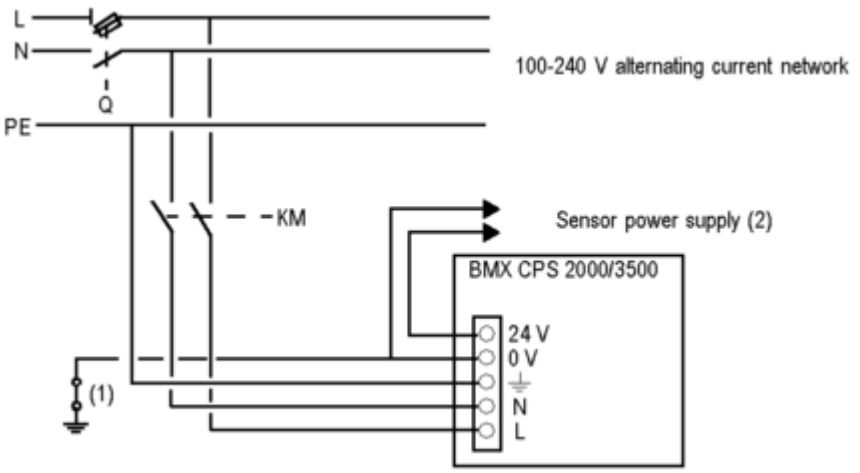
Contractual warranty

Warranty (in months)	18

Connections and Schema

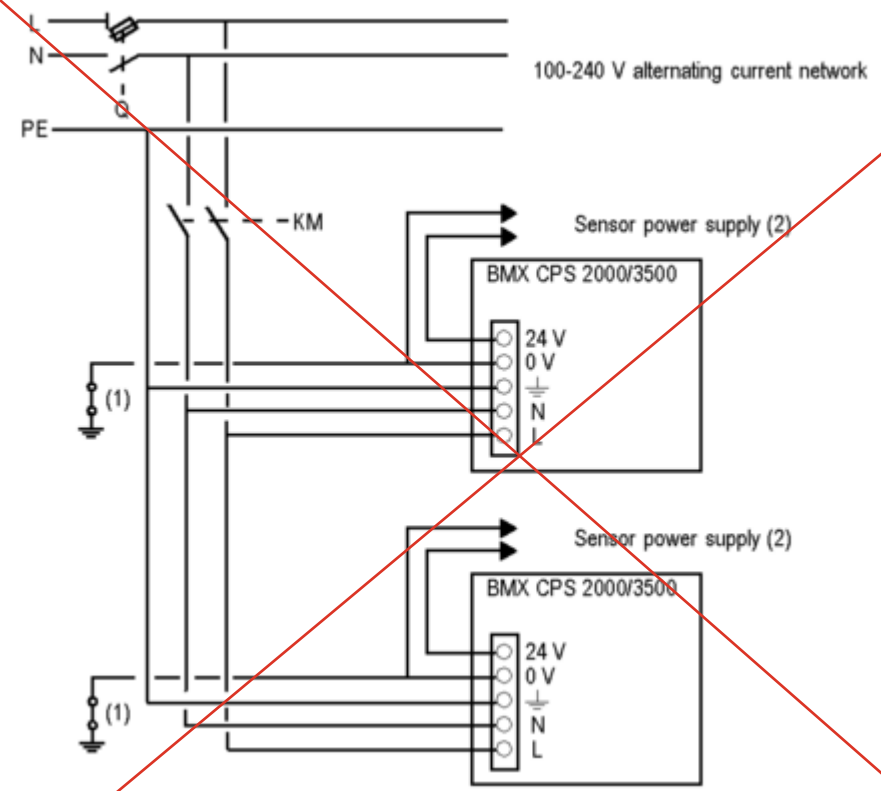
Connection of Alternating Current Power Supply Modules

Connection of a PLC Station Constituted of a Single Rack



Q General isolator
KM Line contactor or circuit breaker
(1) Insulation connector bar for locating grounding errors
(2) Available current of 0.45 A for the BMXCPS2000 module or 0.9 A for the BMXCPS3500 module

Connection of a PLC Station Constituted of Several Racks



Q General isolator
KM Line contactor or circuit breaker
(1) Insulation connector bar for locating grounding errors
(2) Available current of 0.45 A for the BMXCPS2000 module or 0.9 A for the BMXCPS3500 module

Data Sheet 16

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
1	Mfg: Schneider Electric: Operator Interface Terminal Model Number: HMIDT752 Tags/Service: OIT / PLC/RTU Control Panel Specifications: • Harmony GTU product range • Flat, touchscreen display • 24VDC rated supply voltage • 17W power consumption • 30A inrush current • USB 2.0 type A and USB 2.0 type mini B integrated connection types • 2048x2048 touchscreen resolution • 1366x768 pixel resolution • 15.6 inch display size • 4:3 aspect ration • 50,000 hour backlight lifespan
1	Mfg: Schneider Electric: Display Base Unit Model Number: HMIG2U Tags/Service: OIT / PLC/RTU Control Panel Specifications:

	• Harmony GTU product range • 24VDC rated supply voltage • Ethernet TCP/IP 2 RJ45, COM1 serial link RS232C, COM2 serial link RS485 RJ45, 2 USB 2.0 type A, USB 2.0 type mini B integrated connection types • Modbus, Modbus TC/IP, Uni-Telway communication port protocal • ARM Cortex-A9 processor • 1GB built-in flash memory • 1 SD card slot • 256 MB RAM internal, 512KB NVRAM backup memory • Built-in realtime clock
1	Mfg: Schneider Electric: HMI Software Model Number: VJDBTPRO3P Tags/Service: OIT / PLC/RTU Control Panel Specifications: • Vijeo Designer configuration software • 3 station group license • Windows 10 professional 32/64 bit or Windows 11 OS requirement

Product data sheet OPERATOR INTERFACE TERMINAL

Specifications

TAGS: OIT



flat screen, Harmony GTU, 15inch wide display, 1366 x 768pixels FWXGA

HMIDT752

Product availability: Stock - Normally stocked in distribution facility

Main

Range of Product	Harmony GTU
Terminal type	Touchscreen display
Product or Component Type	Flat screen
[Us] rated supply voltage	12...24 V DC power supply
Supply voltage limits	10.8…28.8 V
Power Consumption in W	17 W
Inrush current	30 A

Complementary

Integrated connection type	USB 2.0 type A USB 2.0 type mini B
Touchscreen resolution	2048 x 2048
Pixel resolution	1366 x 768
Display size	15.6 inch
Aspect ratio	4:3
Display type	Colour TFT LCD
Display Resolution	1366 x 768 pixels FWXGA
Display colour	16 million colours
Backlight lifespan	50000 hours
Touch panel	Multi-touch projected capactive technology (PCT) 50000000 cycles
Function Available	Brightness sensor
Type of cooling	Natural convection
Type of Installation	Indoor installation
Cut-out dimensions	396 (+ 1/- 0) x 277 (+ 1/- 0) mm
Width	16.3 in (414 mm)
Height	11.6 in (295 mm)
Depth	2.7 in (69 mm)
Net Weight	11.02 lb(US) (5 kg)

Environment

IP degree of protection	Front panel IP66 Front panel IP67
NEMA degree of protection	NEMA 4X front panel

Price is “List Price” and may be subject to a trade discount – check with your local distributor or retailer for actual price.

Disclaimer: This documentation is not intended as a substitute for and is not to be used for determining suitability or reliability of these products for specific user applications

Standards	UL 508 ANSI/ISA 12-12-01 Class I division 2 CSA C22.2 No 213 CSA C22.2 No 142 IEC 61132-2
Product certifications	KC CE RCM CULus UKCA UKEX
Ambient air temperature for operation	32…131 °F (0…55 °C) IEC 61131-2
Ambient air temperature for storage	-4…140 °F (-20…60 °C)
Relative Humidity	10…90 % non-condensing
Operating altitude	6561.68 ft (2000 m)
Vibration resistance	3.5 mm constant amplitude 5…9 Hz) IEC 61131-2 9.8 m/s² 9…150 Hz) IEC 61131-2
Shock resistance	147 m/s²IEC 61131-2
Electromagnetic compatibility	Electrostatic discharge immunity test level 3 conforming to IEC 61000-4-2

Ordering and shipping details

Category	US10MC222567
Discount Schedule	0MC2
GTIN	3606481203465
Returnability	Yes
Country of origin	ID

Packing Units

Unit Type of Package 1	PCE
Nbr. of units in pkg.	1
Package 1 Height	7.480 in (19.000 cm)
Package 1 Width	16.929 in (43.000 cm)
Package 1 Length	20.866 in (53.000 cm)
Package weight(Lbs)	12.412 lb(US) (5.630 kg)
Unit Type of Package 2	P12
Number of Units in Package 2	16
Package 2 Height	53.150 in (135.000 cm)
Package 2 Width	31.496 in (80.000 cm)
Package 2 Length	47.244 in (120.000 cm)
Package 2 Weight	231.486 lb(US) (105.000 kg)

Contractual warranty

Warranty (in months)	18

Product data sheet DISPLAY BASE UNIT

Specifications

TAGS: OIT



base unit, Harmony GTU, standard box, 1GB built-in flash memory

HMIG2U

Product availability: Stock - Normally stocked in distribution facility

Main

Range of Product	Harmony GTU
Product or Component Type	Base unit
Device short name	standard box
Range Compatibility	Harmony GTU ~~Allen-Bradley~~ ~~ABB~~ ~~Mitsubishi~~ ~~Omron~~ ~~Siemens~~
[Us] rated supply voltage	12...24 V DC power supply

Complementary

Integrated connection type	Ethernet TCP/IP 2 RJ45 COM1 serial link SUB-D 9 RS232C 2400...115200 bps COM2 serial link RJ45 RS485 187.5 kbps USB 2.0 port 2 USB type A USB 2.0 port mini B USB
Communication port protocol	Modbus Modbus TCP/IP Uni-Telway
Software package	Vijeo Designer
Processor name	ARM Cortex-A9
Data storage equipment	1 GB built-in flash memory
Port Ethernet	10BASE-T/100BASE-T
Memory description	256 MB RAM internal 2 x 13 mm board) 512 KB NVRAM backup memory
Number of slots	1 slot(s) for SD card
Realtime clock	Built-in
Local signalling	, green LED steady)operating status (RUN) , green LED faded)stand-by , orange LED flashing)software starting up , red LED steady)power ON , green, orange and red LED alternate)module error (ERR) , red/orange LED alternate)boot status , green/red LED alternate)connection status
Type of cooling	Natural convection
Width	7.4 in (188 mm)
Height	5.2 in (131 mm)
Depth	1.8 in (45 mm)
Net Weight	2.0 lb(US) (0.9 kg)

Price is “List Price” and may be subject to a trade discount – check with your local distributor or retailer for actual price.

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Environment

Standards	CSA C22.2 No 142 ANSI/ISA 12-12-01 UL 508 Class I division 2 CSA C22.2 No 213 IEC 61132-2
Product certifications	CE CULus ATEX IEC-Ex RCM GL KC CUL EAC UKCA UKEX
Resistance to fast transients	1 kV IEC 61000-4-4 signal ports) 2 kV IEC 61000-4-4 power ports)
Resistance to electrostatic discharge	6 kV contact discharge IEC 61000-4-2 level 3 8 kV air discharge IEC 61000-4-2 level 3
Ambient air temperature for operation	32…140 °F (0…60 °C)
Ambient air temperature for storage	-4…140 °F (-20…60 °C)
Pollution degree	2 IEC 61131-2 ed.3
Environmental characteristic	Corrosive gas free
Relative Humidity	10…90 % non-condensing
Operating altitude	6561.68 ft (2000 m)
Vibration resistance	3.5 mm constant amplitude 5…9 Hz) X, Y, Z directions for 10 cycles (approx. 100 min) IEC 61131-2 9.8 m/s² 9…150 Hz) X, Y, Z directions for 10 cycles (approx. 100 min) IEC 61131-2
Shock resistance	147 m/s² X, Y, Z directions for 3 times)IEC 61131-2
Electromagnetic compatibility	Electrostatic discharge immunity test level 3 conforming to IEC 61000-4-2

Ordering and shipping details

Category	US10MC222567
Discount Schedule	0MC2
GTIN	3606489747176
Returnability	Yes
Country of origin	ID

Packing Units

Unit Type of Package 1	PCE
Nbr. of units in pkg.	1
Package 1 Height	5.709 in (14.500 cm)
Package 1 Width	8.465 in (21.500 cm)
Package 1 Length	10.433 in (26.500 cm)
Package weight(Lbs)	2.573 lb(US) (1.167 kg)
Unit Type of Package 2	S04
Number of Units in Package 2	4
Package 2 Height	11.811 in (30.000 cm)

Product data sheet

Specifications

HMI SOFTWARE

TAGS: OIT



Vijeo Designer 6.3, HMI configuration software group license

VJDBTPRO3P

Product availability: Non-Stock - Not normally stocked in distribution facility

Main

Range of Product	Vijeo Designer
Product or Component Type	Configuration software
Format	Digital
Software Type	Build time
Type of License	Group (3 stations)
Software Version	V6.3
Operating system requirement	Windows 10 Professional 32/64 bit Windows 11
User Language	French English German Italian Chinese Spanish Portuguese

Complementary

Provided Equipment	Vijeo Designer software User's documentation in electronic format
Communication Port Protocol	MELSEC Modbus RTU master Modbus TCP master Sysmacway Ethernet/IP Explicit Modbus 32-bit extensions Profibus DP ELAU PacDrive ROC Plus Allen-Bradley UniTE TCP/IP USB Simatic Toyopuc PC3J Uni-Telway Modbus Plus Fipio/Fipway
previous software version	V6.2
Software Designation	Vijeo Designer
Software Function	8000 variables 9999 alarms 4 asian alphabets 2 middle eastern alphabets Sharing up to 300 variables 34 western alphabets

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Range Compatibility	Harmony GTO Harmony GTU Harmony P6 Harmony iPC Harmony GTUX Harmony STU

Ordering and shipping details

Category	US10MC222569
Discount Schedule	0MC2
GTIN	3606486920169
Returnability	No
Country of origin	US

Contractual warranty

Warranty (in months)	18

Data Sheet 17

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
1	Mfg: Modicon: Processor Model Number: BMXP342020H Tags/Service: CONTROLLER / PLC/RTU Control Panel Specifications: • Modicon M340 automation platform • Processor module for severe environments • 1024 discrete I/O processor capacity • 256 analog I/O processor capacity • 2 Ethernet communication ports • 4096kB internal RAM (256kB for data, 3584kB for program constants and symbols) • −13...158°F ambient air temperature range for operation
1	Mfg: Modicon: Modicon PLC Software Model Number: CEXSPUCZSGPAZZ Tags/Service: CONTROLLER / PLC/RTU Control Panel Specifications: • EcoStruxture Control Expert License • Service pack base, V16 • Control expert small • 3 user license group

	• Windows 10/11, Windows Server 2019/2022 operating system requirement

Product data sheet PROCESSOR

Specifications

TAGS: CONTROLLER



processor, Modicon M340, max 1024 discrete, 256 analog IO, Modbus, Ethernet, for severe environment

BMXP342020H

Product availability: Stock - Normally stocked in distribution facility

Main

Range of Product	Modicon M340 automation platform
Product or Component Type	Processor module
Product Specific Application	For severe environments
Concept	Transparent Ready Modbus
Number of racks	4
Local I/O processor capacity (discrete)	1024 I/O
Analogue I/O processor capacity	256 I/O
Number of application specific channel	36
Monitoring	Diagnostic counters Modbus Event counters Modbus

Complementary

Control channels	Programmable loops
Integrated connection type	Non isolated serial link RJ45 character mode asynchronous in baseband, RS232C 2 twisted shielded pairs 0.3...19.2 kbit/s full duplex Non isolated serial link RJ45 character mode asynchronous in baseband, RS485 1 twisted shielded pair 0.3...19.2 kbit/s half duplex Non isolated serial link RJ45, master/slave Modbus, RTU/ASCII asynchronous in baseband, RS232C 1 twisted shielded pair 0.3...19.2 kbit/s half duplex Non isolated serial link RJ45, master/slave Modbus, RTU/ASCII asynchronous in baseband, RS485 1 twisted shielded pair 0.3...19.2 kbit/s half duplex USB port 12 Mbit/s Ethernet TCP/IP RJ45 1 twisted pair 10/100 Mbit/s
Communication module processor capacity	2 Ethernet communication module 4 AS-Interface module
embedded communication service	Bandwidth management, Ethernet TCP/IP Data Editor, Ethernet TCP/IP Modbus TCP messaging, Ethernet TCP/IP Rack Viewer, Ethernet TCP/IP SNMP network administrator, Ethernet TCP/IP
Port Ethernet	10BASE-T/100BASE-TX
Number of devices per segment	0…32character mode) 0…32Modbus)
Number of devices	2 point-to-point character mode 2 point-to-point Modbus

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Bus length	0…32.8 ft (0…10 m) serial link non isolated character mode segment 0…32.8 ft (0…10 m) serial link non isolated Modbus segment 0…3280.8 ft (0…1000 m) serial link isolated character mode segment 0…3280.8 ft (0…1000 m) serial link isolated Modbus segment 0…49.2 ft (0…15 m) character mode point-to-point 0…49.2 ft (0…15 m) Modbus point-to-point
Maximum tap links length	<49.2 ft (15 m) serial link non isolated character mode segment <49.2 ft (15 m) serial link non isolated Modbus segment <131.2 ft (40 m) serial link isolated character mode segment <131.2 ft (40 m) serial link isolated Modbus segment
Number of addresses	0…248 character mode 0…248 Modbus
Requests	1 K data bytes per request character mode 252 data bytes per RTU request Modbus 504 data bytes per ASCII request Modbus
Control parameter	One CRC on each frame (RTU) Modbus One LRC on each frame (ASCII) character mode One LRC on each frame (ASCII) Modbus
Memory description	Supplied memory card (BMXRMS008MP) backup of programs, constants, symbols and data Internal RAM 4096 kB Internal RAM 256 kB data Internal RAM 3584 kB program constants and symbols Supplied memory card (BMXRMS008MP) activation of standard web server, class B10
Maximum size of object areas	32634 %Mi located internal bits 32464 %MWi internal words 32760 %KWi constant words
Default size of object areas	1024 %MWi internal words located internal data 256 %KWi constant words located internal data 512 %Mi located internal bits
Application structure	64 event tasks 1 periodic fast task No auxiliary task 1 cyclic/periodic master task
Execution time per instruction	0.12 µs Boolean 0.17 µs double-length words 0.25 µs single-length words 1.16 µs floating points
Number of instructions per ms	6.4 Kinst/ms 65 % Boolean + 35 % fixed arithmetic 8.1 Kinst/ms 100 % Boolean
System overhead	0.13 ms fast task 0.7 ms master task
Current consumption	95 mA 24 V DC
Supply	Internal power supply via rack
Marking	CE UL CSA RCM EAC UKCA China RoHS WEEE IEC ATEX
Status LED	1 LED (Green) activity on Ethernet network (ETH ACT) 1 LED (Green) processor running (RUN) 1 LED (Green) status of Ethernet network (ETH STS) 1 LED (Green) data rate (ETH 100) 1 LED (Red) I/O module fault (I/O) 1 LED (Red) memory card fault (CARD ERR) 1 LED (Red) processor or system fault (ERR) 1 LED (Yellow) activity on Modbus (SER COM)
Net Weight	0.452 lb(US) (0.205 kg)

Environment

Ambient Air Temperature for Operation	-13…158 °F (-25…70 °C)
Relative Humidity	5…95 % without condensation
IP Degree of Protection	IP20
Protective treatment	Conformal coating
Directives	2014/35/EU - low voltage directive 2014/30/EU - electromagnetic compatibility 2012/19/EU - WEEE directive 2014/34/EU - ATEX directive
Product certifications	CE UL CSA RCM EAC UKCA Merchant Navy ATEX zone 2/22 IEC-Ex zone 2/22 UKEX zone 2/22 CCSAus HazLoc Class I Division 2 Group A CCSAus HazLoc Class I Division 2 Group B CCSAus HazLoc Class I Division 2 Group C CCSAus HazLoc Class I Division 2 Group D
Standards	IEC 61131-2 IEC 61010-2-201 UL 61010-2-201 CSA C22.2 No 61010-2-201 IACS E10 EN/IEC 61000-6-5, interface type 1 and type 2 EN/IEC 61850-3, location G IEC 60079-0
Environmental characteristic	Gas resistant class Gx ISA S71.04 Gas resistant class 3C4 IEC 60721-3-3 Dust resistant class 3S4 IEC 60721-3-3 Sand resistant class 3S4 IEC 60721-3-3 Salt resistant Kb/2 IEC 60068-2-52 Mold growth resistant class 3B2 IEC 60721-3-3 Fungal spore resistant class 3B2 IEC 60721-3-3

Ordering and shipping details

Category	US1PC3418160
Discount Schedule	PC34
GTIN	3595864009766
Returnability	Yes
Country of origin	FR

Packing Units

Unit Type of Package 1	PCE
Nbr. of units in pkg.	1
Package 1 Height	2.205 in (5.600 cm)
Package 1 Width	4.606 in (11.700 cm)
Package 1 Length	4.724 in (12.000 cm)
Package weight(Lbs)	8.995 oz (255.000 g)
Unit Type of Package 2	S02
Number of Units in Package 2	15

Package 2 Height	5.906 in (15.000 cm)
Package 2 Width	11.811 in (30.000 cm)
Package 2 Length	15.748 in (40.000 cm)
Package 2 Weight	8.891 lb(US) (4.033 kg)

Contractual warranty

Warranty (in months)	18

Environmental Data

Schneider Electric aims to achieve Net Zero status by 2050 through supply chain partnerships, lower impact materials, and circularity via our ongoing “Use Better, Use Longer, Use Again” campaign to extend product lifetimes and recyclability.

Environmental Data explained >

How we assess product sustainability >

Environmental footprint

Total lifecycle Carbon footprint	**119 kg CO2 eq.**
Environmental Disclosure	**Product Environmental Profile**
Carbon footprint of the manufacturing phase [A1 to A3]	**17 kg CO2 eq.**
Carbon footprint of the distribution phase [A4]	**0 kg CO2 eq.**
Carbon footprint of the installation phase [A5]	**0.1 kg CO2 eq.**
Carbon footprint of the use phase [B2, B3, B4, B6]	**102 kg CO2 eq.**
Carbon footprint of the end-of-life phase [C1 to C4]	**0.4 kg CO2 eq.**

Use Better

Materials and Substances

Packaging made with recycled cardboard	**Yes**
Packaging without single use plastic	**No**
EU RoHS Directive	**Compliant By Exemption**
REACh Regulation	**Reference contains Substances of Very High Concern above the threshold**
California proposition 65	**WARNING: This product can expose you to chemicals including: Lead and lead compounds, which is known to the State of California to cause cancer and birth defects or other reproductive harm. For more information go to www.P65Warnings.ca.gov**

Use Longer

Lifetime extension

Repair	**No**

Use Again

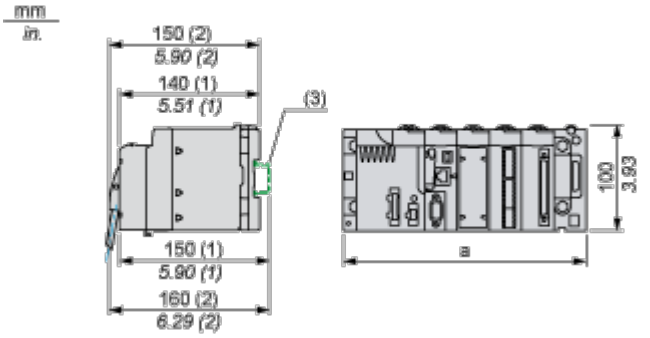
Repack and remanufacture

Recyclability potential, in %	**32**
Circularity Profile	**End of Life Information**
Take-back	**No**

Dimensions Drawings

Modules Mounted on Racks

Dimensions



(1) With removable terminal block (cage, screw or spring).

(2) With FCN connector.

(3) On AM1 ED rail: 35 mm wide, 15 mm deep. Only possible with BMXXBP0400/0400H/0600/0600H/0800/0800H rack.

Rack references	a in mm	a in in.
BMXXBP0400 and BMXXBP0400H	242.4	09.54
BMXXBP0600 and BMXXBP0600H	307.6	12.11
BMXXBP0800 and BMXXBP0800H	372.8	14.68
BMXXBP1200 and BMXXBP1200H	503.2	19.81

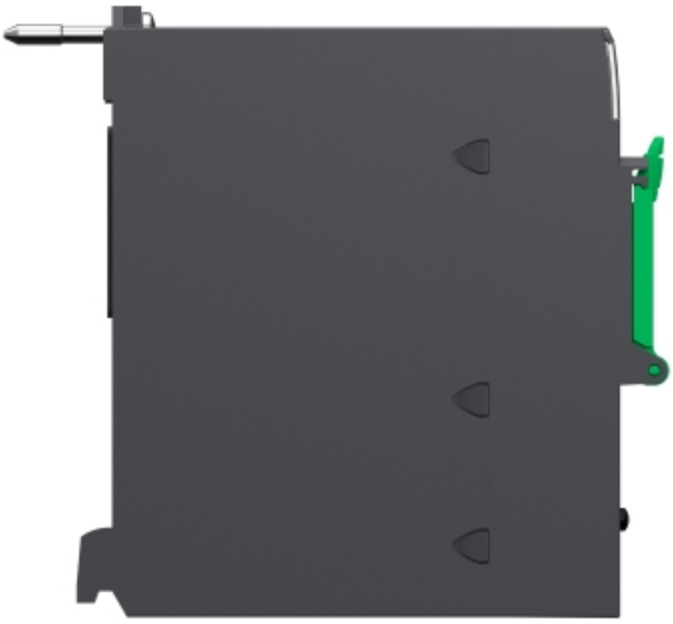
Image of product / Alternate images

Alternative






P342020H
RUN
ERR
I/O
SER COM
CARD ERR
ETH ACT
ETH STS
ETH 100
Eth MAC Address:
XX-XX-XX-XX-XX-XX
M
3
4
0
Ethernet
Modbus VP 5VS






SN:XXXXXXXXXXX
PV:XX SV:XX RL:XX

Product data sheet

MODICON PLC SOFTWARE

Specifications

TAG: CONTROLLER



License, EcoStruxure Control Expert, service pack base, small S, 3 users, node locked, digital license

CEXSPUCZSGPAZZ

Product availability: Non-Stock - Not normally stocked in distribution facility

Main

Range of Product	EcoStruxure Control Expert
Product or Component Type	License
Format	Digital
Software Type	Service pack base
Type of License	Node locked license
Software Version	V16
Operating system requirement	Windows 10 Windows 11 Windows Server 2019 Windows Server 2022
User Language	Italian Chinese Spanish English German French

Complementary

License category	Permanent
Product Compatibility	Modicon controller
Software Designation	Control Expert small
Software Function	Engineering
Hosting option	On-premise
License size	Group (3 users)
Memory requirement	8 GB
Display Resolution	800 x 600 pixels VGA

Ordering and shipping details

Category	US1PC4018220
Discount Schedule	PC40
GTIN	3606482208025
Returnability	No
Country of origin	US

Packing Units

Unit Type of Package 1	PCE

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Nbr. of units in pkg.	1
Package 1 Height	0.4 in (1 cm)
Package 1 Width	0.4 in (1 cm)
Package 1 Length	0.4 in (1 cm)
Package weight(Lbs)	0.04 oz (1 g)

Contractual warranty

Warranty (in months)	18

Data Sheet 18

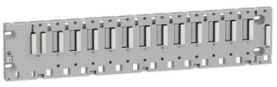
Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
1	Mfg: Modicon: Backplane, 12 Slots Model Number: BMXXBP1200H Tags/Service: BACKPLANE / PLC/RTU Control Panel Specifications: • Ruggedized backplane for severe environments • 12 bus slots • Compatible with BMXP34 processor • 0.83W power consumption

Product data sheet

BACKPLANE, 120 SLOTS

Specifications

TAGS: BACKPLANE



Backplane, Modicon X80, 12 slots, panel, mounting plate mounted, ruggedized

BMXXBP1200H

Product availability: Stock - Normally stocked in distribution facility

Main

Range of Product	Modicon X80
Product or Component Type	Ruggedized backplane
Product Specific Application	For severe environments

Complementary

Number of slots	12 bus X
Product Compatibility	BMXP34 processor ~~BMXCPS power supply~~ ~~I/O module~~ ~~Specific application module~~
Power Consumption in W	0.83 W
Electrical connection	1 connector (XBE) expansion module
Fixing mode	By 4 M4/M5/M6 screws panel ~~By 4 M4/M5/M6 screws mounting plate~~
Height	4.08 in (103.7 mm)
Width	19.8 in (503.2 mm)
Depth	0.7 in (19 mm)
Net Weight	2.80 lb(US) (1.27 kg)

Environment

IP Degree of Protection	IP20
Ambient Air Temperature for Operation	-13…158 °F (-25…70 °C)
Relative Humidity	10…95 % without condensation
Protective treatment	Conformal coating
Directives	2014/35/EU - low voltage directive 2014/30/EU - electromagnetic compatibility 2014/34/EU - ATEX directive
Product certifications	CE UL CSA RCM EAC UKCA Merchant Navy ATEX zone 2/22 IECEx zone 2/22

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Standards	EN/IEC 61131-2 EN/IEC 61010-2-201 UL 61010-2-201 CSA C22.2 No 61010-2-201 IACS E10 EN/IEC 61000-6-5, interface type 1 and type 2 EN/IEC 61850-3, location G EN/IEC 60079-0
Environmental characteristic	Gas resistant class Gx ISA S71.04 Gas resistant class 3C4 IEC 60721-3-3 Dust resistant class 3S4 IEC 60721-3-3 Sand resistant class 3S4 IEC 60721-3-3 Salt resistant level 2 IEC 68252 Mold growth resistant class 3B2 IEC 60721-3-3 Fungal spore resistant class 3B2 IEC 60721-3-3 Hazardous location class I division 2

Ordering and shipping details

Category	US1PC3418160
Discount Schedule	PC34
GTIN	3595864176277
Returnability	Yes
Country of origin	US

Packing Units

Unit Type of Package 1	PCE
Nbr. of units in pkg.	1
Package 1 Height	2.559 in (6.500 cm)
Package 1 Width	5.984 in (15.200 cm)
Package 1 Length	23.031 in (58.500 cm)
Package weight(Lbs)	2.923 lb(US) (1.326 kg)
Unit Type of Package 2	S04
Number of Units in Package 2	8
Package 2 Height	11.811 in (30.000 cm)
Package 2 Width	15.748 in (40.000 cm)
Package 2 Length	23.622 in (60.000 cm)
Package 2 Weight	25.474 lb(US) (11.555 kg)

Contractual warranty

Warranty (in months)	18

Data Sheet 19

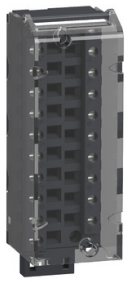
Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
6	Mfg: Modicon: Terminal Block, 20 Pin Model Number: BMXFTB2000 Tags/Service: PLC1502, PLC1521, PLC1602, PLC1701, PLC1901, PLC1925 / PLC/RTU Control Panel Specifications: • 20 terminal cage clamp removable terminal block
1	Mfg: Modicon: Terminal Block, 20 Pin Model Number: BMXFTB2800 Tags/Service: PLC1802 / PLC/RTU Control Panel Specifications: • 28 terminal cage clamp removable terminal block

Product data sheet

TERMINAL BLOCK, 20 PIN

Specifications

TAGS: PLC1502, PLC1521, PLC1602, PLC1701, PLC1901, PLC1925



terminal block, Modicon X80, 20-pin removable caged, 1 x 0.34..1mm2

BMXFTB2000

Product availability: Stock - Normally stocked in distribution facility

Main

Range of Product	Modicon X80
Accessory / Separate Part Type	Removable connection block
Number of terminals	20 removable cage clamp terminal block
Accessory / separate part destination	For module with 20-pin removable terminal block

Complementary

Net Weight	0.205 lb(US) (0.093 kg)

Ordering and shipping details

Category	US1PC3418160
Discount Schedule	PC34
GTIN	3595863920338
Returnability	Yes
Country of origin	CN

Packing Units

Unit Type of Package 1	PCE
Nbr. of units in pkg.	1
Package 1 Height	1.378 in (3.500 cm)
Package 1 Width	1.772 in (4.500 cm)
Package 1 Length	4.331 in (11.000 cm)
Package weight(Lbs)	3.845 oz (109.000 g)
Unit Type of Package 2	S02
Number of Units in Package 2	60
Package 2 Height	5.906 in (15.000 cm)
Package 2 Width	11.811 in (30.000 cm)
Package 2 Length	15.748 in (40.000 cm)
Package 2 Weight	15.302 lb(US) (6.941 kg)
Unit Type of Package 3	P06
Number of Units in Package 3	960
Package 3 Height	29.528 in (75.000 cm)
Package 3 Width	23.622 in (60.000 cm)

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Package 3 Length	31.496 in (80.000 cm)
Package 3 Weight	262.474 lb(US) (119.056 kg)

Contractual warranty

Warranty (in months)	18

Product data sheet

TERMINAL BLOCK, 28 PIN

Specifications

TAGS: PLC1802



terminal block, Modicon X80, 28-pin removable caged, 1 x 0.34..1mm2

BMXFTB2800

Product availability: Stock - Normally stocked in distribution facility

Main

Range of Product	Modicon X80
Accessory / Separate Part Type	Removable connection block
Number of terminals	28 removable cage clamp terminal block
Accessory / separate part destination	For module with 28-pin removable terminal block

Complementary

Net Weight	0.245 lb(US) (0.111 kg)

Ordering and shipping details

Category	US1PC3418160
Discount Schedule	PC34
GTIN	3595864133621
Returnability	Yes
Country of origin	CN

Packing Units

Unit Type of Package 1	PCE
Nbr. of units in pkg.	1
Package 1 Height	1.772 in (4.500 cm)
Package 1 Width	2.165 in (5.500 cm)
Package 1 Length	4.331 in (11.000 cm)
Package weight(Lbs)	5.185 oz (147.000 g)
Unit Type of Package 2	S02
Number of Units in Package 2	40
Package 2 Height	5.906 in (15.000 cm)
Package 2 Width	11.811 in (30.000 cm)
Package 2 Length	15.748 in (40.000 cm)
Package 2 Weight	14.110 lb(US) (6.400 kg)

Contractual warranty

Warranty (in months)	18

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Data Sheet 20

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
3	Mfg: Modicon: Discrete Input Module, 120VAC Model Number: BMXDAI1604H Tags/Service: PLC1502, PLC1521, PLC1602 / PLC/RTU Control Panel Specifications: • 120VAC discrete input voltage • 16 isolated discrete inputs for severe environments • 50/60Hz network frequency • 13kΩ input impedance • 3.8W power dissipation • Capacitive input type • 5mA discrete input current

Product data sheet DISCRETE INPUT MODULE

Specifications

TAGS: PLC1502, PLC1521, PLC1602



discrete input module, Modicon X80, 16 inputs, 100 to 120V AC, for severe environments

BMXDAI1604H

Product availability: Stock - Normally stocked in distribution facility

Main

Range of Product	Modicon X80
Product or Component Type	Discrete input module
Product Specific Application	For severe environments
Discrete input number	16
Discrete input type	Isolated
Input type	Capacitive
Discrete input voltage	100...120 V AC
Discrete input current	5 mA

Complementary

Input compatibility	2-wire proximity sensor IEC 60947-5-2 2-wire proximity sensor IEC 61131-2 type 3
Network Frequency	50/60 Hz
Network frequency limits	47…63 Hz
Sensor power supply	85...132 V
Current state 1 guaranteed	>= 2.5 mA
Current state 0 guaranteed	<= 1 mA
Input impedance	13000 Ohm
Insulation resistance	> 10 MOhm 500 V DC
Power dissipation in W	3.8 W
AC activation response time	10 ms
AC deactivation response time	20 ms
Typical current consumption	76 mA 3.3 V DC
MTBF reliability	1120000 H
Protection type	1 external fuse per group of channel 0.5 A fast blow
Voltage detection threshold	< 40 V AC sensor fault > 82 V AC sensor OK
Status LED	1 LED (Green) module operating (RUN) 1 LED per channel (Green) channel diagnostic 1 LED (Red) module error (ERR) 1 LED (Red) module I/O
Net Weight	0.254 lb(US) (0.115 kg)

Environment

Price is “List Price” and may be subject to a trade discount – check with your local distributor or retailer for actual price.

Disclaimer: This documentation is not intended as a substitute for and is not to be used for determining suitability or reliability of these products for specific user applications

IP Degree of Protection	IP20
Directives	2014/35/EU - low voltage directive 2014/30/EU - electromagnetic compatibility
Environmental characteristic	Gas resistant class Gx Gas resistant class 3C4 Dust resistant class 3S4 Sand resistant class 3S4 Salt resistant level 2 Mold growth resistant class 3B2 Fungal spore resistant class 3B2
Dielectric strength	1500 V AC 50/60 Hz 1 min
Vibration resistance	3 gn
Shock resistance	30 gn
Ambient Air Temperature for Storage	-40…185 °F (-40…85 °C)
Ambient Air Temperature for Operation	-13…158 °F (-25…70 °C)
Relative humidity	5…95 % 131 °F (55 °C) without condensation
Protective treatment	Conformal coating
Operating altitude	0...6561.68 ft (0...2000 m) 2000...5000 m with derating factor

Ordering and shipping details

Category	US1PC3418160
Discount Schedule	PC34
GTIN	3595864006444
Returnability	Yes

Packing Units

Unit Type of Package 1	PCE
Nbr. of units in pkg.	1
Package 1 Height	2.205 in (5.600 cm)
Package 1 Width	4.646 in (11.800 cm)
Package 1 Length	4.646 in (11.800 cm)
Package weight(Lbs)	5.397 oz (153.000 g)
Unit Type of Package 2	S02
Number of Units in Package 2	15
Package 2 Height	5.906 in (15.000 cm)
Package 2 Width	11.811 in (30.000 cm)
Package 2 Length	15.748 in (40.000 cm)
Package 2 Weight	5.785 lb(US) (2.624 kg)

Contractual warranty

Warranty (in months)	18

Data Sheet 21

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
1	Mfg: Modicon: Discrete Output Module Model Number: BMXDRA0815H Tags/Service: PLC1701 / PLC/RTU Control Panel Specifications: • 120VAC discrete output voltage • 8 relay discrete output module for severe environments • 50/60Hz network frequency • 3.6W power dissipation • 1780VAC 50/60Hz 1 min dielectric strength

Product data sheet

DISCRETE OUTPUT MODULE

Specifications

TAGS: PLC1701



Discrete output module, Modicon X80, 8 NO relay outputs, 24 to 240V AC / 24 to 125V DC, for severe environments

BMXDRA0815H

Product availability: Non-Stock - Not normally stocked in distribution facility

Main

Range of Product	Modicon X80
Product or Component Type	Relay discrete output module
Product Specific Application	For severe environments
Discrete output number	8 IEC 61131-2
Discrete output type	Relay
Discrete output logic	Positive
Discrete output voltage	24...240 V 19...264 V AC ~~24...125 V 5...150 V DC~~

Complementary

Electrical connection	20 ways terminal block
Network Frequency	50/60 Hz
Network frequency limits	47…63 Hz
Sensor power supply	5...150 V 19...264 V
[Ith] conventional free air thermal current	3 A
Insulation resistance	> 10 MOhm 500 V DC
Power dissipation in W	3.6 W
Response time on output	<= 10 ms activation <= 13 ms deactivation
Typical current consumption	40 mA 3.3 V DC 101 mA 24 V DC
MTBF reliability	3200000 H
Protection type	External short-circuit protection External overload protection External overvoltage protection, inductive AC External overvoltage protection, inductive DC
Output overload protection	Use 1 fast blow fuse per channel or group of channel
Output overvoltage protection	Use discharge diode on each output DC Use RC circuit on each output AC Use ZNO surge limiter on each output AC
Output short-circuit protection	Use 1 fast blow fuse per channel or group of channel
Minimum switching current	1 mA 5 V DC

Price is “List Price” and may be subject to a trade discount – check with your local distributor or retailer for actual price.

Disclaimer: This documentation is not intended as a substitute for and is not to be used for determining suitability or reliability of these products for specific user applications

Electrical durability	AC-12 200000 cycles 48 VA 24 V -13…140 °F (-25…60 °C) AC-12 200000 cycles 28.8 VA 24 V 158 °F (70 °C) AC-12 300000 cycles 48 VA 48 V -13…140 °F (-25…60 °C) AC-12 300000 cycles 28.8 VA 48 V 158 °F (70 °C) AC-12 150000 cycles 96 VA 48 V -13…140 °F (-25…60 °C) AC-12 150000 cycles 57.6 VA 48 V 158 °F (70 °C) AC-12 300000 cycles 110 VA 100...120 V -13…140 °F (-25…60 °C) AC-12 300000 cycles 66 VA 100...120 V 158 °F (70 °C) AC-12 150000 cycles 220 VA 100...120 V -13…140 °F (-25…60 °C) AC-12 150000 cycles 132 VA 100...120 V 158 °F (70 °C) AC-12 300000 cycles 220 VA 200...250 V -13…140 °F (-25…60 °C) AC-12 300000 cycles 132 VA 200...250 V 158 °F (70 °C) AC-12 150000 cycles 500 VA 200...250 V -13…140 °F (-25…60 °C) AC-12 150000 cycles 300 VA 200...250 V 158 °F (70 °C) AC-15 700000 cycles 10 VA 24 V -13…140 °F (-25…60 °C) 0.4) AC-15 700000 cycles 6 VA 24 V 158 °F (70 °C) 0.4) AC-15 500000 cycles 24 VA 24 V -13…140 °F (-25…60 °C) 0.4) AC-15 500000 cycles 14.4 VA 24 V 158 °F (70 °C) 0.4) AC-15 200000 cycles 48 VA 24 V -13…140 °F (-25…60 °C) 0.4) AC-15 200000 cycles 28.8 VA 24 V 158 °F (70 °C) 0.4) AC-15 700000 cycles 10 VA 48 V -13…140 °F (-25…60 °C) 0.4) AC-15 700000 cycles 6 VA 48 V 158 °F (70 °C) 0.4) AC-15 500000 cycles 24 VA 48 V -13…140 °F (-25…60 °C) 0.4) AC-15 500000 cycles 14.4 VA 48 V 158 °F (70 °C) 0.4) AC-15 300000 cycles 48 VA 48 V -13…140 °F (-25…60 °C) 0.4) AC-15 300000 cycles 28.8 VA 48 V 158 °F (70 °C) 0.4) AC-15 100000 cycles 96 VA 48 V -13…140 °F (-25…60 °C) 0.4) AC-15 100000 cycles 57.6 VA 48 V 158 °F (70 °C) 0.4) AC-15 1000000 cycles 10 VA 100...120 V -13…140 °F (-25…60 °C) 0.4) AC-15 1000000 cycles 6 VA 100...120 V 158 °F (70 °C) 0.4) AC-15 300000 cycles 50 VA 100...120 V -13…140 °F (-25…60 °C) 0.4) AC-15 300000 cycles 30 VA 100...120 V 158 °F (70 °C) 0.4) AC-15 200000 cycles 110 VA 100...120 V -13…140 °F (-25…60 °C) 0.4) AC-15 200000 cycles 66 VA 100...120 V 158 °F (70 °C) 0.4) AC-15 70000 cycles 220 VA 100...120 V -13…140 °F (-25…60 °C) 0.4) AC-15 70000 cycles 132 VA 100...120 V 158 °F (70 °C) 0.4) AC-15 1000000 cycles 10 VA 200...250 V -13…140 °F (-25…60 °C) 0.4) AC-15 1000000 cycles 6 VA 200...250 V 158 °F (70 °C) 0.4) AC-15 500000 cycles 50 VA 200...250 V -13…140 °F (-25…60 °C) 0.4) AC-15 500000 cycles 30 VA 200...250 V 158 °F (70 °C) 0.4) AC-15 200000 cycles 110 VA 200...250 V -13…140 °F (-25…60 °C) 0.4) AC-15 200000 cycles 66 VA 200...250 V 158 °F (70 °C) 0.4) AC-15 150000 cycles 220 VA 200...250 V -13…140 °F (-25…60 °C) 0.4) AC-15 150000 cycles 132 VA 200...250 V 158 °F (70 °C) 0.4) ~~DC-12 200000 cycles 24 W 24 V -13…140 °F (-25…60 °C)~~ ~~DC-12 200000 cycles 14.4 W 24 V 158 °F (70 °C)~~ ~~DC-12 150000 cycles 48 W 24 V -13…140 °F (-25…60 °C)~~ ~~DC-12 150000 cycles 28.8 W 24 V 158 °F (70 °C)~~ ~~DC-12 150000 cycles 40 W 48...60 V -13…140 °F (-25…60 °C)~~ ~~DC-12 150000 cycles 24 W 48...60 V 158 °F (70 °C)~~ ~~DC-12 100000 cycles 45 W 100...125 V -13…140 °F (-25…60 °C)~~ ~~DC-12 60000 cycles 45 W 100...125 V 158 °F (70 °C)~~ ~~DC-13 100000 cycles 10 W 24 V -13…140 °F (-25…60 °C)~~ ~~DC-13 100000 cycles 6 W 24 V 158 °F (70 °C)~~ ~~DC-13 60000 cycles 24 W 24 V -13…140 °F (-25…60 °C)~~ ~~DC-13 60000 cycles 14.4 W 24 V 158 °F (70 °C)~~ ~~DC-13 40000 cycles 48 W 24 V -13…140 °F (-25…60 °C)~~ ~~DC-13 40000 cycles 28.8 W 24 V 158 °F (70 °C)~~ ~~DC-13 40000 cycles 40 W 48...60 V -13…140 °F (-25…60 °C)~~ ~~DC-13 40000 cycles 24 W 48...60 V 158 °F (70 °C)~~ ~~DC-13 100000 cycles 15 W 100...125 V -13…140 °F (-25…60 °C)~~ ~~DC-13 40000 cycles 15 W 100...125 V 158 °F (70 °C)~~
Status LED	1 LED (Green) RUN 1 LED per channel (Green) channel diagnostic 1 LED (Red) ERR 1 LED (Red) I/O
Net Weight	0.373 lb(US) (0.169 kg)

Environment

IP Degree of Protection	IP20

Environmental characteristic	Gas resistant class Gx Gas resistant class 3C4 Dust resistant class 3S4 Sand resistant class 3S4 Salt resistant level 2 Mold growth resistant class 3B2 Fungal spore resistant class 3B2
Dielectric strength	1780 V AC 50/60 Hz 1 min
Vibration resistance	3 gn
Shock resistance	30 gn
Ambient Air Temperature for Storage	-40…185 °F (-40…85 °C)
Ambient Air Temperature for Operation	-13…158 °F (-25…70 °C)
Relative humidity	5…95 % 140 °F (60 °C) without condensation
Protective treatment	Conformal coating
Operating altitude	0...6561.68 ft (0...2000 m) 2000...5000 m with derating factor

Ordering and shipping details

Category	US1PC3418160
Discount Schedule	PC34
GTIN	3606481275615
Returnability	No
Country of origin	US

Packing Units

Unit Type of Package 1	PCE
Nbr. of units in pkg.	1
Package 1 Height	2.165 in (5.500 cm)
Package 1 Width	4.567 in (11.600 cm)
Package 1 Length	4.843 in (12.300 cm)
Package weight(Lbs)	7.372 oz (209.000 g)
Unit Type of Package 2	S02
Number of Units in Package 2	15
Package 2 Height	5.906 in (15.000 cm)
Package 2 Width	11.811 in (30.000 cm)
Package 2 Length	15.748 in (40.000 cm)
Package 2 Weight	7.588 lb(US) (3.442 kg)

Contractual warranty

Warranty (in months)	18

Data Sheet 22

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
1	Mfg: Modicon: Analog Input Module Model Number: BMXAMI0810H Tags/Service: PLC1802 / PLC/RTU Control Panel Specifications: • 8 analog input module for severe environments • 4...20mA analog input type • 16 bit analog/digital conversion • 15 bits + sign analog input resolution • ± 30mA permitted overload on inputs • First order digital filtering • 0.15% of full scale measurement error, 4…20mA 25°C

Product data sheet ANALOG INPUT MODULE

Specifications

TAGS: PLC1802



Analog isolated high level input module, Modicon X80, 8 I, 0 to 20mA, 4 to 20mA, 10V positive or negative, severe environment

BMXAMI0810H

Product availability: Stock - Normally stocked in distribution facility

Main

Range of Product	Modicon X80
Product or Component Type	Analog input module
Product Specific Application	For severe environments
Electrical connection	28 ways 1 connector
Isolation between channels	Isolated
Input level	High level
Analogue input number	8
Analogue input type	~~Current +/- 20 mA~~ ~~Current 0...20 mA~~ Current 4...20 mA ~~Voltage +/- 10 V~~ ~~Voltage +/- 5 V~~ ~~Voltage 0...10 V~~ ~~Voltage 0...5 V~~ ~~Voltage 1...5 V~~

Complementary

Analog/digital conversion	16 bits
Analogue input resolution	15 bits + sign
Permitted overload on inputs	~~+/- 30 mA 0...20 mA~~ +/- 30 mA 4...20 mA ~~+/- 30 V +/- 10 V~~ ~~+/- 30 V +/- 5 V~~ ~~+/- 30 V 0...10 V~~ ~~+/- 30 V 0...5 V~~ ~~+/- 30 V 1...5 V~~ ~~+/- 30 mA +/- 20 mA~~
Input impedance	10 MOhm in voltage mode 250 Ohm + 3.6...50 Ohm internal protective resistor in current mode
Precision of internal conversion resistor	0.1 % - 15 ppm/°C
Type of filter	First order digital filtering
Fast read cycle time	1 ms + 1 ms x number of channels used
Nominal read cycle time	9 ms for 8 channels

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Measurement error	~~0.15 % of full scale +/- 20 mA 25 °C~~ ~~0.15 % of full scale 0...20 mA 25 °C~~ 0.15 % of full scale 4...20 mA 25 °C ~~0.075 % of full scale +/- 10 V 25 °C~~ ~~0.075 % of full scale 0...10 V 25 °C~~ ~~0.075 % of full scale 0...5 V 25 °C~~ ~~0.075 % of full scale 1...5 V 25 °C~~ ~~0.075 % of full scale +/- 5 V 25 °C~~ ~~<= 0.2 % of full scale +/- 10 V - 25...70 °C~~ ~~<= 0.2 % of full scale +/- 5 V - 25...70 °C~~ ~~<= 0.2 % of full scale 0...10 V - 25...70 °C~~ ~~<= 0.2 % of full scale 0...5 V - 25...70 °C~~ ~~<= 0.2 % of full scale 1...5 V - 25...70 °C~~ ~~<= 0.55 % of full scale +/- 20 mA - 25...70 °C~~ ~~<= 0.55 % of full scale 0...20 mA - 25...70 °C~~ <= 0.55 % of full scale 4...20 mA - 25...70 °C
Temperature drift	~~30 ppm/°C +/- 10 V~~ ~~30 ppm/°C +/- 5 V~~ ~~30 ppm/°C 0...10 V~~ ~~30 ppm/°C 0...5 V~~ ~~30 ppm/°C 1...5 V~~ ~~50 ppm/°C +/- 20 mA~~ ~~50 ppm/°C 0...20 mA~~ 50 ppm/°C 4...20 mA
Minimum crosstalk attenuation	80 dB
EMI/RFI Noise rejection (100 kHz to 10 MHz)	80 dB
Digital value format	- 32768 to + 32767 in maximum user scale +/- 10000 by default
Isolation voltage	300 V DC between channels 1400 V DC between channels and ground 1400 V DC between channels and bus
Measurement resolution	~~0.36 mV +/- 10 V~~ ~~0.36 mV 0...10 V~~ ~~0.36 mV 0...5 V~~ ~~0.36 mV 1...5 V~~ ~~0.36 mV +/- 5 V~~ ~~1.4 µA +/- 20 mA~~ ~~1.4 µA 0...20 mA~~ 1.4 µA 4...20 mA
Maximum conversion value	~~+/- 11.4 V +/- 10 V~~ ~~+/- 11.4 V 0...10 V~~ ~~+/- 11.4 V 0...5 V~~ ~~+/- 11.4 V 1...5 V~~ ~~0...30 mA +/- 20 mA~~ ~~0...30 mA 0...20 mA~~ 0...30 mA 4...20 mA ~~0...30 mA +/- 5 V~~
MTBF reliability	900000 H
Operating altitude	0...6561.68 ft (0...2000 m) 2000...5000 m with derating factor
Status LED	1 LED (Green) RUN 1 LED per channel (Green) channel diagnostic 1 LED (Red) ERR 1 LED (Red) I/O
Net Weight	0.386 lb(US) (0.175 kg)
Power consumption in W	1.06 W 24 V DC typical 1.50 W 24 V DC maximum ~~0.32 W 3.3 V DC typical~~ ~~0.48 W 3.3 V DC maximum~~
Current consumption	~~150 mA 3.3 V DC~~ 54 mA 24 V DC

Environment

Vibration resistance	3 gn

Shock resistance	30 gn
Ambient Air Temperature for Storage	-40…185 °F (-40…85 °C)
Ambient Air Temperature for Operation	-13…158 °F (-25…70 °C)
Relative humidity	5…95 % 131 °F (55 °C) without condensation
IP Degree of Protection	IP20
Directives	2014/35/EU - low voltage directive 2014/30/EU - electromagnetic compatibility
Product Certifications	CE RCM CSA EAC Merchant Navy UL ATEX IEC-Ex
Standards	IEC 61010-2-201 IEC 61131-2 UL 61010-2-201 CSA C22.2 No 61010-2-201
Environmental characteristic	Gas resistant class Gx Gas resistant class 3C4 Dust resistant class 3S4 Sand resistant class 3S4 Salt resistant level 2 Mold growth resistant class 3B2 Fungal spore resistant class 3B2 Hazardous location
Protective treatment	Conformal coating

Ordering and shipping details

Category	US1PC3418160
Discount Schedule	PC34
GTIN	3595864156620
Returnability	Yes
Country of origin	US

Packing Units

Unit Type of Package 1	PCE
Nbr. of units in pkg.	1
Package 1 Height	2.165 in (5.500 cm)
Package 1 Width	4.331 in (11.000 cm)
Package 1 Length	4.567 in (11.600 cm)
Package weight(Lbs)	5.926 oz (168.000 g)
Unit Type of Package 2	S02
Number of Units in Package 2	15
Package 2 Height	5.906 in (15.000 cm)
Package 2 Width	11.811 in (30.000 cm)
Package 2 Length	15.748 in (40.000 cm)
Package 2 Weight	6.294 lb(US) (2.855 kg)

Data Sheet 23

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
1	Mfg: Modicon: Analog Output Module Model Number: BMXAMO0410H Tags/Service: PLC1901 / PLC/RTU Control Panel Specifications: • 4 analog input module for severe environments • 4...20mA analog output type • 15 bits + sign analog input resolution • 3W power consumption typical at 24VDC • 140mA current consumption at 24VDC • Open circuit detection

Product data sheet

ANALOG OUTPUT MODULE

Specifications

TAGS: PLC1901



Analog isolated high level output module, Modicon X80, 4 O, 0 to 20mA, 4 to 20mA, 10V positive or negative, severe environment

BMXAMO0410H

Product availability: Stock - Normally stocked in distribution facility

Main

Range of Product	Modicon X80
Product or Component Type	Analog output module
Product Specific Application	For severe environments
Electrical connection	20 ways 1 connector
Isolation between channels	Isolated

Complementary

Measurement error	0.1 % of full scale 25 °C <= 0.45 % of full scale - 25...70 °C
Temperature drift	~~45 ppm/°C +/- 10 V~~ ~~45 ppm/°C 0...20 mA~~ 45 ppm/°C 4...20 mA
Minimum crosstalk attenuation	80 dB
Common mode rejection	100 dB
Isolation voltage	1400 V DC between channels and ground 1400 V DC between channels and bus 750 V DC between channels
Detection type	~~Open circuit 0...20 mA~~ Open circuit 4...20 mA ~~Short circuit +/- 10 V~~
Load impedance ohmic	~~>= 1000 Ohm +/- 10 V~~ ~~<= 500 Ohm 0...20 mA~~ <= 500 Ohm 4...20 mA
Output level	High level
Analogue output number	4
Analogue output type	~~Current 0...20 mA~~ Current 4...20 mA ~~Voltage +/- 10 V~~
Analogue output resolution	15 bits + sign
Supply	Internal power supply via rack
Conversion time	<= 1 ms
Maximum conversion value	~~0...21 mA 0...20 mA~~ 0...21 mA 4...20 mA ~~+/- 10.5 V +/- 10 V~~
Fallback mode	Configurable Predefined
MTBF reliability	1000000 H

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Operating altitude	0...6561.68 ft (0...2000 m) 2000...5000 m with derating factor
Status LED	1 LED (Green) RUN 1 LED per channel (Green) channel diagnostic 1 LED (Red) ERR 1 LED (Red) I/O
Net Weight	0.33 lb(US) (0.15 kg)
Power consumption in W	3 W 24 V DC typical 3.6 W 24 V DC maximum ~~0.45 W 3.3 V DC typical~~ ~~0.51 W 3.3 V DC maximum~~
Current consumption	~~150 mA 3.3 V DC~~ 140 mA 24 V DC

Environment

Vibration resistance	3 gn
Shock resistance	30 gn
Ambient Air Temperature for Storage	-40…185 °F (-40…85 °C)
Ambient Air Temperature for Operation	-13…158 °F (-25…70 °C)
Relative humidity	5…95 % 131 °F (55 °C) without condensation
IP Degree of Protection	IP20
Directives	2014/35/EU - low voltage directive 2014/30/EU - electromagnetic compatibility
Product Certifications	CE RCM CSA EAC Merchant Navy UL ATEX IEC-Ex
Standards	IEC 61010-2-201 IEC 61131-2 UL 61010-2-201 CSA C22.2 No 61010-2-201
Environmental characteristic	Gas resistant class Gx Gas resistant class 3C4 Dust resistant class 3S4 Sand resistant class 3S4 Salt resistant level 2 Mold growth resistant class 3B2 Fungal spore resistant class 3B2 Hazardous location
Protective treatment	Conformal coating

Ordering and shipping details

Category	US1PC3418160
Discount Schedule	PC34
GTIN	3595864156637
Returnability	Yes

Packing Units

Unit Type of Package 1	PCE
Nbr. of units in pkg.	1

Package 1 Height	2.165 in (5.500 cm)
Package 1 Width	4.567 in (11.600 cm)
Package 1 Length	4.646 in (11.800 cm)
Package weight(Lbs)	7.831 oz (222.000 g)
Unit Type of Package 2	S02
Number of Units in Package 2	15
Package 2 Height	5.906 in (15.000 cm)
Package 2 Width	11.811 in (30.000 cm)
Package 2 Length	15.748 in (40.000 cm)
Package 2 Weight	7.990 lb(US) (3.624 kg)

Contractual warranty

Warranty (in months)	18

Data Sheet 24

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
1	Mfg: Modicon: Discrete Input Module, 24VDC Model Number: BMXDDI1602H Tags/Service: PLC1925 / PLC/RTU Control Panel Specifications: • 24VDC discrete input voltage • 16 isolated discrete inputs for severe environments • Reverse polarity input protection • 6.8kΩ input impedance • 2.5W power dissipation • Paralleling of inputs • 3.5mA discrete input current
1	Mfg: Phoenix Contact: IS Barrier/Splitter Model Number: 2865366 Tags/Service: ISB1432 / PLC/RTU Control Panel Specifications: • Single channel repeater power supply • 3-way electrical isolation • Category II overvoltage protection • 24VDC nominal supply voltage • < 1.45W power dissipation

	• ≤ 1.8W power consumption

Product data sheet

DISCRETE INPUT MODULE, 24VDC

Specifications

TAGS: PLC1925



discrete input module, Modicon X80, 16 inputs, 24V DC positive, for severe environments

BMXDDI1602H

Product availability: Stock - Normally stocked in distribution facility

Main

Range of Product	Modicon X80
Product or Component Type	Discrete input module
Device Application	Industrial application in severe environment
Discrete input number	16
Discrete input voltage	24 V DCisolated
Discrete input logic	Positive logic (sink)

Complementary

Discrete input number	16 isolated conforming to IEC 60947-5-2 at 24 V DC 16 isolated conforming to IEC 61131-2 type 3 at 24 V DC
Input wiring mode	2 or 3 wires
Discrete input current	3.5 mA at 24 V DC
Voltage state 1 guaranteed	>= 11 V
Voltage state 0 guaranteed	< 5 V
Current state 1 guaranteed	>= 2 mA
Current state 0 guaranteed	<= 1.5 mA
Input impedance	6800 Ohm
Paralleling of inputs	Yes
Input protection type	Reverse polarity protection
Insulation resistance	> 10 MOhm 500 V DC
DC typical response time	4 ms
DC maximum response time	7 ms
Sensor power supply	19…30 V 60 °C 21.1…26.4 V 70 °C
Typical current consumption	76 mA DC
Current consumption	<= 107 mA 3.3 V DC
Power dissipation in W	2.5 W
Isolation between channels and bus	1500 V for 1 minute
Hot swapping fallback	Yes
MTBF reliability	775000 H
Voltage detection threshold	< 14 V DC sensor fault > 18 V DC sensor OK

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Local signalling	1 LED (green) for module operating (RUN) 1 LED per channel (green) for channel diagnostic 1 LED (red) for module error (ERR) 1 LED (red) for module I/O
Height	4.08 in (103.7 mm)
Width	1.3 in (32 mm)
Depth	3.4 in (86 mm)
Product Weight	0.254 lb(US) (0.115 kg)

Environment

IP Degree of Protection	IP20
Protective treatment	Conformal coating
Marking	CE UKCA CULus RCM EAC
Product certifications	ATEX zone 2 IECEx zone 2 CCC Ex zone 2 (pending) CULus HazLoc
Standards	IEC 61131-2
Directives	2014/35/EU - low voltage directive 2014/30/EU - electromagnetic compatibility
Dielectric strength	1500 V AC 50/60 Hz 1 minute, primary/secondary
Ambient air temperature for operation	-13…158 °F (-25…70 °C)
Ambient air temperature for storage	-40…185 °F (-40…85 °C)
Relative humidity	5…95 % 131 °F (55 °C) without condensation
Operating altitude	0...6561.68 ft (0...2000 m) 2000...5000 m with derating
Pollution degree	2
Environmental characteristic	Gas resistant class Gx Gas resistant class 3C4 Dust resistant class 3S4 Sand resistant class 3S4 Salt resistant level 2 Mold growth resistant class 3B2 Fungal spore resistant class 3B2
Vibration resistance	3 gn
Shock resistance	30 gn

Ordering and shipping details

Category	US1PC3418160
Discount Schedule	PC34
GTIN	3595864025605
Returnability	Yes
Country of origin	US

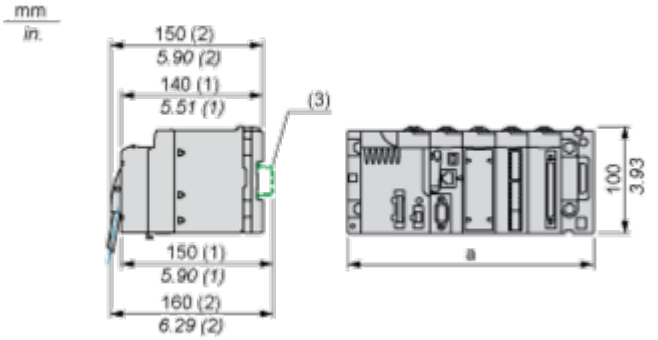
Packing Units

Unit Type of Package 1	PCE

Dimensions Drawings

Modules Mounted on Racks

Dimensions



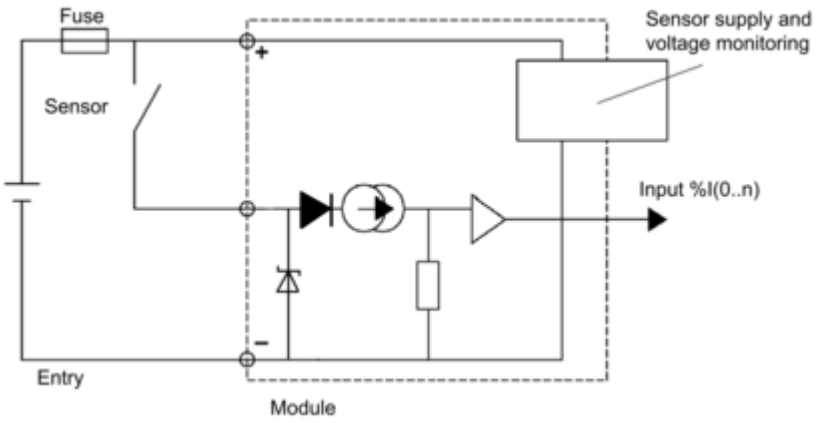
(1) With removable terminal block (cage, screw or spring).
(2) With FCN connector.
(3) On AM1 ED rail: 35 mm wide, 15 mm deep. Only possible with BMXXBP0400/0400H/0600/0600H/0800/0800H rack.

Rack references	a in mm	a in in.
BMXXBP0400 and BMXXBP0400H	242.4	09.54
BMXXBP0600 and BMXXBP0600H	307.6	12.11
BMXXBP0800 and BMXXBP0800H	372.8	14.68
BMXXBP1200 and BMXXBP1200H	503.2	19.81

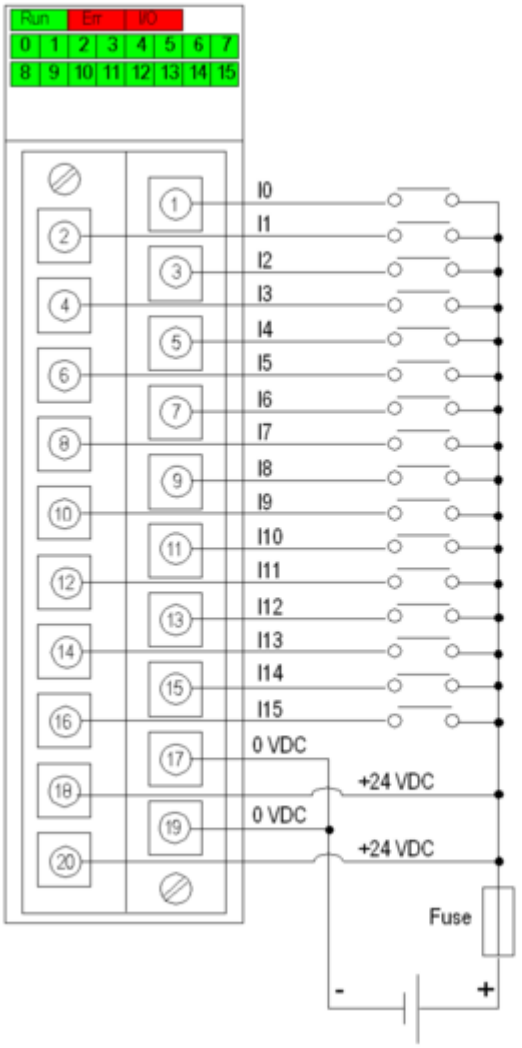
Connections and Schema

Connecting the Module

Input Circuit Diagram



Module Connection



IS BARRIER/SPLITTER

MACX MCR-EX-SL-RPSSI-2I - Repeater power supply

2865366

https://www.phoenixcontact.com/us/products/2865366

TAGS: ISB1432

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Ex i-Measuring transducer repeater power supply and input signal conditioner transmits supplied or active 0 mA/4 mA ... 20 mA electrically isolated signals from the Ex area to two loads in the safe area. number of channels: 1, HART transparency, Standard configuration, 3-way isolation, Safety Integrity Level (SIL, IEC 61508): 2, Performance Level (ISO 13849): d / KAT 2, Systematic Capability: 3, Screw connection

Your advantages

- 0/4 mA ... 20 mA input, intrinsically safe, [Ex ia], powered and not powered
- Measuring transducer supply voltage > 16 V
- Two electrically isolated 0/4 mA ... 20 mA (active) outputs
- Bidirectional HART transmission (both outputs)
- Error indication according to NAMUR NE 43
- SIL 2 according to IEC/EN 61508
- Safe electrical isolation between input, outputs, and supply
- Power supply possible via DIN rail connector
- Installation in zone 2 permitted
- Plug-in connection terminal blocks, screw connection technology, with integrated sockets for HART communicators
- Housing width: 12.5 mm
- Minimal power dissipation
- High transmission accuracy

Technical data

Product properties

Product family	MACX Analog
No. of channels	1

Electrical properties

Electrical isolation	3-way isolation
Electrical isolation between input and output	yes
Signal transmission behavior	In = Out
Step response (10-90%)	1.3 ms (for jump 4 mA ... 20 mA, typical)
Maximum temperature coefficient	< 0.01 %/K
Maximum transmission error	< 0.1 % (of final value 20 mA)
Transmission error, typical	< 0.05 % (of final value 20 mA)

Electrical isolation

Overvoltage category	II
Pollution degree	2

Electrical isolation Input/output/power supply IEC/EN 61010-1

Standards/regulations	IEC/EN 61010-1
Rated insulation voltage	300 V_{rms}
Test voltage	2.5 kV AC (50 Hz, 60 s)
Insulation	Safe isolation

Electrical isolation Input/output IEC/EN 60079-11

Standards/regulations	IEC/EN 60079-11
Rated insulation voltage	265 V_{rms}

Electrical isolation Input/power supply IEC/EN 60079-11

Standards/regulations	IEC/EN 60079-11
Rated insulation voltage	265 V_{rms}

Electrical isolation Output 1/output 2

Test voltage	1.5 kV AC (50 Hz, 60 s)

Supply

Designation	Repeater power supply operation
Nominal supply voltage	24 V DC -20 % ... +25 %
Supply voltage range	19.2 V DC ... 30 V DC
Max. current consumption	< 75 mA (24 V DC / 20 mA)
Power dissipation	< 1.45 W (24 V DC / 20 mA)
Power consumption	≤ 1.8 W

Supply

Designation	Signal conditioner operation

Nominal supply voltage	24 V DC -20 % ... +25 %
Supply voltage range	19.2 V DC ... 30 V DC
Max. current consumption	< 46 mA (24 V DC / 20 mA)
Power dissipation	< 1.2 W (24 V DC / 20 mA)

Input data

Signal: Repeater power supply operation

Description of the input	Repeater power supply operation
Number of inputs	1
Current input signal	4 mA ... 20 mA
Transmitter supply voltage	> 16 V (20 mA)
	> 15.1 V (23 mA)
Polarization and surge protection	Yes
Underload/overload signal range	0 mA ... 24 mA (Extended transmission range for diagnostics)

Signal: Signal conditioner operation

Description of the input	Signal conditioner operation
Current input signal	0 mA ... 20 mA
	4 mA ... 20 mA
Voltage drop	< 3.9 V (in input isolating amplifier operation)
Underload/overload signal range	0 mA ... 24 mA (Extended transmission range for diagnostics)

Output data

Signal: Repeater power supply operation

Output description	Repeater power supply operation
Number of outputs	2
Current output signal	4 mA ... 20 mA (Output 1 and output 2 active)
Load/output load current output	< 450 Ω (20 mA)
	< 380 Ω (23 mA)
Output ripple	< 20 mV_{rms}
Output behavior in the event of an error	0 mA (Cable break in the input)
	≥ 23 mA (Cable short-circuit in the input)
Underload/overload signal range	0 mA ... 24 mA (Extended transmission range for diagnostics)

Signal: Signal conditioner operation

Output description	Signal conditioner operation
Current output signal	0 mA ... 20 mA (active)
	4 mA ... 20 mA (active)
Load/output load current output	< 450 Ω (20 mA)
	< 380 Ω (23 mA)
Output ripple	< 20 mV_{rms}
Output behavior in the event of an error	0 mA (Cable break in the input)
	0 mA (Cable short-circuit in the input)
Underload/overload signal range	0 mA ... 24 mA (Extended transmission range for diagnostics)

Connection data

Connection method	Screw connection
Stripping length	7 mm
Screw thread	M3
Conductor cross-section rigid	0.2 mm² ... 2.5 mm²
Conductor cross-section flexible	0.2 mm² ... 2.5 mm²
Conductor cross-section AWG	24 ... 14
Tightening torque	0.5 Nm ... 0.6 Nm

Test socket

Max. diameter	2.3 mm

Ex data

Ex installation (EPL)	Gc
	Div. 2
Ex i circuits (EPL)	[Ga]
	[Da]
	[Ma]
	[Div. 1]

Safety data: Repeater power supply operation

Max. output voltage U_o	25.2 V
Max. output current I_o	93 mA
Max. output power P_o	587 mW
Safety-related maximum voltage U_m	253 V AC
	125 V DC
I (simple circuit): Max. external inductivity L_o / Max. external capacitance C_o	40 mH / 4.8 µF
IIA (simple circuit): Max. external inductivity L_o / Max. external capacitance C_o	26 mH / 2.9 µF
IIB (simple circuit): Max. external inductivity L_o / Max. external capacitance C_o	14 mH / 820 nF
IIC (simple circuit): Max. external inductivity L_o / Max. external capacitance C_o	3 mH / 107 nF
IIA (mixed circuit): Max. external inductivity L_o / Max. external capacitance C_o	26 mH / 470 nF, 20 mH / 570 nF, 1 mH / 630 nF, 0.5 mH / 720 nF, 0.1 mH / 1.1 µF, 0.005 mH / 2.9 µF
IIB/IIIC (mixed circuit): Max. external inductivity L_o / Max. external capacitance C_o	16 mH / 370 nF, 1 mH / 430 nF, 500 µH / 510 nF, 200 µH / 660 nF, 100 µH / 820 nF
IIC (mixed circuit): Max. external inductivity L_o / Max. external capacitance C_o	2.2 mH / 47 nF, 2 mH / 49 nF, 1 mH / 63 nF, 500 µH / 80 nF, 200 µH / 107 nF
I (mixed circuit): Max. external inductivity L_o / Max. external capacitance C_o	37 mH / 0.54 µF, 0.2 mH / 1.1 µF, 10 µH / 2.8 µF, 0.001 mH / 4. 15 µF

Safety data: Signal conditioner operation

Input voltage U_i	≤ 30 V
Input current I_i	≤ 150 mA
Max. internal inductance L_i	negligible

Max. internal capacitance C_i	negligible
Safety-related maximum voltage U_m	253 V AC
	125 V DC

Interfaces

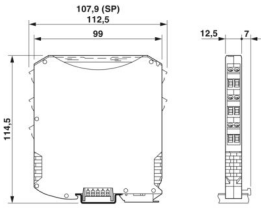
Data communication (bypass)

HART function	HART transparency
Protocols supported	HART

Signaling

Status display	Green LED (supply voltage)

Dimensions

Dimensional drawing	
Width	12.5 mm
Height	112.5 mm
Depth	113.7 mm
Depth NS 35/7,5	114.5 mm (Snapped onto DIN rail NS 35/7,5 in accordance with EN 60715)

Material specifications

Color	gray (RAL 7042)
Flammability rating according to UL 94 (Housing)	V0 (Housing)
Housing material	PA 6.6-FR

Environmental and real-life conditions

Ambient conditions

Degree of protection	IP20 (not assessed by UL)
Ambient temperature (operation)	-40 °C ... 60 °C (Any mounting position)
	-40 °C ... 70 °C (Derating)
Ambient temperature (storage/transport)	-40 °C ... 80 °C
Permissible humidity (operation)	10 % ... 95 % (non-condensing)

Altitude range (≤ 2000 m)

Altitude	≤ 2000 m (The technical data refers to altitudes ≤2000 m above mean sea level. For altitudes >2000 m above mean sea level, refer to the data sheet.)
Ambient temperature (operation)	-40 °C ... 60 °C
	-40 °C ... 70 °C (Derating)

Rated insulation voltage	375 V_{PP} (Power supply, input / output)

Altitude range (≤ 3000 m)

Height range	> 2000 m ... 3000 m
Ambient temperature (operation)	-40 °C ... 54 °C
	-40 °C ... 63 °C (Derating)
Rated insulation voltage	190 V AC (Power supply, input / output)
	110 V DC (Power supply, input / output)

Altitude range (≤ 4000 m)

Height range	> 3000 m ... 4000 m
Ambient temperature (operation)	-40 °C ... 48 °C
	-40 °C ... 56 °C (Derating)
Rated insulation voltage	60 V AC/DC (Power supply, input / output)

Altitude range (≤ 5000 m)

Height range	> 4000 m ... 5000 m
Ambient temperature (operation)	-40 °C ... 42 °C
	-40 °C ... 49 °C (Derating)
Rated insulation voltage	60 V AC/DC (Power supply, input / output)

Approvals

CE

Certificate	CE-compliant
Note	and EN 61326

ATEX

Identification	Ⓔx II (1) G [Ex ia Ga] IIC
	Ⓔx II (1) D [Ex ia Da] IIIC
	Ⓔx II 3(1) G Ex ec [ia Ga] IIC T4 Gc
	Ⓔx I (M1) [Ex ia Ma] I
Certificate	BVS 10 ATEX E 143 X

IECEx

Identification	[Ex ia Ga] IIC
	[Ex ia Da] IIIC
	Ex ec [ia Ga] IIC T4 Gc
	[Ex ia Ma] I
Certificate	IECEx BVS 10.0097X

CCC / China-Ex

Identification	[Ex ia Ga] IIC
	[Ex ia Da] IIIC
	Ex ec [ia Ga] IIC T4 Gc
Certificate	2022122316115971

UL, USA/Canada

Identification	IS for Class I,II,III, Division 1 and Zone 0
	Installation in Class I, Division 2 and Zone 2
Certificate	C.D.-No 83104549

Shipbuilding approval

Certificate	DNV GL TAA000020C

Safety Integrity Level (SIL, IEC 61508)

Identification	2
Certificate	IN-AT-AS-MRL-23-00432A

Systematic Capability

Identification	3

Performance Level (ISO 13849)

Identification	d / KAT 2

INMETRO

Identification	[Ex ia Ga] IIC
	[Ex ia Da] IIIC
	Ex ec [ia Ga] IIC T4 Gc
	[Ex ia Ma] I
Certificate	DNV 18.0139 X

Shipbuilding data

Temperature	B
Humidity	B
Vibration	A
EMC	A
Enclosure	Required protection according to the Rules shall be provided upon installation on board

EMC data

Electromagnetic compatibility	Conformance with EMC directive
Noise immunity	EN 61000-6-2
Note	When being exposed to interference, there may be minimal deviations.

Noise emission

Standards/regulations	EN 61000-6-4

Standards and regulations

Electrical isolation	3-way isolation

GB Standard

Standards/regulations	GB/T 3836.1
	GB/T 3836.3
	GB/T 3836.4

Mounting

Mounting type	DIN rail mounting

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Middletown, PA 17057, United States
(+717) 944-1300
info@phoenixcon.com

Data Sheet 25

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
1	Mfg: Polyphaser: Coaxial RF Surge Protector Model Number: IS-50NX-C2 Tags/Service: SP1402 / PLC/RTU Control Panel Specifications: • 125...1,000MHz frequency range • 50Ω impedance • 0.1dB insertion loss • 50kA surge current • 220µJ throughput energy at 3kA • 375W at 125VAC input power to 220MHz • N female to N female connectors

PolyPhaser
an INFINITE brand

COAXIAL RF SURGE PROTECTOR

TAGS: SP1402

Type N F/F Coaxial RF Surge Protector, 125MHz - 1GHz, DC Block, 375W, 220uJ, 50kA, Blocking Cap, Bracket Up, Hole Mount



IS-50NX-C2

Features

- Surge current of 50kA
- Max Power 375W
- Frequency range from 125 MHz to 1000 MHz
- N Female to N Female connectors
- VSWR <1.1:1
- Multi-strike capability
- CE & RoHS compliant

Applications

- HF, UHF and VHF radios
- Ham radios
- Remote industrial monitoring

Description

RF surge protector (also known as lightning arrester or surge arrestor) IS-50NX-C2 from PolyPhaser, integrating a RF blocking capacitor with a gas tube (GT or GDT). This RF surge protector component is manufactured in a coaxial in-line design with wide operating frequency range. All PolyPhaser RF surge protector products are available in stock with same day shipping.

Electrical Specifications

Surge Protector Type	Blocking Cap and Gas Tube
DC Handling	DC Block

Description	Minimum	Typical	Maximum	Units
Frequency Range	125		1,000	MHz
Impedance		50		Ohms
VSWR			1.1:1	
Insertion Loss			0.1	dB
Input Power, CW 375W @ 125 to 220MHz 125W @ 220 to 700MHz 50W @ 700 to 1000MHz			375	Watts
Surge Current IEC 61000-4-5 8/20µs WAVEFORM			50	kA
Turn On Voltage		600 20%		Volts
Throughput Energy @ 3kA, 8/20µs WAVEFORM			220	uJ

Click the following link (or enter part number in “SEARCH” on website) to obtain additional part information including price, inventory and certifications:
Type N F/F Coaxial RF Surge Protector, 125MHz - 1GHz, DC Block, 375W, 220uJ, 50kA, Blocking Cap, Bracket Up, Hole Mount IS-50NX-C2

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Type N F/F Coaxial RF Surge Protector, 125MHz - 1GHz, DC Block, 375W, 220uJ, 50kA, Blocking Cap, Bracket Up, Hole Mount



IS-50NX-C2

Mechanical Specifications

Size

Length	2.45 in [62.23 mm]
Width/Diameter	1.75 in [44.45 mm]
Height	1.5 in [38.1 mm]
Weight	0.3 lbs [136.08 g]

Configuration

Input Connector	N Female
Output Connector	N Female

Environmental Specifications

Temperature

Operating Range	-50 to +50 deg C
Storage Range	-55 to +85 deg C
Ingress Protection (IP) Rating	None
Vibration	1G up to 100Hz

Compliance Certifications

CE Compliant	Yes
UL	UL497E
Certifications	NOM

Plotted and Other Data

Notes:

Click the following link (or enter part number in “SEARCH” on website) to obtain additional part information including price, inventory and certifications:
Type N F/F Coaxial RF Surge Protector, 125MHz - 1GHz, DC Block, 375W, 220uJ, 50kA, Blocking Cap, Bracket Up, Hole Mount IS-50NX-C2

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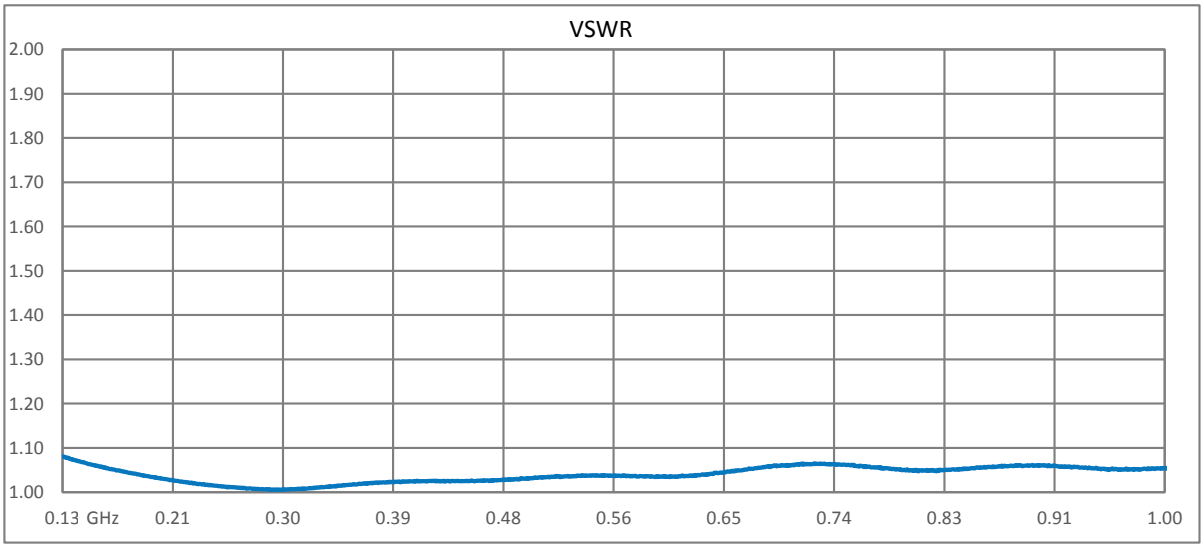
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Type N F/F Coaxial RF Surge Protector, 125MHz - 1GHz, DC Block, 375W, 220uJ, 50kA, Blocking Cap, Bracket Up, Hole Mount



IS-50NX-C2

Typical Performance Data



Click the following link (or enter part number in “SEARCH” on website) to obtain additional part information including price, inventory and certifications:
Type N F/F Coaxial RF Surge Protector, 125MHz - 1GHz, DC Block, 375W, 220uJ, 50kA, Blocking Cap, Bracket Up, Hole Mount IS-50NX-C2

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Type N F/F Coaxial RF Surge Protector, 125MHz - 1GHz, DC Block, 375W, 220uJ, 50kA, Blocking Cap, Bracket Up, Hole Mount



IS-50NX-C2



PolyPhaser protects and increases the reliability of global RF communications networks, including transportation, telecommunications, defense, security and industrial applications, with superior RF surge protection technologies including DC Block, DC Pass and Ultra Low PIM. Backed by responsive service and expert technical support PolyPhaser continually expands its product offering and services to serve engineers' urgent needs for RF components in mission critical communication networks.

Click the following link (or enter part number in "SEARCH" on website) to obtain additional part information including price, inventory and certifications: Type N F/F Coaxial RF Surge Protector, 125MHz - 1GHz, DC Block, 375W, 220uJ, 50kA, Blocking Cap, Bracket Up, Hole Mount IS-50NX-C2

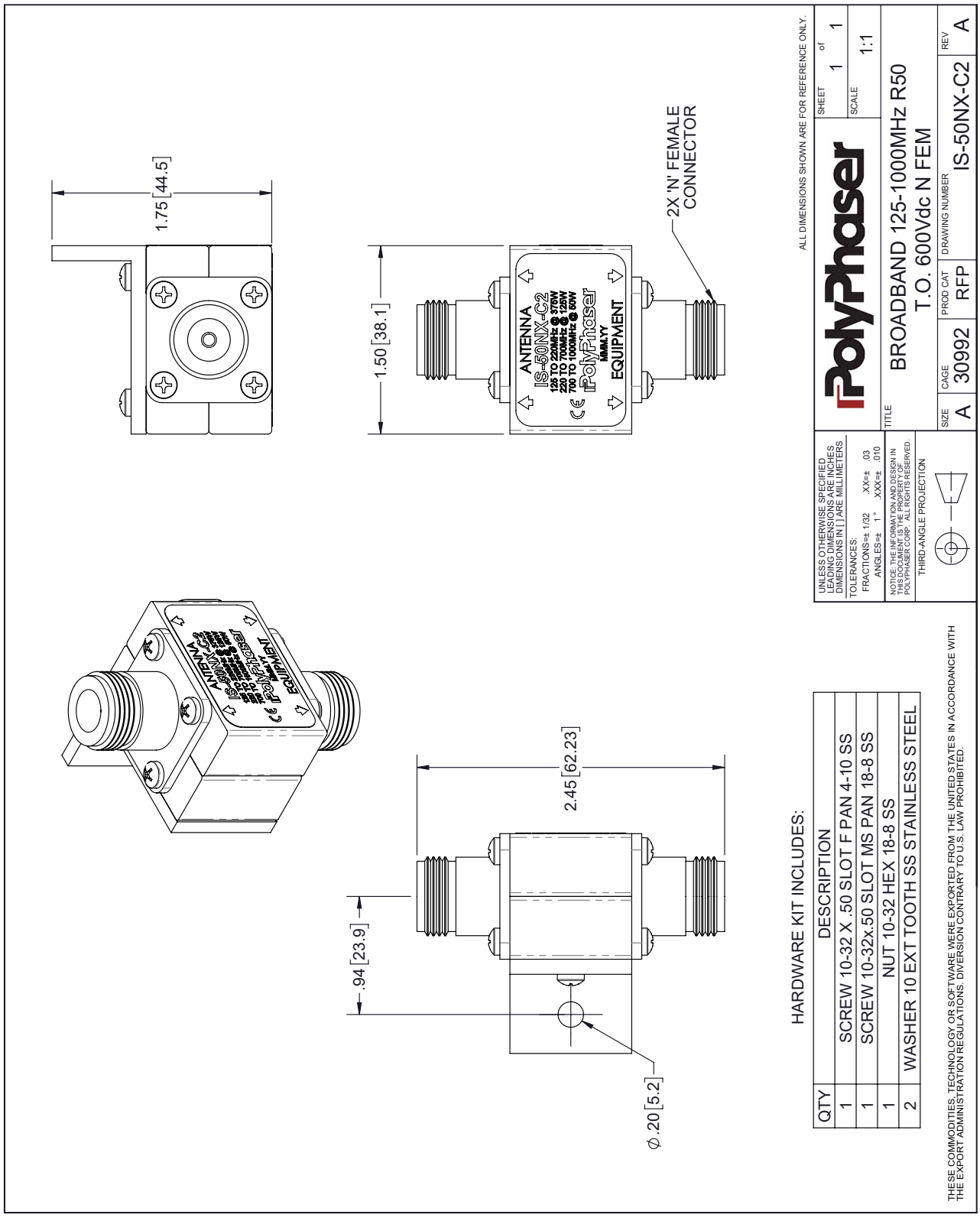
URL: https://www.polyphaser.com/type-n-surge-protector-1ghz-blocking-cap-gas-tube-is-50nx-c2-p.aspx

The information contained in this document is accurate to the best of our knowledge and representative of the part described herein. It may be necessary to make modifications to the part and/or the documentation of the part, in order to implement improvements. PolyPhaser reserves the right to make such changes as required. Unless otherwise stated, all specifications are nominal. PolyPhaser does not make any representation or warranty regarding the suitability of the part described herein for any particular purpose, and PolyPhaser does not assume any liability arising out of the use of any part or documentation.

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Type N F/F Coaxial RF Surge Protector, 125MHz - 1GHz, DC Block, 375W, 220uJ, 50kA, Blocking Cap, Bracket Up, Hole Mount

IS-50NX-C2 CAD Drawing



IS-50NX-C2 REV 1.1 | © 2020 Infinite Electronics, Inc. PolyPhaser is a registered trademark of Infinite Electronics, Inc.

Data Sheet 27

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
1	Mfg: Ventev: SMA Jumper Cable Model Number: LMR195NMSM-6 Tags/Service: CBL1402 / PLC/RTU Control Panel Specifications: • N male to SMA male connector • General purpose fire rating • No fire retardant • Black jacket color • Straight-straight connector body style • Not UV resistant or rated for direct burial • 6ft jumper length



SMA JUMPER CABLE

TAGS: CBL1402

Ventev

6 ft LMR®-195 Cable Assembly with N Male - SMA Male Connectors

GSA **NCNR** **TESSCO SKU:** 322085 **UPC:** 322085 **QTY/UOM:** 1/each

MFG PART #: TRFC-5620-72

Ventev's 6-foot LMR®-195 cable assembly with N Male - SMA Male connectors.

Technical Specs

Connector 1	N Male
Coaxial Cable Series	195

Cable Type	LMR-195
Connector 2	SMA Male
Jumper Length	~~5 - 7.99 ft~~ 6ft
Cable Manufacturer	Times Microwave
Connector 1 Manufacturer	RF Industries
Connector 2 Manufacturer	RF Industries
Fire Rating	General Purpose
Fire Retardant	No
Jacket Color	Black
Low PIM	No
RF Connector Body Style	Straight, Straight
UV Resistant	No
Environment	Indoor/Outdoor
Direct Burial	No

Data Sheet 26

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
1	Mfg: Amphenol: Coaxial Cable Model Number: LDF4-50A Tags/Service: CBL1402A / PLC/RTU Control Panel Specifications: • Black jacket color • 50Ω ± 1Ω cable impedance • 23.104 pF/ft capacitance • 0.451Ω/kft inner conductor resistance • 0.82Ω/kft outer conductor resistance • 0.058μH/ft inductance • 100000 MΩ-km insulation resistance • 1-8800MHz operating frequency band • 40kW peak power • Attenuation values typical, guaranteed within 5%

COAXIAL CABLE

LDF4-50A

TAGS: CBL1402A



LDF4-50A, HELIAX® Low Density Foam Coaxial Cable, corrugated copper, 1/2 in, black PE jacket Halogen free jacketing non-fire-retardant (General propose cable for outdoor use only)

Product Classification

Product Type	Coaxial wireless cable
Product Brand	HELIAX®
Product Series	LDF4-50A
Ordering Note	ANDREW® standard product (Global)

General Specifications

Product Number	520094002/00 | SZ520094902/00
Flexibility	Standard
Jacket Color	Black
Performance Note	Attenuation values typical, guaranteed within 5%

Dimensions

Diameter Over Dielectric	12.954 mm | 0.51 in
Diameter Over Jacket	15.875 mm | 0.625 in
Inner Conductor OD	4.826 mm | 0.19 in
Outer Conductor OD	13.97 mm | 0.55 in
Nominal Size	1/2 in

Electrical Specifications

Cable Impedance	50 ohm ±1 ohm
Capacitance	75.8 pF/m | 23.104 pF/ft
dc Resistance, Inner Conductor	1.48 ohms/km | 0.451 ohms/kft
dc Resistance, Outer Conductor	2.69 ohms/km | 0.82 ohms/kft
dc Test Voltage	4000 V
Inductance	0.19 µH/m | 0.058 µH/ft
Insulation Resistance	100000 MOhms-km
Jacket Spark Test Voltage (rms)	8000 V

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LDF4-50A

Operating Frequency Band	1 – 8800 MHz
Peak Power	40 kW
Velocity	88 %

Attenuation

Frequency (MHz)	Attenuation (dB/100 m)	Attenuation (dB/100 ft)	Average Power (kW)
1.0	0.211	0.064	36.11
1.5	0.259	0.079	29.46
2.0	0.299	0.091	25.5
10.0	0.672	0.205	11.35
20.0	0.954	0.291	7.99
30.0	1.172	0.357	6.51
50.0	1.521	0.463	5.02
85.0	1.995	0.608	3.82
88.0	2.031	0.619	3.76
100.0	2.169	0.661	3.52
108.0	2.256	0.688	3.38
150.0	2.673	0.815	2.85
174.0	2.887	0.88	2.64
200.0	3.103	0.946	2.46
204.0	3.135	0.956	2.43
300.0	3.835	1.169	1.99
400.0	4.462	1.36	1.71
450.0	4.749	1.447	1.61
460.0	4.804	1.464	1.59
500.0	5.021	1.53	1.52
512.0	5.085	1.55	1.5
600.0	5.533	1.686	1.38
700.0	6.009	1.831	1.27
800.0	6.456	1.968	1.18
824.0	6.56	1.999	1.16
894.0	6.855	2.089	1.11
960.0	7.124	2.171	1.07
1000.0	7.284	2.22	1.05
1218.0	8.11	2.472	0.94

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1250.0	8.226	2.507	0.93
1500.0	9.093	2.771	0.84
1700.0	9.744	2.97	0.78
1794.0	10.039	3.06	0.76
1800.0	10.058	3.066	0.76
2000.0	10.666	3.251	0.72
2100.0	10.961	3.341	0.7
2200.0	11.251	3.429	0.68
2300.0	11.535	3.516	0.66
2500.0	12.09	3.685	0.63
2700.0	12.627	3.849	0.6
3000.0	13.407	4.086	0.57
3400.0	14.401	4.389	0.53
3600.0	14.882	4.536	0.51
3700.0	15.118	4.608	0.5
3800.0	15.353	4.679	0.5
3900.0	15.585	4.75	0.49
4000.0	15.815	4.82	0.48
4100.0	16.042	4.889	0.48
4200.0	16.268	4.958	0.47
4300.0	16.492	5.027	0.46
4400.0	16.714	5.094	0.46
4500.0	16.934	5.161	0.45
4600.0	17.153	5.228	0.44
4700.0	17.37	5.294	0.44
4800.0	17.585	5.36	0.43
4900.0	17.798	5.425	0.43
5000.0	18.01	5.489	0.42
6000.0	20.055	6.113	0.38
8000.0	23.826	7.262	0.32
8800.0	25.244	7.694	0.3

Material Specifications

Dielectric Material	Foam PE
Jacket Material	PE

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LDF4-50A

Inner Conductor Material	Copper-clad aluminum wire
Outer Conductor Material	Corrugated copper

Mechanical Specifications

Minimum Bend Radius, multiple Bends	127 mm | 5 in
Minimum Bend Radius, single Bend	50.8 mm | 2 in
Number of Bends, minimum	15
Number of Bends, typical	50
Tensile Strength	113 kg | 249.122 lb
Bending Moment	3.8 N-m | 33.633 in lb
Flat Plate Crush Strength	2 kg/mm | 111.995 lb/in

Environmental Specifications

Installation temperature	-40 °C to +60 °C (-40 °F to +140 °F)
Operating Temperature	-55 °C to +85 °C (-67 °F to +185 °F)
Storage Temperature	-70 °C to +85 °C (-94 °F to +185 °F)
Attenuation, Ambient Temperature	68 °F | 20 °C
Average Power, Ambient Temperature	104 °F | 40 °C
Average Power, Inner Conductor Temperature	212 °F | 100 °C

Packaging and Weights

Cable weight	0.22 kg/m | 0.148 lb/ft

Regulatory Compliance/Certifications

Agency	Classification
CENELEC	EN 50575 compliant, Declaration of Performance (DoP) available
CHINA-ROHS	Below maximum concentration value
ISO 9001:2015	Designed, manufactured and/or distributed under this quality management system
REACH-SVHC	Compliant as per SVHC revision on www.andrew.com/ProductCompliance
ROHS	Compliant
UK-ROHS	Compliant

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Revised: August 12, 2025

Data Sheet 28

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
1	Mfg: Finder: Enclosure Heater Thermostat, NC Model Number: 7T.81.0.000.2403 Tags/Service: THERMOSTAT / PLC/RTU Control Panel Specifications: • 1 NC SPST contact configuration • 10A rated current, 20A maximum peak current • 0.3A breaking capacity at 110VAC • Silver Nickel standard contact material • 0...60°C setting range • 7±4K switch differential • $100 * 10^{13}$ cycle electrical life at 2500VA load • IP20 protection category • −45...80°C ambient temperature range
1	Mfg: SCE: Enclosure Heater Model Number: SCE-TSH75 Tags/Service: HTR1206 / PLC/RTU Control Panel Specifications: • Positive temperature coefficient application • Designed for protection against condensation and corrosion • 35mm DIN rail clip for mounting

	• 120VAC supply voltage • Touch safe, with 2 screw terminals for standard 14 AWG wire

ENCLOSURE HEATER THERMOSTAT, NC

finder
RELAYING INNOVATION

TAGS: THERMOSTAT

Panel Thermo-Hygrostat and Thermostats

7T
SERIES

- Drying kilns
- Industrial refrigeration
- Road / tunnel lighting
- Industrial furnaces and ovens
- Automatic car-washes
- Panels for electrical distribution
- Control panels
- Forced-air ventilators



Panel Thermo-Hygrostat

- Small, compact size (17.5 mm wide)
- Electronic control
- 4 functions
- Nominal voltage 110…240 V AC/DC
- Temperature ranges from +10 ° to +60°C
- Humidity range up to 90%
- LED status indication contact ON
- 35 mm rail (EN 60715) mount

Panel thermostat

- Small, compact size (17.5 mm wide)
- Snap action themostatic Bimetal sensor
- Wide temperature setting range
- Long electrical life
- 35 mm rail (EN 60715) mount

7T.51/7T.81
Box clamp



		NEW 7T.51	7T.81.0.000.240x	7T.81.0.000.230x
				
		• Thermo-Hygrostatic control • Nominal voltage 110…240 V AC/DC	• Heating control	• Ventilation control
Contact specification				
Contact configuration		1 NO (SPST-NO)	1 NC (SPST-NC)	1 NO (SPST-NO)
Rated current/Maximum peak current	A	10/15	10/20	10/20
Rated voltage/ Maximum switching voltage	V AC	250/400	250/250	250/250
Rated load AC1	VA	2500	2500	2500
Rated load AC15 (230 V AC)	VA	500	250	250
Single phase motor rating AC3 (230 V AC)	kW	—	1.1	1.1
Breaking capacity DC1: 24/110/220 V	A	10/0.3/0.12	1/0.3/0.15	1/0.3/0.15
Minimum switching load	mW (V/mA)	500 (12/10)	500 (12/10)	500 (12/10)
Standard contact material		AgNi	AgNi	AgNi
Coil specifications				
Nominal Voltage	V AC/DC	110…240	—	—
Rated Power	VA (50Hz)/W	1.8/0.44	—	—
Operating range	V AC/DC	88…264	—	—
Temperature specifications *				
Setting range (ventilation)	°C	+10…+60	~~–20…+40~~ ~~–20…+60~~ 0…+60	–20…+40 –20…+60 0…+60
Switch differential	K	4 ± 2	7 ± 4	7 ± 4
Setting accuracy full range	K	–1…+3	—	—
Humidity specifications **				
Setting range (humidity)	%	50…90	—	—
Hysteresis	%	4 ± 2	—	—
Setting accuracy	%	5	—	—
Technical data				
Electrical life at rated load AC1	cycles	$100 \cdot 10^3$	$100 \cdot 10^3$	$100 \cdot 10^3$
Ambient temperature range	°C	–25…+60	–45…+80	–45…+80
Protection category		IP 20	IP 20	IP 20
Approvals (according to type)		CE UKCA EAC cULus	CE UKCA EAC 03 cULus	

* Measured with 0.3 K/min
** Measured with 0.5 %/min

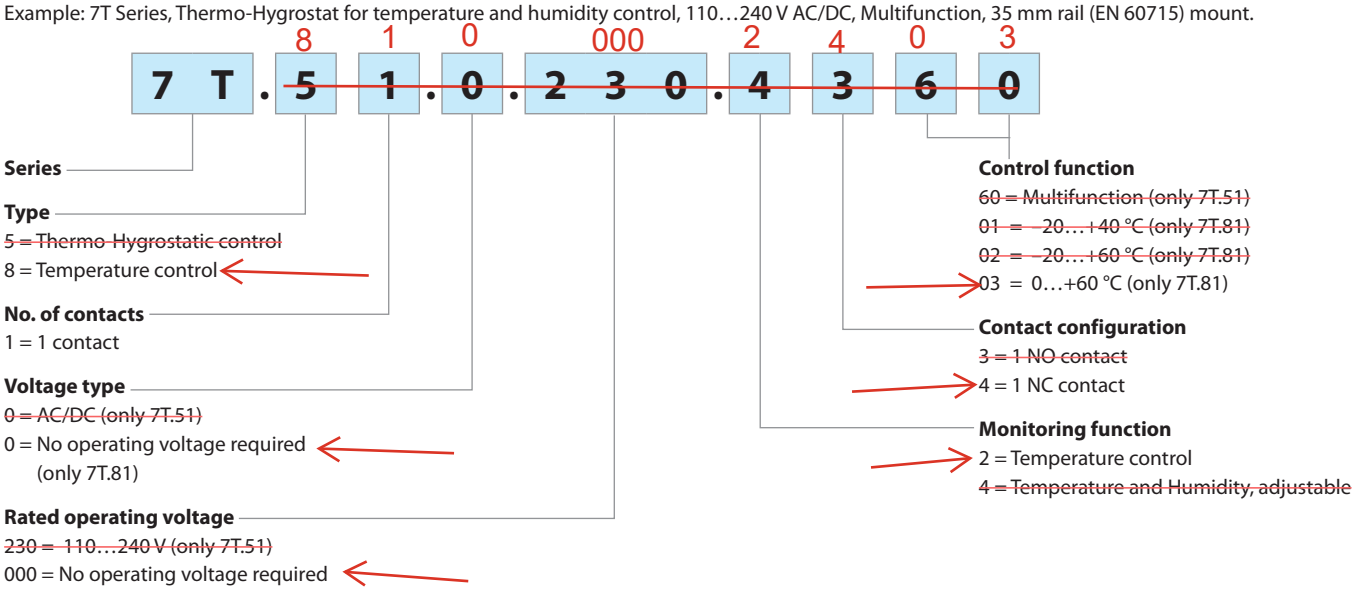
For outline drawing see page 6

G

XI-2025, www.findernet.com

Ordering information

Example: 7T Series, Thermo-Hygrostat for temperature and humidity control, 110…240 V AC/DC, Multifunction, 35 mm rail (EN 60715) mount.

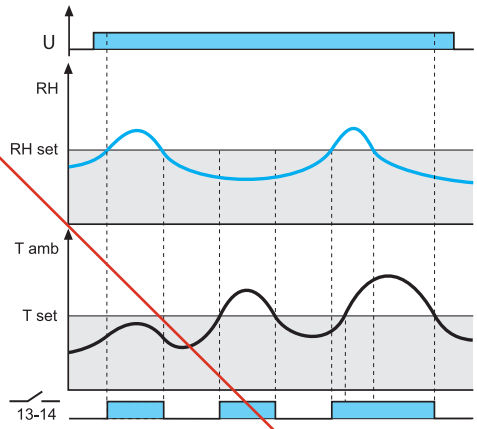


Technical data

Insulation		7T.51	7T.81
Dielectric strength between open contacts	V AC	1000	500
Dielectric strength between supply and contact	V AC	2000	—
Other data		**7T.51**	**7T.81**
Vibration resistance	g	—	16
Shock resistance	g	—	20
		7T.51/7T.81	
Screw torque	Nm	0.5	0.5
Max. wire size		solid cable	stranded cable
	mm²	1 x 4 / 2 x 2.5	1 x 4 / 2 x 2.5
	AWG	1 x 12 / 2 x 14	1 x 12 / 2 x 14
Wire strip length	mm	9	9

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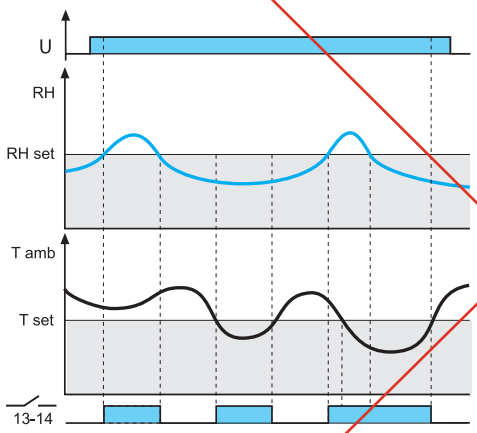
Functions 7T.51



HT: RH > RHset or Tamb > Tset

Power is permanentely applied to the thermo-hygrostat.
The contact 13-14 closes if the ambient humidity (RH) is > of set humidity (RHset) or if the ambient temperature (Tamb) is > of set temperature (Tset).

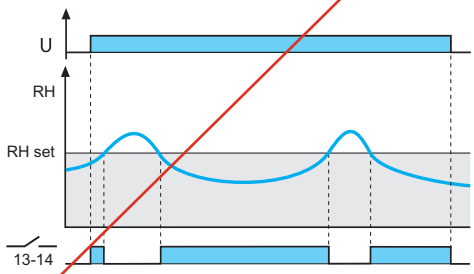
When contact is close, LED is ON



TH: RH > RHset or Tamb < Tset

The contact 13-14 closes if the ambient humidity (RH) is > of set humidity (RHset) or if the ambient temperature (Tamb) is < of set temperature (Tset).

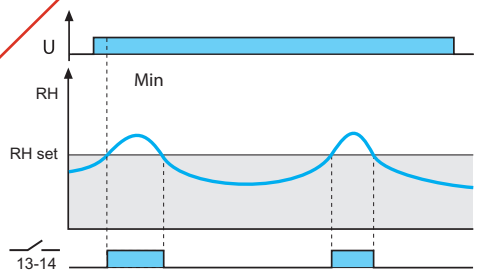
When contact is close, LED is ON



HL: RH < RHset

The contact 13-14 closes if the ambient humidity (RH) is < of set humidity (RHset)

When contact is close, LED is ON

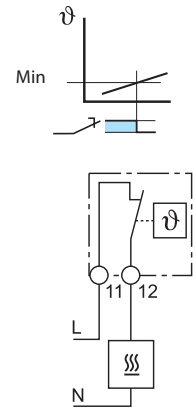


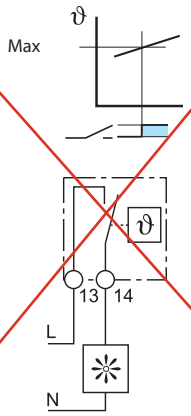
HM: RH > RHset

The contact 13-14 closes if the ambient humidity (RH) is > of set humidity (RHset)

When contact is close, LED is ON

Functions 7T.81





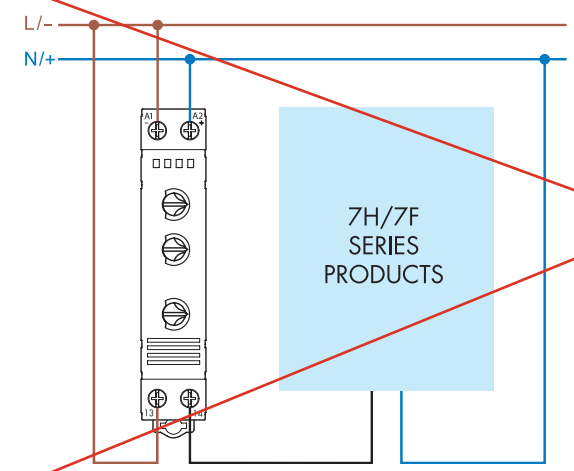
Heating control - Should the panel temperature fall below the (minimum) set temperature the contact will close to call for heat.
The contact will open when this set temperature is exceeded.

Ventilation control - Should the panel temperature exceed the (maximum) set temperature then the contact will close to call for cooling.
The contact will open when the temperature falls below this set temperature.

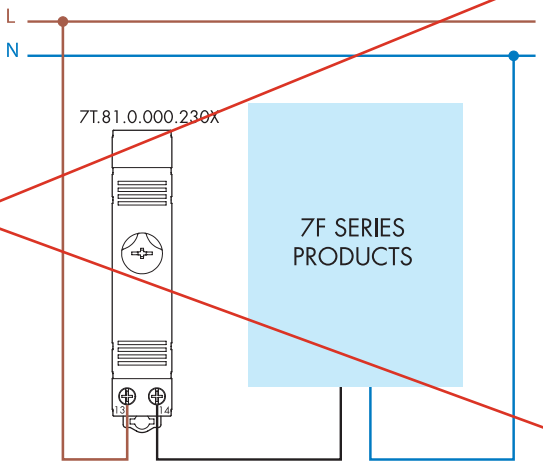
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Wiring diagrams

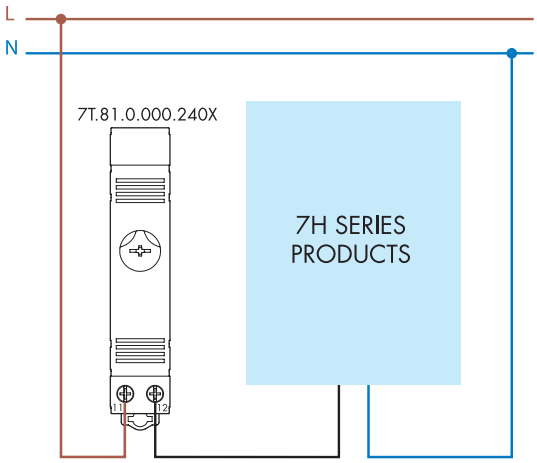
Type 7T.51



Type 7T.81…230x

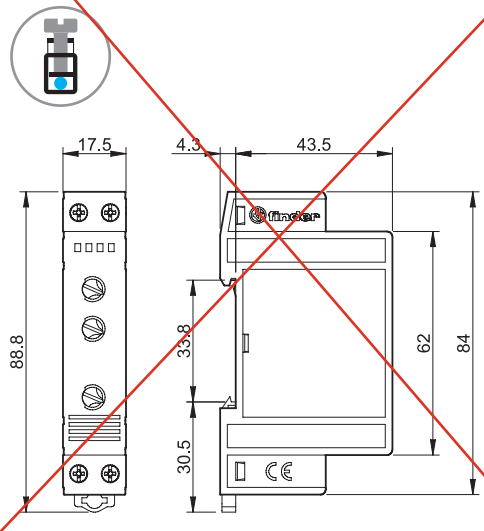


Type 7T.81…240x

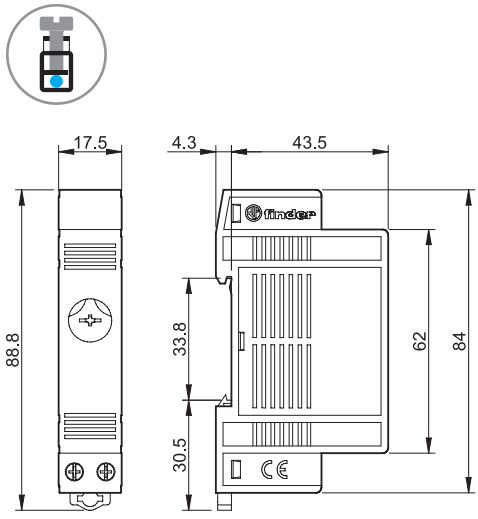


Outline drawings

Type 7T.51
Box clamp



Type 7T.81
Box clamp



Please see general technical information

XI-2025, www.findernet.com

SAGINAW CONTROL & ENGINEERING
Your Enclosure Source ®

ENCLOSURE HEATER

TAGS: HTR1207

Saginaw Control and Engineering
95 Midland Road Saginaw, MI 48638-5770
(800) 234-6871 - Fax: (989) 799-4524
SCE@SaginawControl.com

SCE-TSH75



Product Specifications:

Part Number: SCE-TSH75
Description: Heater - 75W
Height: 5.51"
Width: 2.76"
Depth: 2.48"
Price Code: P1
List Price: $182.71
Catalog Page: 402
Est. Ship Weight: 0.86 lbs
Max. Current : 4 Amp
Model Number: 501075

Application
PTC (Positive Temperature Coefficient)
Din rail clip design for 35mm Din Rail. Designed for protection from low temperatures, condensation and corrosion. Touch Safe with 2 screw terminal for standard AWG 14 wire. 110-240V AC.

For low temperatures management use with thermostat

For condensate management no thermostat is necessary - continuous heat is most effective see installaton Instruction for sizing.

Industry Standards - (IS24)
- UL Component Recognized

Notes
UL File #E358386

Optional Accessories
SCE-TEMD Thermostat (Dual)
SCE-TEMNC Thermostat (Normally Closed)

Similar Part Numbers
SCE-TSH100Heater - 100W
SCE-TSH150Heater - 150W
SCE-TSH25Heater - 25W
SCE-TSH50Heater - 50W

Installation Information
- Touch Safe Heaters

Data Sheet 29

Customer: Reference: Date:	Johnston Construction Charles County Pomonkey Pump Station Improvements 11/11/2026
Qty	Description
A/R	Mfg: Phoenix Contact: Terminal Block Model Number: 3044076
A/R	Mfg: Phoenix Contact: Grounding Terminal Block Model Number: 3044092
A/R	Mfg: Phoenix Contact: Terminal End Barrier Model Number: 3047028
A/R	Mfg: Phoenix Contact: Terminal Anchor Model Number: 0800886 Tags / Service: TB-AC, TB-AI, TB-AO, TB-DC, TB-DI, TB-DO / PLC/RTU Control Panel Specifications: • Feed through and grounding terminals • Screw clamps • End plates and anchors • DIN rail mount

TAGS: TB-AC, TB-AI, TB-AO, TB-DC, TB-DI, TB-DO

Extract from the online catalog

UT 2,5

Order No.: 3044076



http://eshop.phoenixcontact.de/phoenix/treeViewClick.do?UID=3044076

Universal terminal block, Type of connection: Screw connection, Screw connection, Cross section: 0.14 mm² - 4 mm², AWG 26 - 12, Width: 5.2 mm, Color: gray, Mounting type: NS 35/7,5, NS 35/15

Product notes

WEEE/RoHS-compliant since: 01/01/2003

http://www.download.phoenixcontact.com
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Commercial data	
EAN	4017918960377
Pack	50 pcs.
Customs tariff	85369010
Weight/Piece	0.008227 KG
Catalog page information	Page 26 (CL-2009)

Technical data

General

Number of levels	1
Number of connections	2
Color	gray

Insulating material	PA
Inflammability class acc. to UL 94	V0

Dimensions

Width	5.2 mm
Length	47.7 mm
Height NS 35/7,5	47.5 mm
Height NS 35/15	55 mm

Technical data

Maximum load current	32 A (with 4 mm² conductor cross section)
Rated surge voltage	8 kV
Pollution degree	3
Surge voltage category	III
Insulating material group	I
Connection in acc. with standard	IEC 60947-7-1
Nominal current I_N	24 A
Nominal voltage U_N	1000 V
Open side panel	ja

Connection data

Conductor cross section solid min.	0.14 mm²
Conductor cross section solid max.	4 mm²
Conductor cross section stranded min.	0.14 mm²
Conductor cross section stranded max.	4 mm²
Conductor cross section AWG/kcmil min.	26
Conductor cross section AWG/kcmil max	12
Conductor cross section stranded, with ferrule without plastic sleeve min.	0.14 mm²
Conductor cross section stranded, with ferrule without plastic sleeve max.	2.5 mm²
Conductor cross section stranded, with ferrule with plastic sleeve min.	0.14 mm²
Conductor cross section stranded, with ferrule with plastic sleeve max.	2.5 mm²
2 conductors with same cross section, solid min.	0.14 mm²
2 conductors with same cross section, solid max.	1.5 mm²
2 conductors with same cross section, stranded min.	0.14 mm²

2 conductors with same cross section, stranded max.	1.5 mm²
2 conductors with same cross section, stranded, TWIN ferrules with plastic sleeve, min.	0.5 mm²
2 conductors with same cross section, stranded, TWIN ferrules with plastic sleeve, max.	1.5 mm²
2 conductors with same cross section, stranded, ferrules without plastic sleeve, min.	0.14 mm²
2 conductors with same cross section, stranded, ferrules without plastic sleeve, max.	1.5 mm²
Type of connection	Screw connection
Stripping length	9 mm
Internal cylindrical gage	A3
Screw thread	M3
Tightening torque, min	0.6 Nm
Tightening torque max	0.8 Nm

Certificates / Approvals

Certification	CB, CSA, CUL, DNV, GL, GOST, LR, UL, VDE-PZI
Certification Ex:	IECEx, KEMA-EX

CSA

Nominal voltage U_N	600 V
Nominal current I_N	20 A
AWG/kcmil	26-12

CUL

Nominal voltage U_N	600 V
Nominal current I_N	20 A
AWG/kcmil	26-12

UL

Nominal voltage U_N	600 V
Nominal current I_N	20 A
AWG/kcmil	26-12

Extract from the online catalog

UT 2,5-PE

Order No.: 3044092



http://eshop.phoenixcontact.de/phoenix/treeViewClick.do?UID=3044092

Feed-through modular terminal block, Type of connection: Screw connection, Screw connection, Cross section: 0.14 mm² - 4 mm², AWG 26 - 12, Width: 5.2 mm, Color: green-yellow, Mounting type: NS 35/7,5, NS 35/15

Product notes

WEEE/RoHS-compliant since: 01/01/2003

Commercial data	
EAN	4017918960360
Pack	50 pcs.
Customs tariff	85369010
Weight/Piece	0.012156 KG
Catalog page information	Page 32 (CL-2009)

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Technical data

General

Number of levels	1
Number of connections	2
Color	green-yellow

Insulating material	PA
Inflammability class acc. to UL 94	V0

Dimensions

Width	5.2 mm
Length	47.7 mm
Height NS 35/7,5	47.5 mm
Height NS 35/15	55 mm

Technical data

Rated surge voltage	8 kV
Pollution degree	3
Surge voltage category	III
Insulating material group	I
Connection in acc. with standard	IEC 60947-7-2
Open side panel	ja

Connection data

Conductor cross section solid min.	0.14 mm²
Conductor cross section solid max.	4 mm²
Conductor cross section stranded min.	0.14 mm²
Conductor cross section stranded max.	4 mm²
Conductor cross section AWG/kcmil min.	26
Conductor cross section AWG/kcmil max	12
Conductor cross section stranded, with ferrule without plastic sleeve min.	0.14 mm²
Conductor cross section stranded, with ferrule without plastic sleeve max.	2.5 mm²
Conductor cross section stranded, with ferrule with plastic sleeve min.	0.14 mm²
Conductor cross section stranded, with ferrule with plastic sleeve max.	2.5 mm²
2 conductors with same cross section, solid min.	0.14 mm²
2 conductors with same cross section, solid max.	1.5 mm²
2 conductors with same cross section, stranded min.	0.14 mm²
2 conductors with same cross section, stranded max.	1.5 mm²

2 conductors with same cross section, stranded, TWIN ferrules with plastic sleeve, min.	0.5 mm²
2 conductors with same cross section, stranded, TWIN ferrules with plastic sleeve, max.	1.5 mm²
2 conductors with same cross section, stranded, ferrules without plastic sleeve, min.	0.14 mm²
2 conductors with same cross section, stranded, ferrules without plastic sleeve, max.	1.5 mm²
Type of connection	Screw connection
Stripping length	9 mm
Internal cylindrical gage	A3
Screw thread	M3
Tightening torque, min	0.6 Nm
Tightening torque max	0.8 Nm

Certificates / Approvals

Certification	CB, CSA, CUL, DNV, GL, LR, UL, VDE-PZI
Certification Ex:	IECEx, KEMA-EX

CSA

AWG/kcmil	26-12

CUL

AWG/kcmil	26-12

UL

AWG/kcmil	26-12

Extract from the online catalog

D-UT 2,5/10

Order No.: 3047028



http://eshop.phoenixcontact.de/phoenix/treeViewClick.do?UID=3047028

End cover, Length: 47.7 mm, Width: 2.2 mm, Height: 48.4 mm, Color: gray

Commercial data	
EAN	4017918960346
Pack	50 pcs.
Customs tariff	85389099
Weight/Piece	0.00248 KG
Catalog page information	Page 26 (CL-2009)

Product notes

WEEE/RoHS-compliant since: 01/01/2003

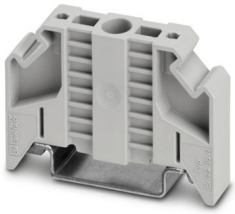
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E/NS 35 N - End bracket

0800886

https://www.phoenixcontact.com/us/products/0800886

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End clamp, width: 9.5 mm, color: gray

Your advantages

- Large-surface marking

Commercial data

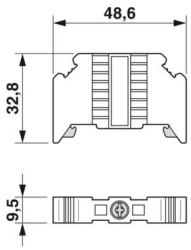
Item number	0800886
Packing unit	50 pc
Minimum order quantity	50 pc
Sales key	BE07
Product key	BE7111
GTIN	4017918129309
Weight per piece (including packing)	14.16 g
Weight per piece (excluding packing)	13.708 g
Customs tariff number	39269097
Country of origin	DE

Technical data

Connection data

Screw thread	M3
Tightening torque	0.5 ... 0.8 Nm

Dimensions

Dimensional drawing	
Width	9.5 mm
Height	48.6 mm
Depth	32.8 mm
Depth on NS 35/7,5	34.3 mm
Length	48.6 mm

Material specifications

Color	gray (RAL 7042)
Material	PA
Flammability rating according to UL 94	V0
Static insulating material application in cold	-60 °C
Temperature index of insulation material (DIN EN 60216-1 (VDE 0304-21))	125 °C
Relative insulation material temperature index (Elec., UL 746 B)	130 °C

Environmental and real-life conditions

Ambient conditions

Ambient temperature (operation)	-60 °C ... 105 °C (max. short-term operating temperature RTI Elec.)
Ambient temperature (storage/transport)	-25 °C ... 60 °C (for a short time, not exceeding 24 h, -60 °C to +70 °C)
Ambient temperature (assembly)	-5 °C ... 70 °C
Ambient temperature (actuation)	-5 °C ... 70 °C
Permissible humidity (storage/transport)	30 % ... 70 %

Mounting

Mounting type	NS 35/7,5
	NS 35/15

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info@phoenixcon.com

32384 CHARLES COUNTY POMONKEY PUMP STATION IMPROVEMENTS

Heating and Cooling Sizes:

Device	HEAT DISSIPATED (BTU/Hr)
PLC	95.5
Ethernet Switch	20.5
24VDC Power Supplies	72
PLC Power Supply	29.0
Total	217

Product selection result

Project

- Title: Charles County Pomonkey PLC Panel
- Create date: Mar 11, 2026
- Amend date: Mar 25, 2026

User

- Company:
- Name:
- Address:
- City:

Enclosure

Type	Wall Mount Enclosure
Height	1,524 mm
Width	914 mm
Depth	406 mm

Wall

Material	Stainless Steel
k	3.7 W/m²K
Insulation	R0 = 0 mm
Enclosure Color	Gray

Location

for Outdoor enviroments / NEMA Type 3R/4/4x

Single housing, free-standing on all sides

Wind speed	No Wind
Surface according to VDE 0660 part 600	4.1 m²

Temperature

Min. ambient temperature	-18 °C
Max. ambient temperature	38 °C
Min. admissible temperature	-10 °C
Max. admissible temperature	55 °C

Rated connection value

Rated voltage:	115 V
Frequency	60 Hz

Dissipation based on manual input 64 W

Switch cabinet acclimatisation

Total dissipation in enclosure	64 W
Dissipation due to solar radiation	0 W
Heat Dissipation from environment	0 W
Heat Dissipation to environment	261 W
Required cooling capacity	**0 W**
Required heating capacity	116 W

Product selection result

Warning

Ensure there is sufficient clearance for air flow around the cooling unit to prevent a so-called air short-circuit. There should be a minimum distance of approximately 200 mm (7.9 inches) to the next component in the housing and 400 mm (15.7 inches) outside the housing

32384 CHARLES COUNTY POMONKEY PUMP STATION IMPROVEMENTS

Drawings Index:

Drawing Number	Drawing Description
D32384-10	Pomonkey Pump Station Improvements PLC/RTU Control Panel EXT. Layout
D32384-11	Pomonkey Pump Station Improvements PLC/RTU Control Panel INT. Layout
D32384-12	Pomonkey Pump Station Improvements 120VAC Wiring Diagram
D32384-13	Pomonkey Pump Station Improvements 120VAC Wiring Diagram
D32384-14	Pomonkey Pump Station Improvements 24VDC Wiring Diagram
D32384-15	Pomonkey Pump Station Improvements Digital Input Wiring Diagram
D32384-16	Pomonkey Pump Station Improvements Digital Input Wiring Diagram
D32384-17	Pomonkey Pump Station Improvements Digital Input Wiring Diagram
D32384-18	Pomonkey Pump Station Improvements Analog Input Wiring Diagram
D32384-19	Pomonkey Pump Station Improvements Analog Output Wiring Diagram

32384 CHARLES COUNTY POMONKEY PUMP STATION IMPROVEMENTS

Drawings:

WARNING NAMEPLATE DETAIL

S-L CONTROLS, DWG D32384-10
VOLTAGE: 120VAC, 60HZ, SINGLE PHASE
AMPS: 0.93A

SHORT CIRCUIT CURRENT: N/A

ENCLOSURE RATING: UL TYPE 4X

CAUTION: MULTIPLE POWER SOURCES ARE PROVIDED FROM FIELD TERMINALS. EACH SOURCE SHOULD BE DISCONNECTED FOR SERVICING.

WARNING: TO MAINTAIN OVERCURRENT, SHORT CIRCUIT, AND GROUND FAULT PROTECTION, THE MANUFACTURER'S INSTRUCTIONS FOR SELECTING CURRENT ELEMENTS MUST BE FOLLOWED

NAMEPLATES SIZED AS REQUIRED

NAME PLATE LEGEND TABLE

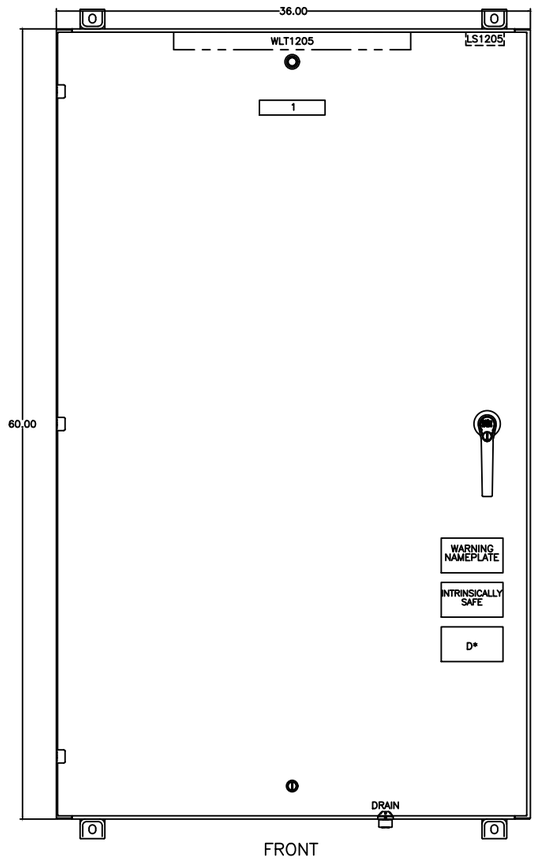
ITEM	QTY	DESCRIPTION
1	1	PLC/RTU / CONTROL PANEL
2	1	CONTROL POWER
3	1	ALARM TEST
4	1	FU1501, FU1520, FU1601: MERSEN MI5HF25V0.5/100 0.5A FU1758, FU1759, FU1760, FU1761, FU1762, FU1763, FU1764, FU1765: MERSEN GGM3 3A
5	1	FU1401, FU1405, FU1407: MERSEN MI5HF25V2/100 2A

NOTES:
ALL PLATES SHALL BE ENGRAVED ON WHITE BACKGROUND WITH BLACK CHARACTERS. LETTER HEIGHT SHOULD BE MINIMUM 1/4"

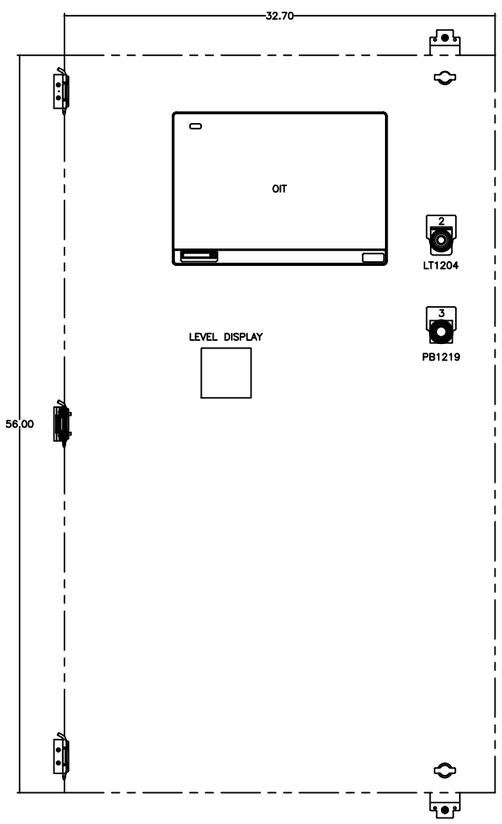
WIRING:
1) □ – INDICATES CABINET TERMINAL BLOCK
2) DASHED LINES INDICATE WIRING REMOTE TO CONTROL PANEL.
3) PROVIDE 20% SPARE TERMINALS
4) UL APPROVED NO. TO BE PROVIDED AT TIME OF MANUFACTURING
5) ELECTRICAL CONNECTION TORQUES:
   TERMINAL: 12 INCH POUNDS
   CIRCUIT BREAKER: 45 INCH POUNDS

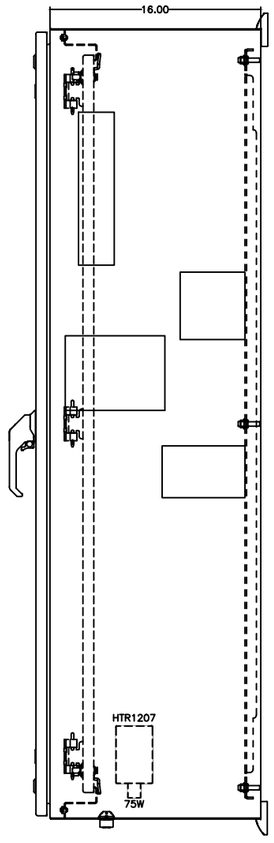
ADDITIONAL PANEL LABELS:

A* = "INTRINSICALLY SAFE FIELD WIRING TERMIANLS"
B* = "WARNING – EXPLOSION HAZARD – SUBSTITUTION OF COMPONENTS MAY IMPAIR INTRINSIC SAFETY"
C* = "INTRINSICALLY SAFE CIRCUITS" (TO BE PLACED BY CONTRACTOR AT CONDUIT OPENING)
D* = "WARNING – EXPLOSION HAZARD, TO PREVENT IGNITION OF FLAMMABLE OR COMBUSTIBLE ATMOSPHERES, DISCONNECT POWER BEFORE SERVICING.



FRONT VIEW



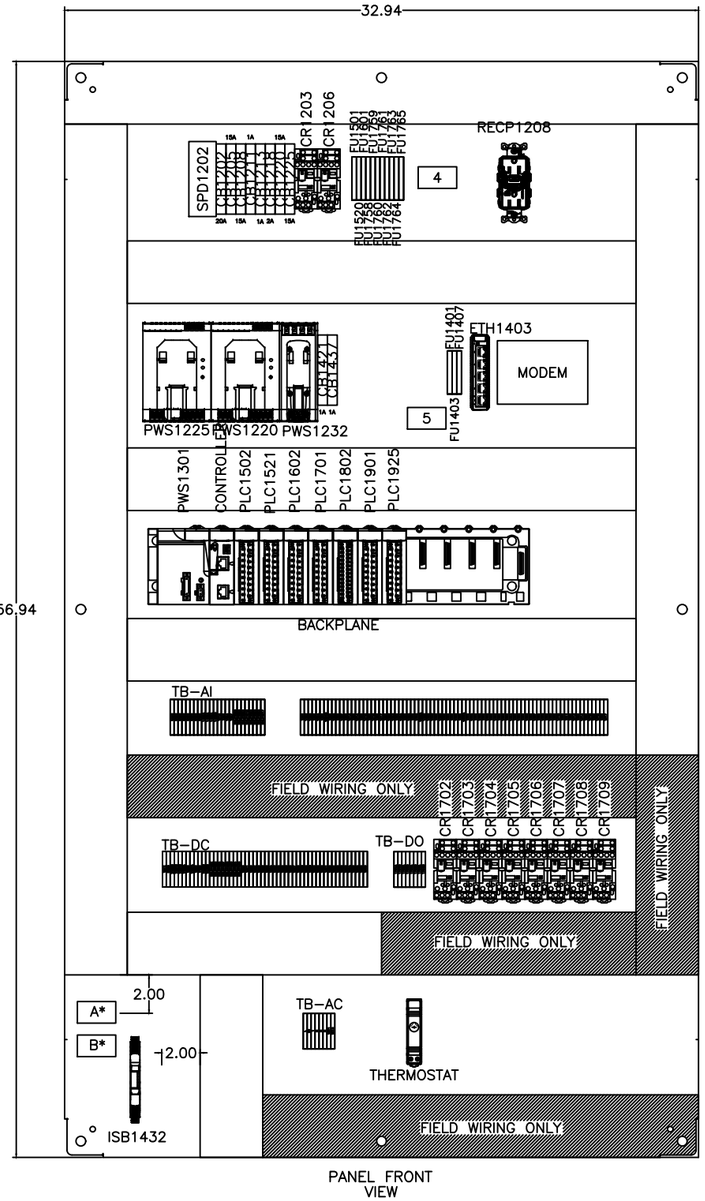


RIGHT SIDE VIEW

PANEL WIRING

AREA	VOLTAGE	NOUN NAME	ID	GAGE	COLOR	TYPE
INTERNAL	120VAC	HOT	(1), (1A), (X1)	14	BLACK	MTW
INTERNAL	120VAC	NEUTRAL	(N)	14	WHITE	MTW
CONTROL	120VAC	HOT	VARIOUS	16	RED	MTW
CONTROL	120VAC	NEUTRAL	N	16	WHITE	MTW
CONTROL	DC	DC+	VARIOUS	16	BLUE	MTW
CONTROL	DC	DC–	VARIOUS	16	BLUE W/ WHITE STRIPE	MTW
EXTERNALLY ENERGIZED		HOT	VARIOUS	16	YELLOW	MTW
ANALOG	24VDC	4–20MA	VARIOUS	18	BLACK/RED	TSP

TOLERANCES	REVISIONS			POMONKEY PUMP STATION IMPROVEMENTS PLC/RTU CONTROL PANEL EXT. LAYOUT		
	NO.	DATE	BY	S-L CONTROLS		
DECIMAL	1					
	2					
FRACTIONAL	3					
	4					
ANGULAR	5					
	6					
				DRAWN BY ZBB	SCALE NONE	MATERIAL
				CH'D MN	DATE 11-11-2026	DRAWING NUMBER 32384-10
				TRACED	APP'D	



TAGS	QTY	CATALOG	DESCRIPTION	MANUFACTURER
ENC	1	SCE-60EL3616SS6LPPL	NEMA 4X WALL MOUNT ENCLOSURE	SCE
ENC	1	SCE-DF60EL36LP	DEAD FRONT PANEL	SCE
ENC	1	SCE-60P36	ENCLOSURE BACK PANEL	SCE
ENC	1	SCE-ELMFK4	WALL MOUNT FOOT KIT	SCE
DRAIN	1	SCE-BVKD	NEMA 4X VENT/DRAIN	SCE
WLT1205	1	26741	LED LIGHT FIXTURE	JASCO
LT1204	1	9001SKT38LWW31	PTT LIGHT, 120VAC LED, WHITE	SCHNEIDER ELECTRIC
PB1219	1	9001SKR1B	PUSHBUTTON, BLACK	SCHNEIDER ELECTRIC
PB1219	1	9001KA2	1 NO CONTACT	SCHNEIDER ELECTRIC
SPD1202	1	DS42S-120/G	TRANS. VOLTAGE SURGE SUPRESSOR	CITEL
PWS1220, PWS1225	2	2866776	24VDC POWER SUPPLY	PHOENIX CONTACT
PWS1232	1	2320157	24VDC POWER SUPPLY REDUNDANCY MODULE	PHOENIX CONTACT
CB1202	1	7940102584	CIRCUIT BREAKER, 120VAC, 20A	WEIDMULLER
CB1211, CB1213	2	7940125379	CIRCUIT BREAKER, 120VAC, 1A	WEIDMULLER
CB1205, CB1208, CB1220, CB1225	4	7940102582	CIRCUIT BREAKER, 120VAC, 15A	WEIDMULLER
CB1218	1	7940102572	CIRCUIT BREAKER, 120VAC, 2A	WEIDMULLER
CB1421, CB1437	2	7940102624	CIRCUIT BREAKER, 24VDC, 1A	WEIDMULLER
RECP1208	1	GFRST15W	DUPLEX GFCI RECEPTACLE, 15A	HUBBEL
CR1203, CR1206, CR1702, CR1703, CR1704, CR1705, CR1706, CR1707, CR1708, CR1709	10	RXM2AB2F7	MINIATURE RELAY, 120VAC, 8 BLADE	SCHNEIDER ELECTRIC
CR1203, CR1206, CR1702, CR1703, CR1704, CR1705, CR1706, CR1707, CR1708, CR1709	10	RXZE2M114	MINIATURE RELAY SOCKET, 120VAC, DIN	SCHNEIDER ELECTRIC
LEVEL DISPLAY	1	RIA452-C211A11A	DIGITAL PROCESS METER	E+H
ANTENNA	1	T09060010006HM	CELLULAR ANTENNA	VENTEV
ETH1403	1	EDS-205	UNMANAGED ETHERNET SWITCH, 5 PORT	MOXA
FU1403, FU1407	2	MI5HF25V2/100	MINIATURE FUSE, FAST ACTING, 2A	MERSEN
FU1501, FU1520, FU1601, FU1920	4	MI5HF25V0.5/100	MINIATURE FUSE, FAST ACTING, 0.5A	MERSEN
FU1758, FU1759, FU1760, FU1761, FU1762, FU1763, FU1764, FU1765	8	GGM3	FAST BLOW FUSE, 3A	MERSEN
FU1403, FU1407, FU1501, FU1520, FU1601, FU1758, FU1759, FU1760, FU1761, FU1762, FU1763, FU1764, FU1765, FU1920	14	3046100	FUSE TERMINAL BLOCK, 120VAC	PHOENIX CONTACT
LS1205	1	ALFSWD	REMOTE DOOR SWITCH	HOFFMAN
PWS1301	1	BMXCPS3500H	POWER SUPPLY MODULE	MODICON
OIT	1	HMIDT752	OPERATOR INTERFACE TERMINAL	SCHNEIDER ELECTRIC
OIT	1	HMIG2U	DISPLAY BASE UNIT	SCHNEIDER ELECTRIC
CONTROLLER	1	BMXP342020H	PROCESSOR	MODICON
BACKPLANE	1	BMXXBP1200H	BACKPLANE, 12 SLOTS	MODICON
PLC1502, PLC1521, PLC1602, PLC1701, PLC1901, PLC1925	6	BMXFTB2000	TERMINAL BLOCK, 20 PIN	MODICON
PLC1802	1	BMXFTB2800	TERMINAL BLOCK, 28 PIN	MODICON
PLC1502, PLC1521, PLC1602	3	BMXDAI1604H	DISCRETE INPUT MODULE, 120VAC	MODICON
PLC1701	1	BMXDRA0815H	DISCRETE OUTPUT MODULE	MODICON
PLC1802	1	BMXAMI0810H	ANALOG INPUT MODULE	MODICON
PLC1901	1	BMXAMO0410H	ANALOG OUTPUT MODULE	MODICON
PLC1925	1	BMXDDI1602H	DISCRETE INPUT MODULE, 24VDC	MODICON
ISB1432	1	2865366	IS BARRIER / SPLITTER	PHOENIX CONTACT
SP1402	1	IS-50NX-C2	COAXIAL RF SURGE PROTECTOR	POLYPHASER
CBL1402	1	LMR195NMSM-6	SMA JUMPER CABLE	VENTEV
CBL1402A	1	LDF4-50A	COAXIAL CABLE	AMPHENOL
THERMOSTAT	1	7T.81.0.000.2403	ENCLOSURE HEATER THERMOSTAT, NC	FINDER
HTR1207	1	SCE-TSH75	ENCLOSURE HEATER	SCE
TB-AC, TB-AI, TB-AO, TB-DC, TB-DI, TB-DO	163	3044076	UNIVERSAL TERMINAL BLOCK – UT 2,5	PHOENIX CONTACT
TB-AC, TB-AI, TB-AO, TB-DC, TB-DI, TB-DO	A/R	3047028	TERMINAL END BARRIER	PHOENIX CONTACT
TB-AC, TB-AI, TB-AO, TB-DC, TB-DI, TB-DO	A/R	3044092	GROUNDING TERMINAL BLOCK	PHOENIX CONTACT
TB-AC, TB-AI, TB-AO, TB-DC, TB-DI, TB-DO	A/R	0800886	TERMINAL ANCHOR	PHOENIX CONTACT

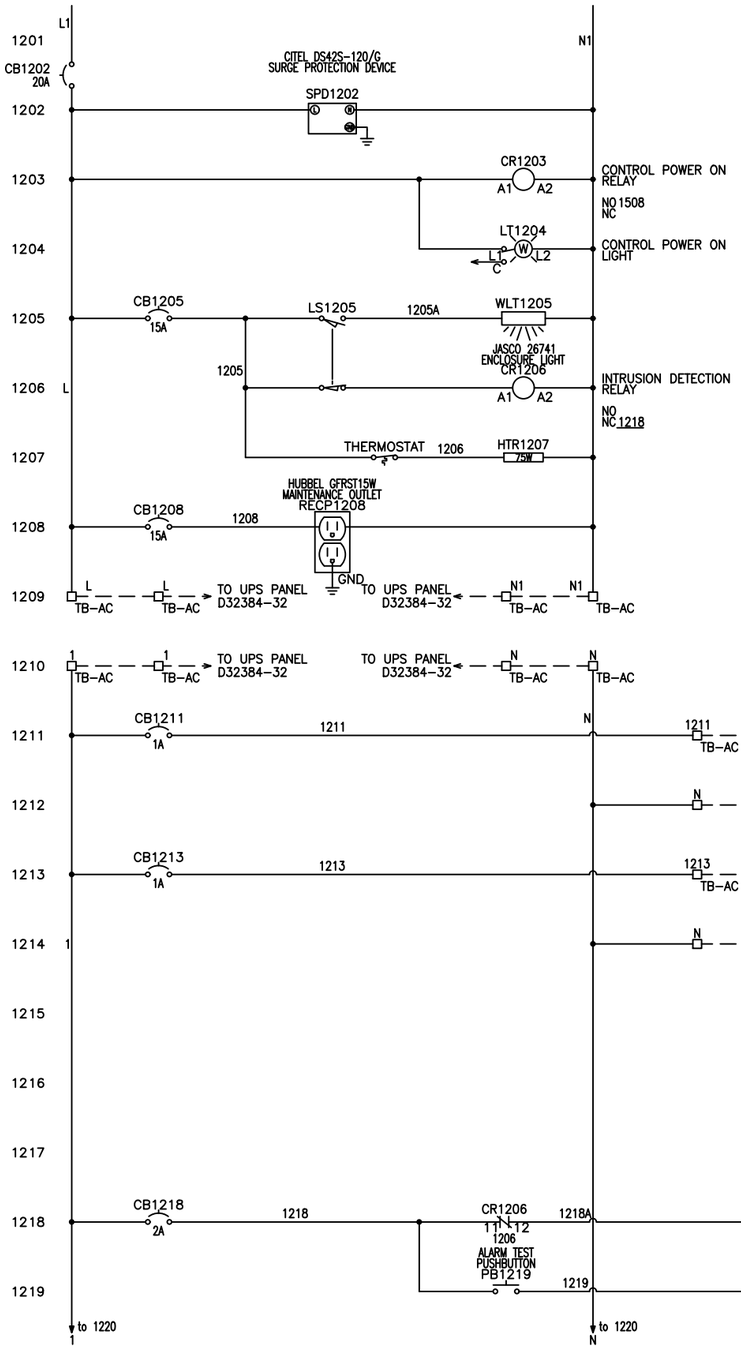
TOLERANCES
DECIMAL
FRACTIONAL
ANGULAR

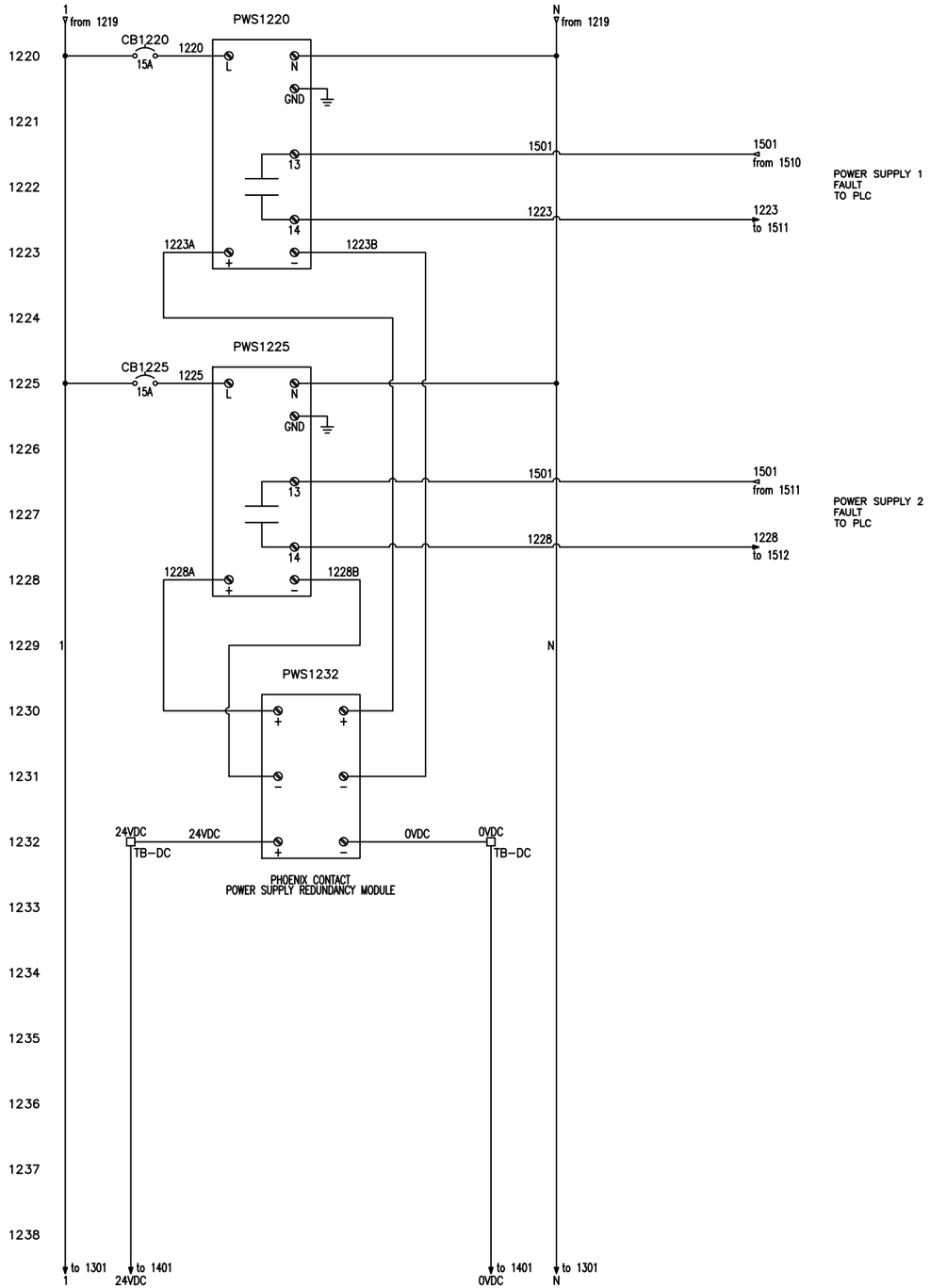
REVISIONS		
NO.	DATE	BY
1		
2		
3		
4		
5		
6		

POMONKEY PUMP STATION IMPROVEMENTS
PLC/RTU CONTROL PANEL INT. LAYOUT

S4 CONTROLS

DRAWN BY ZBB	SCALE NONE	MATERIAL
CHK'D MN	DATE 11-11-2026	DRAWING NUMBER 32384-11
TRACED	APP'D	



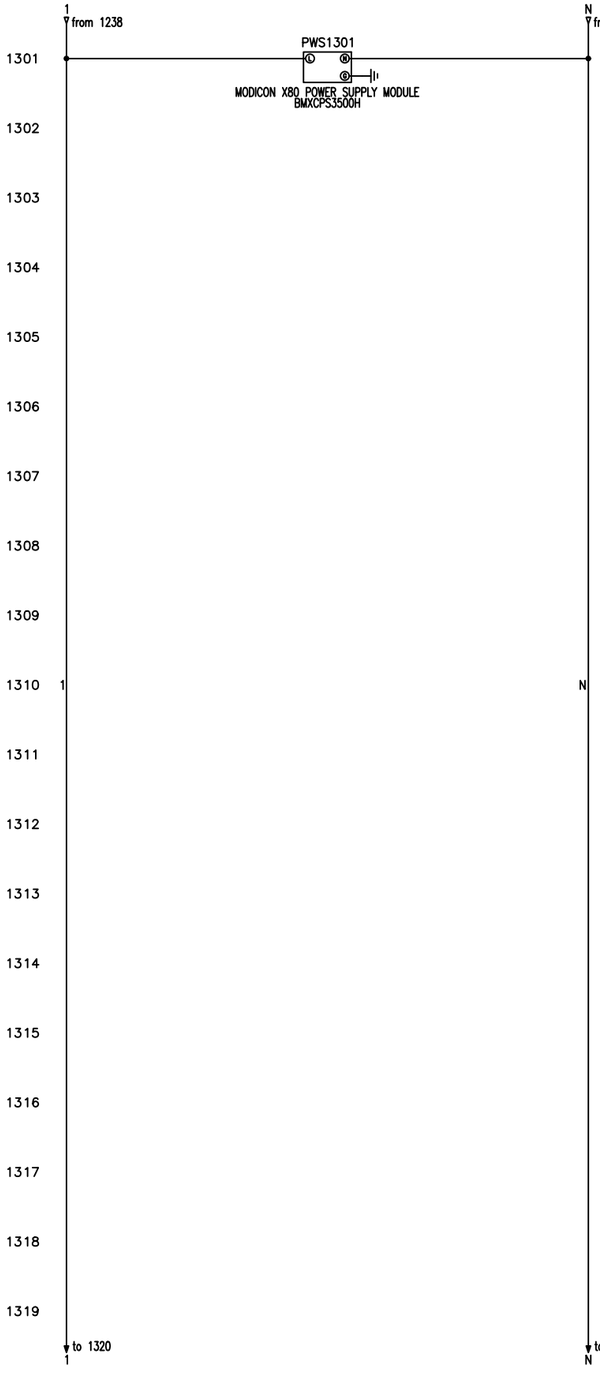


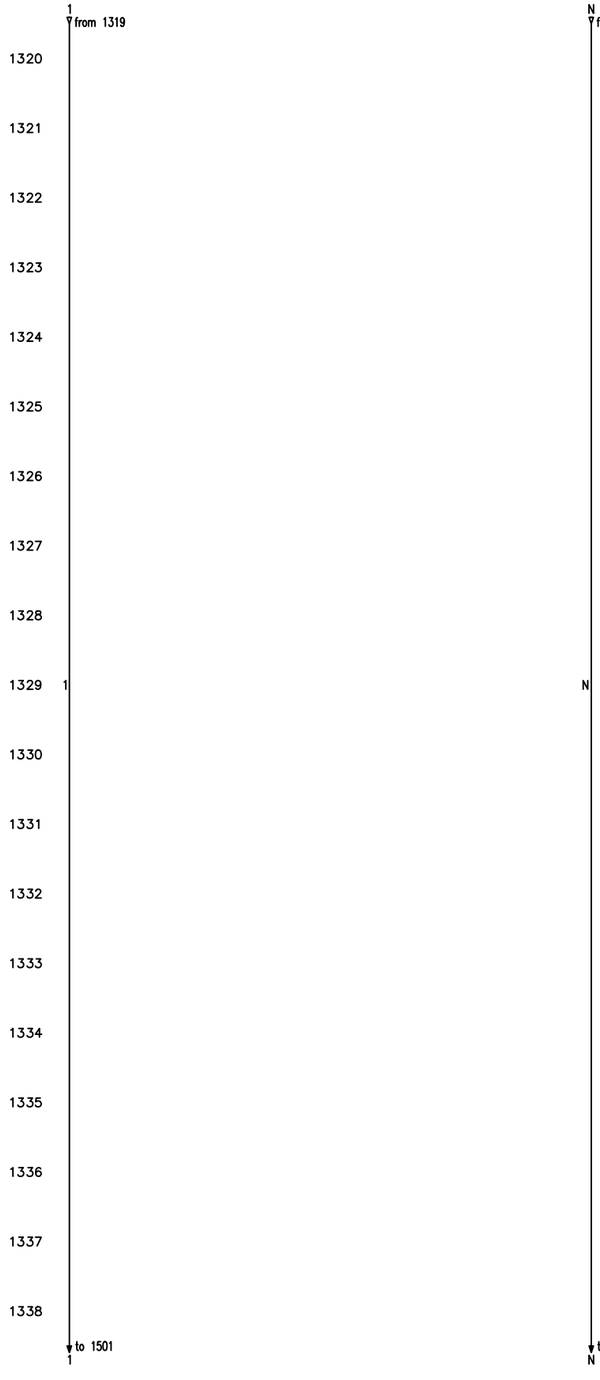
TOLERANCES	REVISIONS		
	NO.	DATE	BY
DECIMAL	1		
	2		
FRACTIONAL	3		
	4		
ANGULAR	5		
	6		

POMONKEY PUMP STATION IMPROVEMENTS
120VAC WIRING DIAGRAM

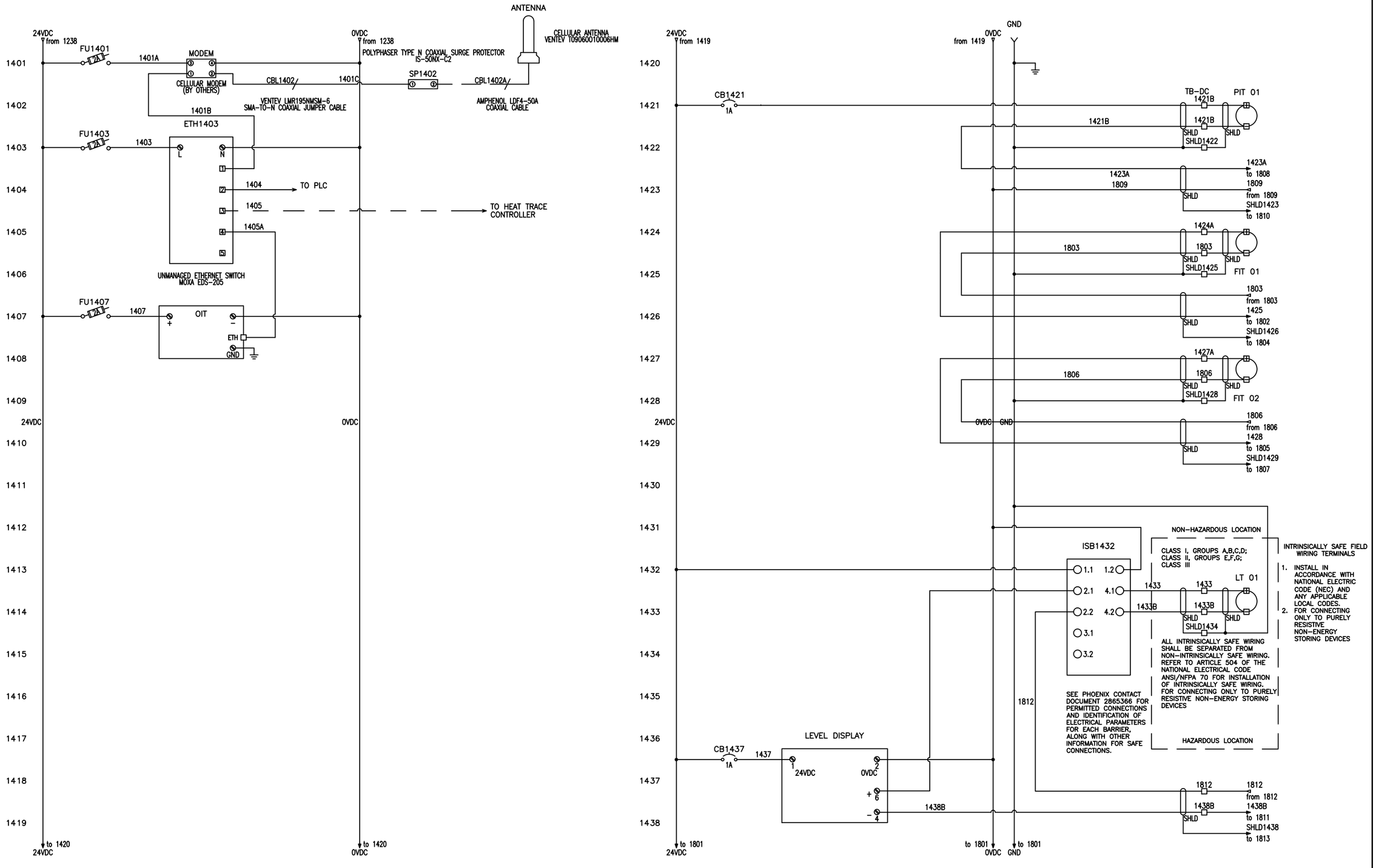
S-L CONTROLS

DRAWN BY ZBB	SCALE NONE	MATERIAL
CHK'D MN	DATE 11-11-2026	DRAWING NUMBER 32384-12
TRACED	APP'D	

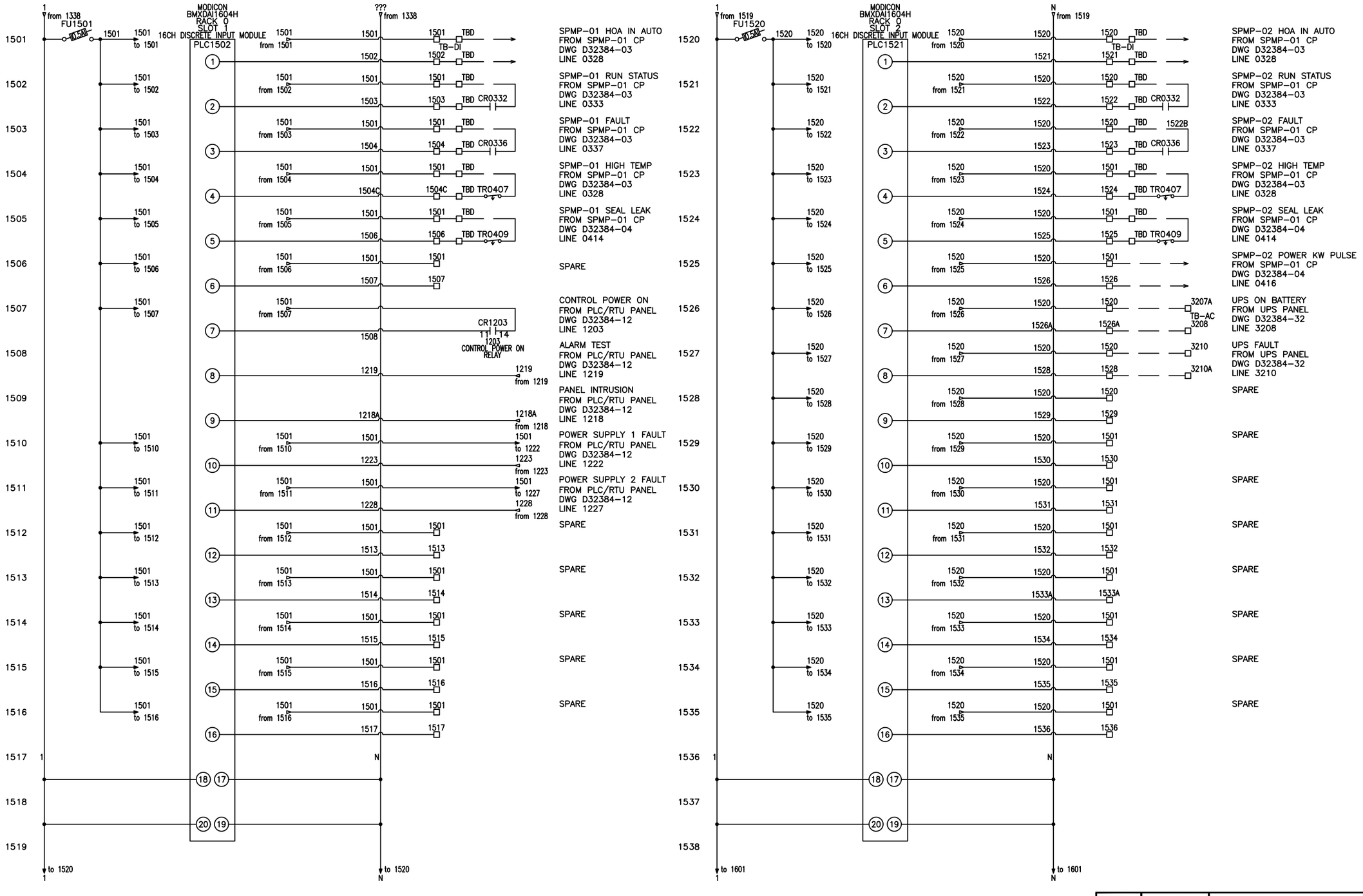




TOLERANCES	REVISIONS	POMONKEY PUMP STATION IMPROVEMENTS	
DECIMAL	NO. DATE BY	120VAC WIRING DIAGRAM	
FRACTIONAL	1 2 3 4 5 6	S-L CONTROLS	
ANGULAR		DRAWN BY ZBB / SCALE NONE / MATERIAL	
		CHK'D MN / DATE 11-11-2026 / DRAWING NUMBER 32384-13	
		TRACED / APP'D	



TOLERANCES	REVISIONS			POMONKEY PUMP STATION IMPROVEMENTS 24VDC WIRING DIAGRAM		
	NO.	DATE	BY	S-L CONTROLS		
DECIMAL	1			DRAWN BY ZBB	SCALE NONE	MATERIAL
	2			CHK'D MN	DATE 11-11-2026	DRAWING NUMBER
FRACTIONAL	3			TRACED	APP'D	32384-14
	4					
ANGULAR	5					
	6					

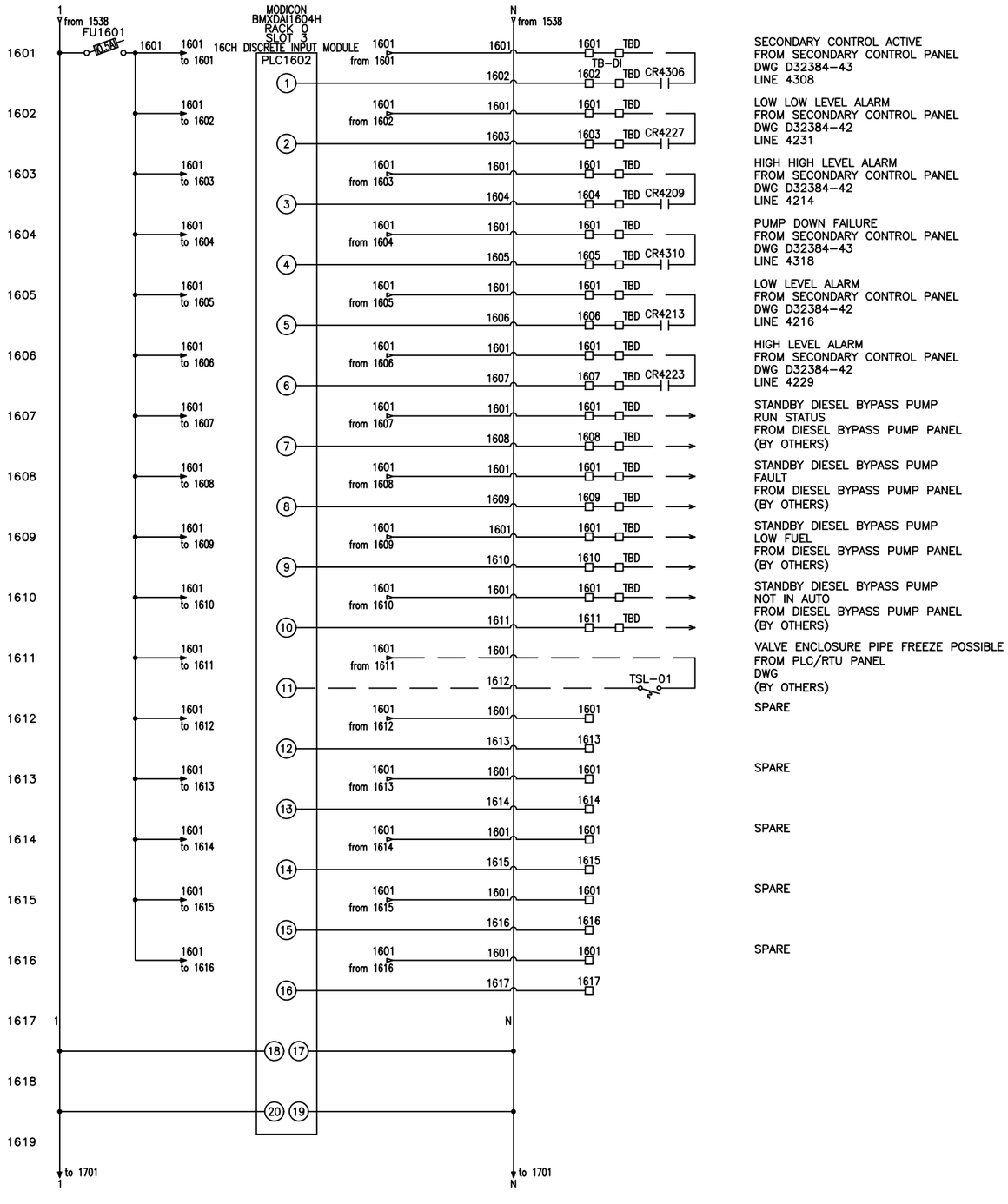


TOLERANCES	REVISIONS NO.	DATE	BY
DECIMAL	1		
	2		
FRACTIONAL	3		
	4		
ANGULAR	5		
	6		

POMONKEY PUMP STATION IMPROVEMENTS
DIGITAL INPUT WIRING DIAGRAM

SJ CONTROLS

DRAWN BY ZBB	SCALE NONE	MATERIAL
CHK'D MN	DATE 11-11-2026	DRAWING NUMBER 32384-15
TRACED	APP'D	



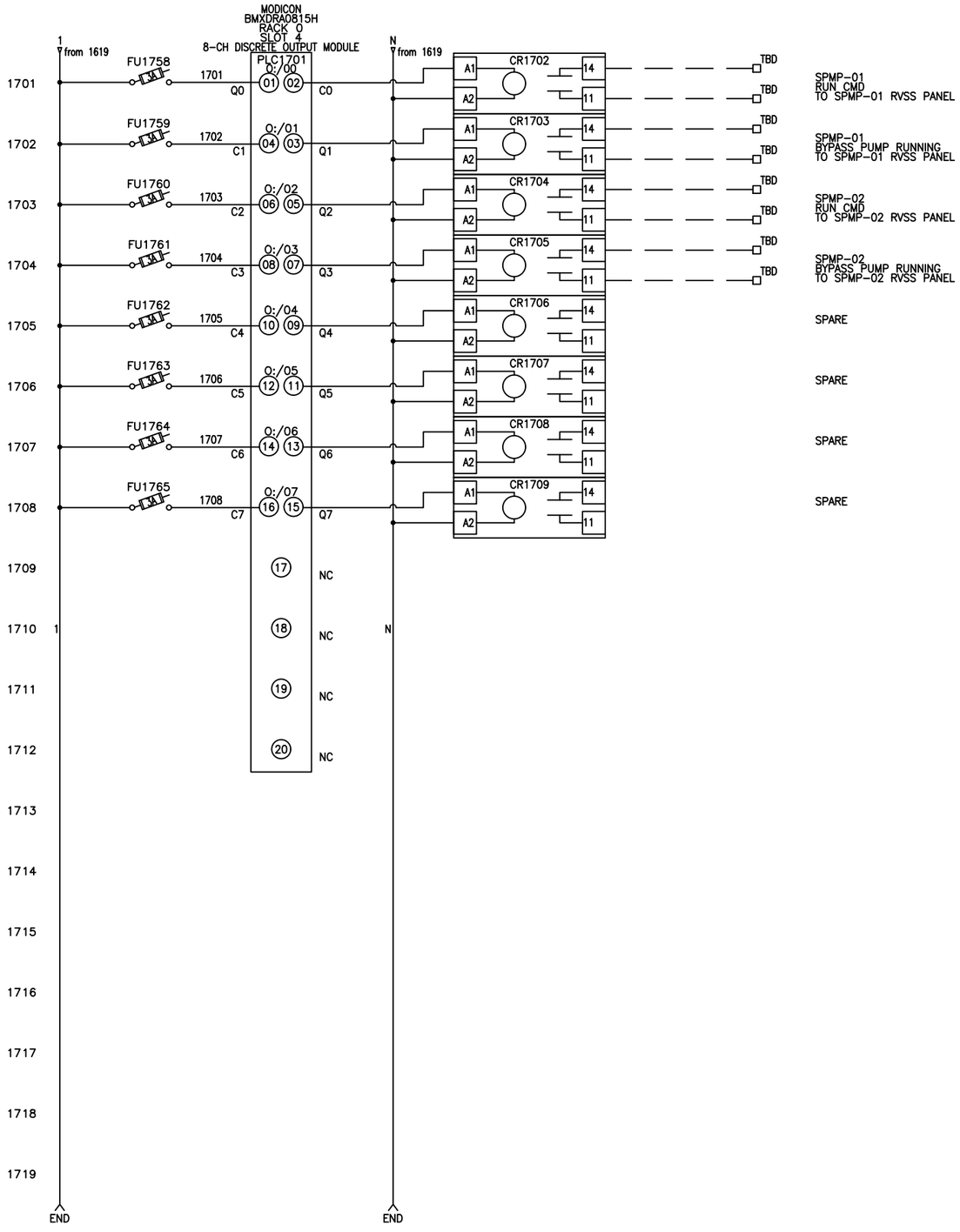
1601	SECONDARY CONTROL ACTIVE FROM SECONDARY CONTROL PANEL DWG D32384-43 LINE 4308
1602	LOW LOW LEVEL ALARM FROM SECONDARY CONTROL PANEL DWG D32384-42 LINE 4231
1603	HIGH HIGH LEVEL ALARM FROM SECONDARY CONTROL PANEL DWG D32384-42 LINE 4214
1604	PUMP DOWN FAILURE FROM SECONDARY CONTROL PANEL DWG D32384-43 LINE 4318
1605	LOW LEVEL ALARM FROM SECONDARY CONTROL PANEL DWG D32384-42 LINE 4216
1606	HIGH LEVEL ALARM FROM SECONDARY CONTROL PANEL DWG D32384-42 LINE 4229
1607	STANDBY DIESEL BYPASS PUMP RUN STATUS FROM DIESEL BYPASS PUMP PANEL (BY OTHERS)
1608	STANDBY DIESEL BYPASS PUMP FAULT FROM DIESEL BYPASS PUMP PANEL (BY OTHERS)
1609	STANDBY DIESEL BYPASS PUMP LOW FUEL FROM DIESEL BYPASS PUMP PANEL (BY OTHERS)
1610	STANDBY DIESEL BYPASS PUMP NOT IN AUTO FROM DIESEL BYPASS PUMP PANEL (BY OTHERS)
1611	VALVE ENCLOSURE PIPE FREEZE POSSIBLE FROM PLC/RTU PANEL DWG (BY OTHERS)
1612	SPARE
1613	SPARE
1614	SPARE
1615	SPARE
1616	SPARE

TOLERANCES	REVISIONS		
	NO.	DATE	BY
DECIMAL	1		
	2		
FRACTIONAL	3		
	4		
ANGULAR	5		
	6		

POMONKEY PUMP STATION IMPROVEMENTS
DIGITAL INPUT WIRING DIAGRAM

SJL CONTROLS

DRAWN BY ZBB	SCALE NONE	MATERIAL
CHK'D MN	DATE 11-11-2026	DRAWING NUMBER 32384-16
TRACED	APP'D	

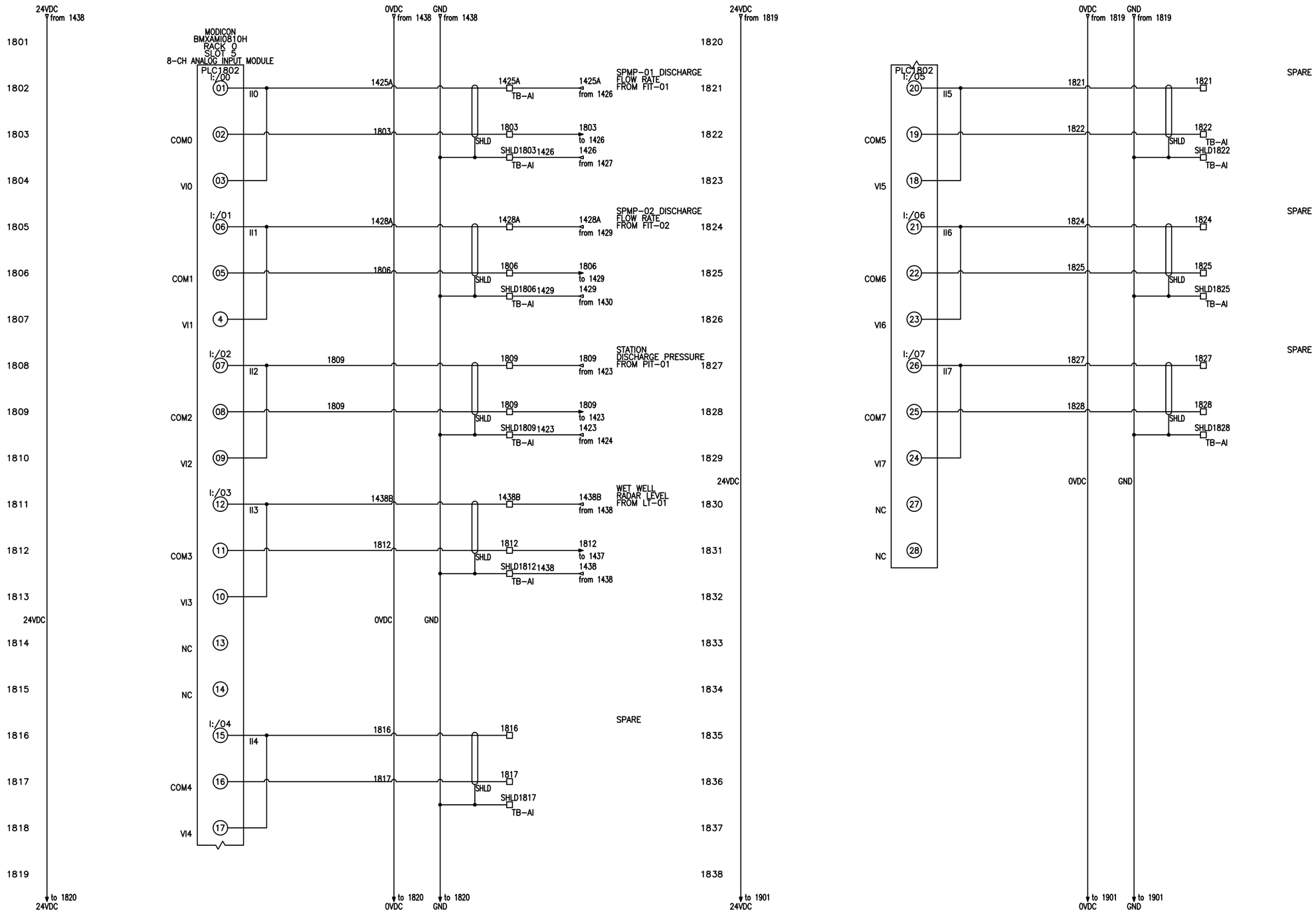


TOLERANCES	REVISIONS		
	NO.	DATE	BY
DECIMAL	1		
	2		
FRACTIONAL	3		
	4		
ANGULAR	5		
	6		

POMONKEY PUMP STATION IMPROVEMENTS
DIGITAL OUTPUT WIRING DIAGRAM

SJ CONTROLS

DRAWN BY ZBB	SCALE NONE	MATERIAL
CHK'D MN	DATE 11-11-2026	DRAWING NUMBER 32384-17
TRACED	APP'D	

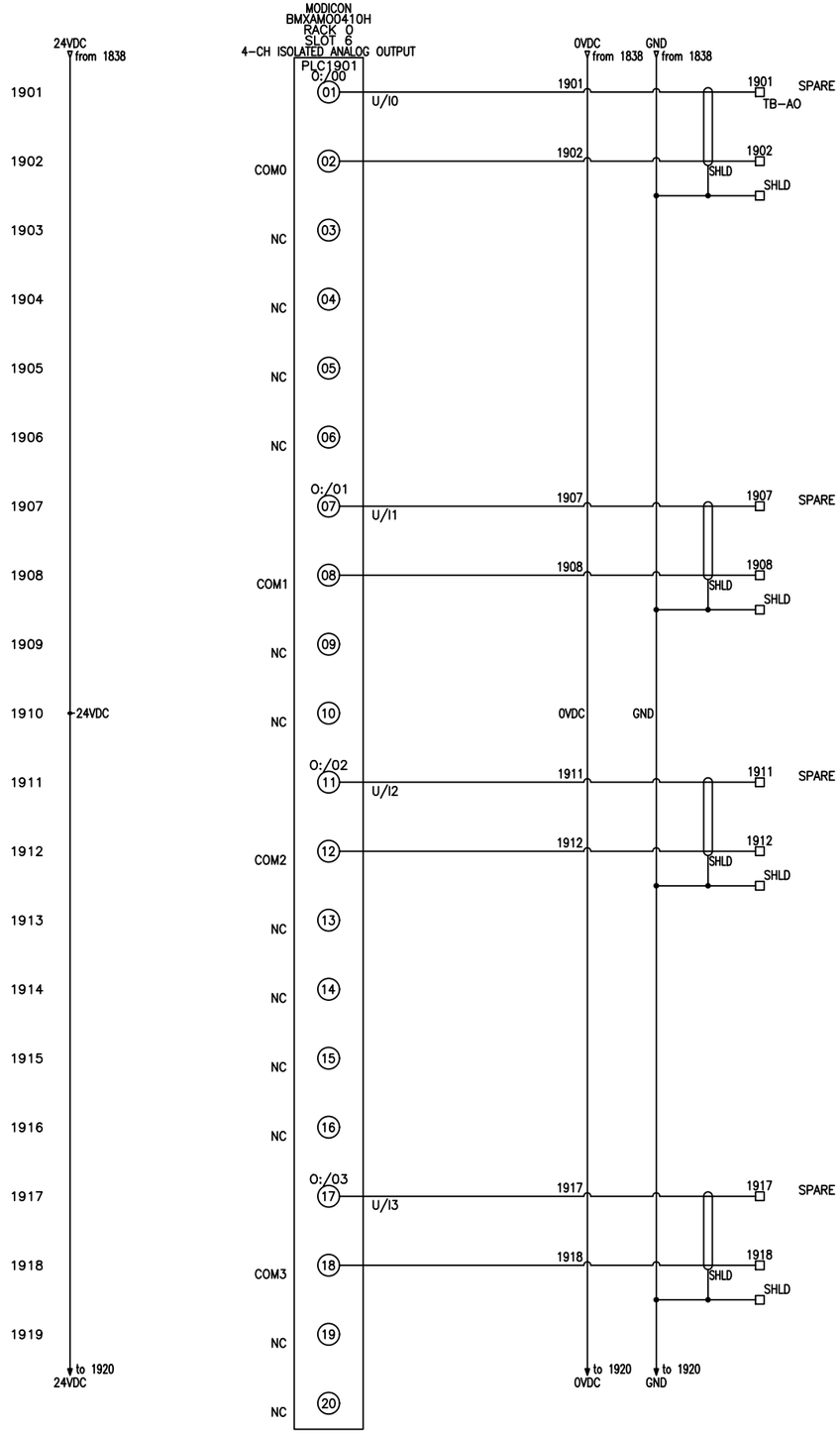


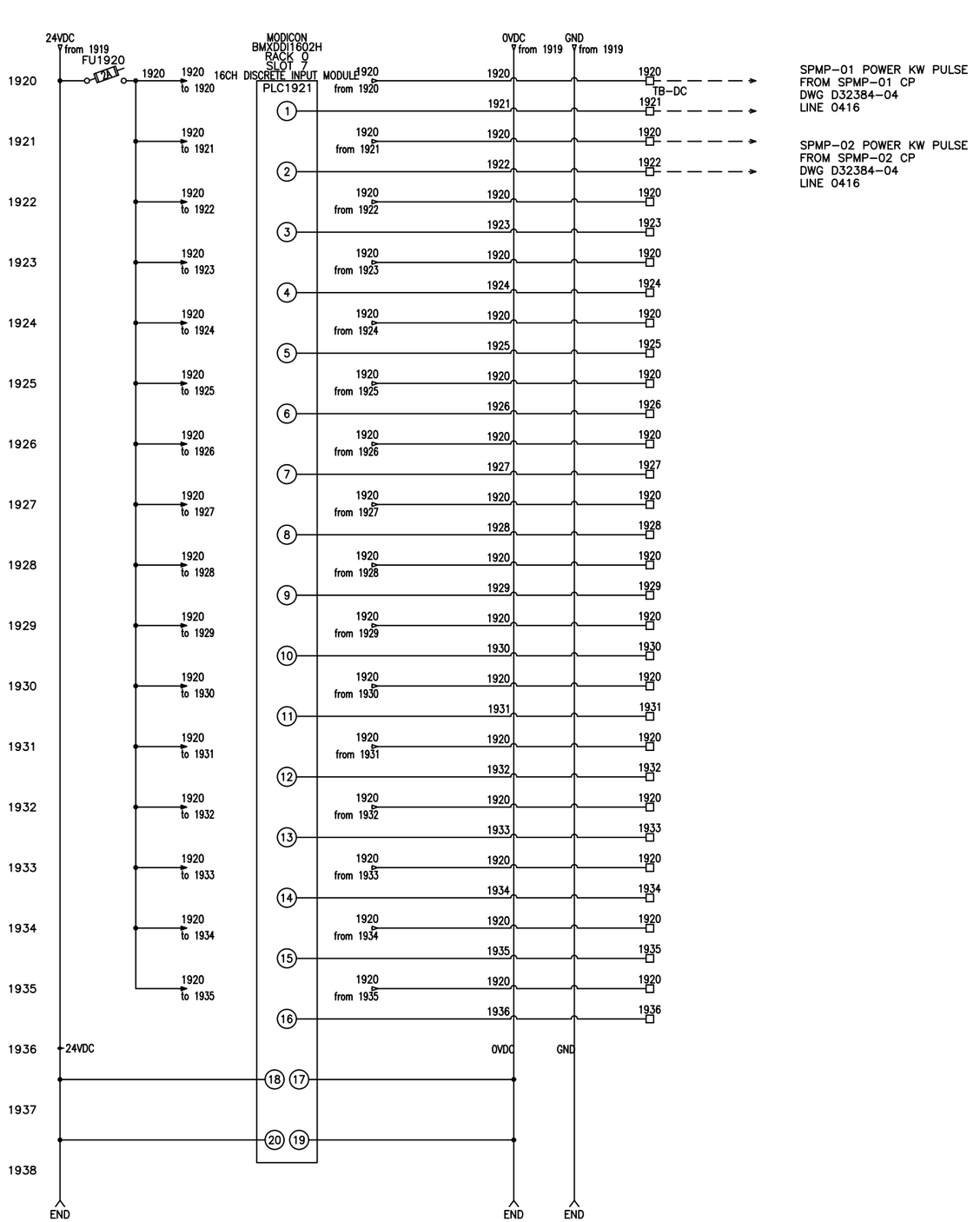
TOLERANCES	REVISIONS		
	NO.	DATE	BY
DECIMAL	1		
	2		
FRACTIONAL	3		
	4		
ANGULAR	5		
	6		

POMONKEY PUMP STATION IMPROVEMENTS
ANALOG INPUT WIRING DIAGRAM

S-1 CONTROLS

DRAWN BY ZBB	SCALE NONE	MATERIAL
CHK'D MN	DATE 11-11-2026	DRAWING NUMBER 32384-18
TRACED	APP'D	





TOLERANCES	REVISIONS			POMONKEY PUMP STATION IMPROVEMENTS		
	NO.	DATE	BY	ANALOG OUTPUT WIRING DIAGRAM		
DECIMAL	1			S&L CONTROLS		
	2					
FRACTIONAL	3			DRAWN BY ZBB	SCALE NONE	MATERIAL
	4			CHK'D MN	DATE 11-11-2026	DRAWING NUMBER 32384-19
ANGULAR	5			TRACED	APP'D	
	6					